

Private Pilot Questions

for the Knowledge Test

**Includes AC 61-117:
Private and Recreational Pilot
Knowledge Test Guide Instructions**



U.S. Department
of Transportation

**Federal Aviation
Administration**

Flight Standards Service
Washington, DC 20591

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*Private Pilot Questions
for the Knowledge Test*

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All material herein has been taken from United States government sources and contains information current as of July 13, 1995.

None of the material in this publication supersedes any documents, procedures or regulations issued by the Federal Aviation Administration. AC 61-117 has been reprinted in its entirety except for the exclusion of Sample Test Questions and Answers, and Appendix 2 sample figures.

Important: This question book should be used in conjunction with *Computerized Testing Supplement for Private and Recreational Pilots* (FAA-CT-8080-2A).

ASA reprints the FAA test figures contained within this government document, and it is available through aviation retailers nationwide. Order #ASA-CT-8080-2A.

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Private and Recreational Pilot Knowledge Test Guide

Preface

The Flight Standards Service of the Federal Aviation Administration (FAA) has developed this guide to help applicants meet the knowledge requirements for recreational pilot and private pilot certification.

This guide contains information about eligibility requirements, test descriptions, testing and retesting procedures. In addition, Appendix 1 provides a list of reference materials and subject matter knowledge codes, and computer testing designees. The list of subject matter knowledge codes should be referred to when reviewing areas of deficiency on the airman test report. Changes to the subject matter knowledge code list will be published as a separate advisory circular.

The recreational pilot and private pilot test question bank and subject matter knowledge code list for all airmen certificates and ratings, with changes, may be obtained by computer modem from FedWorld at (703) 321-8020. This bulletin board service is provided by the U.S. Department of Commerce, 24 hours a day, 7 days per week. For technical assistance regarding computer software and modem requirements for this service, contact the FedWorld help desk at (703) 487-4608 from 7:30 a.m. to 5 p.m. EST, Monday through Friday.

This publication may be purchased from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-9325 or from U.S. Government Printing Office bookstores located in major cities throughout the United States.

Comments regarding this guide should be sent to:

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Operations Support Branch, AFS-630
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Introduction

The FAA has available hundreds of computer testing centers nationwide. These testing centers offer the full range of airman knowledge tests including military competence, instrument foreign pilot, and pilot examiner predesignated tests. Refer to Appendix 1 in this guide for a list of computer testing designees.

This knowledge test guide was developed to be used by applicants preparing to take a knowledge test for the following ratings:

Recreational Pilot — Airplane

Recreational Pilot — Rotorcraft/Helicopter

Recreational Pilot — Rotorcraft/Gyroplane

Private Pilot — Airplane/Recreational Pilot–Transition

Private Pilot — Helicopter/Recreational Pilot–Transition

Private Pilot — Gyroplane/Recreational Pilot–Transition

Private Pilot — Airplane

Private Pilot — Rotorcraft/Helicopter

Private Pilot — Rotorcraft/Gyroplane

Private Pilot — Glider

Private Pilot — Free Balloon–Hot Air

Private Pilot — Free Balloon–Gas

Private Pilot — Lighter-Than-Air–Airship

This guide is not offered as a quick and easy way to obtain the necessary information for passing the knowledge tests. There is no quick and easy way to obtain this knowledge in addition to the skills needed to transform a student into a pilot capable of operating safely in our complex national airspace system. Rather, the intent of this guide is to define and narrow the field of study, while directing the applicant to the required knowledge for obtaining a recreational pilot or a private pilot certificate.

Eligibility Requirements

An applicant for a recreational pilot or private pilot certificate should review FAR Section 61.83, *Eligibility requirements: Student pilots*, for detailed information pertaining to eligibility.

An applicant for a recreational pilot certificate should review FAR Section 61.96, *Eligibility requirements: Recreational pilots*, for additional detailed information pertaining to eligibility.

An applicant for a private pilot certificate should review FAR Section 61.103, *Eligibility requirements: General*, for additional detailed information pertaining to eligibility.

Knowledge Areas on the Tests

The tests are comprehensive as they must test an applicant's knowledge in many subject areas.

An applicant for a recreational pilot certificate or added rating should review FAR Section 61.97, *Aeronautical knowledge*, for the knowledge areas on the tests.

An applicant for a private pilot certificate or added rating should review FAR Section 61.105, *Aeronautical knowledge*, for the knowledge areas on the tests.

Description of the Tests

All test questions are the objective, multiple-choice type, with three choices of answers. Each question can be answered by the selection of a single response. Each test question is independent of other questions, that is, a correct response to one does not depend upon, or influence the correct response to another.

The maximum time allowed for taking each test is based on previous experience and educational statistics. This amount of time is considered adequate for applicants with proper preparation and instruction.

The following tests each contain 50 questions and 2 hours is allowed to take each test:

- Recreational Pilot — Airplane
- Recreational Pilot — Rotorcraft/Helicopter
- Recreational Pilot — Rotorcraft/Gyroplane

The following tests each contain 30 questions and 1.5 hours is allowed to take each test:

- Private Pilot — Airplane/Recreational Pilot-Transition
- Private Pilot — Helicopter/Recreational Pilot-Transition
- Private Pilot — Gyroplane/Recreational Pilot-Transition

The following tests each contain 60 questions and 2.5 hours is allowed to take each test:

- Private Pilot — Airplane
- Private Pilot — Rotorcraft/Helicopter
- Private Pilot — Rotorcraft/Gyroplane
- Private Pilot — Glider
- Private Pilot — Free Balloon — Hot Air
- Private Pilot — Free Balloon — Gas
- Private Pilot — Lighter-Than-Air — Airship

Communication between individuals through the use of words is a complicated process. In addition to being an exercise in the application and use of aeronautical knowledge, a knowledge test is also an exercise in communication since it involves the use of the written language. Since the tests involve written rather than spoken words, communication between the test writer and the person being tested may become a difficult matter if care is not exercised by both parties. Consequently, considerable effort is expended to write each question in a clear, precise manner. Make sure you carefully read the instructions given with each test, as well as the statements in each test item.

When taking a test, keep the following points in mind:

1. Answer each question in accordance with the latest regulations and procedures.
2. Read each question carefully before looking at the possible answers. You should clearly understand the problem before attempting to solve it.
3. After formulating an answer, determine which choice most nearly corresponds with that answer. The answer chosen should completely resolve the problem.
4. From the answer given, it may appear that there is more than one possible answer. However, there is only one answer that is correct and complete. The other answers are either incomplete or are derived from popular misconceptions.
5. If a certain question is difficult for you, it is best to mark it for **RECALL** and proceed to the other questions. After you answer the less difficult questions, return to those which you marked for recall and answer them. The recall marking procedure will be explained to you prior to starting the test. Although the computer should alert you to unanswered questions, make sure every question has an answer recorded. This procedure will enable you to use the available time to the maximum advantage.
6. When solving a calculation problem, select the answer nearest to your solution. The problem has been checked with various types of calculators; therefore, if you have solved it correctly, your answer will be closer to the correct answer than any of the other choices.

Taking a Knowledge Test by Computer

You must determine what authorization requirements are necessary before going to the computer testing center. Testing center personnel cannot begin the test until you provide them with the proper authorization, if one is required. A limited number of tests require no authorization. In the case of retesting, you must present either a passed, expired passed (24 months old) or failed test report for that particular test. This policy is covered in FAA Order 8080.6, Conduct of Airmen Knowledge Tests via the Computer Medium. However, you should always check with your instructor or your local Flight Standards District Office if you are unsure of what kind of authorization to bring to the testing facility.

The next step is the actual registration process. Most computer testing centers require that all applicants contact a central 1-800 phone number. At this time, you should select a testing site of your choice, schedule a test date, and make financial arrangements for test payment.

You may register for tests several weeks in advance of the proposed testing date. You may also cancel your appointment up to 2 business days before test time, without financial penalty. After that time, you may be subject to a cancellation fee as determined by the testing center.

You are now ready to take the test. Remember, you always have an opportunity to take a sample test before the actual test begins. Your actual test is under a time limit, but if you know the material, there should be sufficient time to complete and review your test. Within moments of completing the test, you will receive an airman test report, which contains your score. It also lists those subject matter knowledge areas where questions were answered incorrectly. **The total number of subject matter knowledge codes shown on the test report is not necessarily an indication of the total number of questions answered incorrectly.** These codes refer to a list of knowledge areas that can be found in Appendix 1 of this guide. You can study these knowledge areas to improve your understanding of the subject matter. Your instructor is required to review each of the knowledge areas listed on your airman test report with you, and complete an endorsement that remedial study was conducted in these deficient areas. The examiner may also quiz you on these areas of deficiency during the practical test.

The airman test report, which must show the computer testing company's embossed seal, is an important document. **DO NOT LOSE THE AIRMAN TEST REPORT** as you will need to present it to the examiner prior to taking the practical test. Loss of this report means that you will have to request a duplicate copy from the FAA in Oklahoma City. This will be costly and time consuming.

Cheating or Other Unauthorized Conduct

Computer testing centers follow rigid testing procedures established by the FAA. This includes test security. When entering the test area, you are permitted to take only scratch paper furnished by the test administrator and an authorized aviation computer, plotter, etc., approved for use in accordance with FAA Order 8080.6, *Conduct of Airmen Knowledge Testing via the Computer Medium*, and AC 60-11, *Aids Authorized for Use by Airman Written Test Applicants*. The FAA has directed testing centers to stop a test any time a test administrator suspects a cheating incident has occurred. An FAA investigation will then follow. If the investigation determines that cheating or other unauthorized conduct has occurred, any airman certificate that you hold may be revoked, and you may not be allowed to take a test for 1 year.

Retesting Procedures

If the score on the airman test report is 70 percent or above, in most cases the report is valid for 24 calendar months. You may elect to retake the test, in anticipation of a better score, after 30 days from the date your last test was taken. Prior to retesting, you must give your current airman test report to the computer testing administrator. Remember, the score of the **latest** test you take will become the official test score. The FAA will not consider allowing anyone to retake a valid test before the 30-day remedial study period.

A person who fails a knowledge test may apply for retesting before 30 days of the last test providing that person presents the failed test report and an endorsement from an authorized instructor certifying that additional instruction has been given, and the instructor finds the person competent to pass the test. A person may retake a failed test after 30 days without an endorsement from an authorized instructor.

Appendix 1

List of Reference Materials and Subject Matter Knowledge Codes

The publications listed in the following pages contain study material you need to be familiar with when preparing for recreational pilot and private pilot knowledge tests. All of these publications can be purchased through U.S. Government bookstores, commercial aviation supply houses, or industry organizations. The latest revision of the listed references should be requested. Additional study material is also available through these sources that may be helpful in preparing for recreational pilot and private pilot knowledge tests.

The subject matter knowledge codes establish the specific reference for the knowledge standard. When reviewing results of your knowledge test, you should compare the subject matter knowledge code(s) on your airman test report to the ones found below. This will be helpful for both review and preparation for the practical test.

FAR 1	Definitions and Abbreviations	G12	Preservation of Aircraft Wreckage, Mail, Cargo, and Records
A01	General Definitions	G13	Reporting of Aircraft Accidents, Incidents, and Overdue Aircraft
A02	Abbreviations and Symbols		
FAR 43	Maintenance, Preventive Maintenance Rebuilding, and Alteration	AC 61-23	Pilot's Handbook of Aeronautical Knowledge
A15	General	H01	Principles of Flight
A16	Appendixes	H02	Airplanes and Engines
		H03	Flight Instruments
FAR 61	Certification: Pilots and Flight Instructors	H04	Airplane Performance
A20	General	H05	Weather
A21	Aircraft Ratings and Special Certificates	H06	Basic Calculations Using Navigational Computers or Electronic Calculators
A22	Student Pilots	H07	Navigation
A23	Private Pilots	H09	Appendix 1: Obtaining FAA Publications
FAR 71	Designation of Class A, Class B, Class C, Class D, and Class E Airspace Areas; Airways; Routes; and Reporting Points	AC 61-21	Flight Training Handbook
A60	General – Class A Airspace	H50	Introduction To Flight Training
		H52	Introduction to the Basics of Flight
FAR 91	General Operating and Flight Rules	H54	Ground Operations
B07	General	H58	Landing Approaches and Landings
B08	Flight Rules – General	H60	Proficiency Flight Maneuvers
B09	Visual Flight Rules	H63	Night Flying
B11	Equipment, Instrument, and Certification Requirements	H66	Principles of Flight and Performance Characteristics
B12	Special Flight Operations		
B13	Maintenance, Preventive Maintenance, and Alterations	AC 61-13	Basic Helicopter Handbook
NTSB 830	Rules Pertaining to the Notification and Reporting of Aircraft Accidents or Incidents and Overdue Aircraft, and Preservation of Aircraft Wreckage, Mail, Cargo, and Records	H70	General Aerodynamics
G10	General	H71	Aerodynamics of Flight
G11	Initial Notification of Aircraft Accidents, Incidents, and Overdue Aircraft	H73	Function of the Controls
		H74	Other Helicopter Components and Their Functions
		H76	Weight and Balance
		H78	Some Hazards of Helicopter Flight
		H79	Precautionary Measures and Critical Conditions
		H80	Helicopter Flight Maneuvers
		H81	Confined Area, Pinnacle, and Ridgeline Operations
		Gyroplane Flight Training Manual – McCulloch	
		H91	Gyroplane Terms
		H94	Basic Flight Maneuvers (Gyroplane)

AC 00-6 Aviation Weather

I21	Temperature
I22	Atmospheric Pressure and Altimetry
I23	Wind
I24	Moisture, Cloud Formation, and Precipitation
I25	Stable and Unstable Air
I26	Clouds
I27	Air Masses and Fronts
I28	Turbulence
I29	Icing
I30	Thunderstorms
I31	Common IFR Producers
I33	Arctic Weather
I35	Soaring Weather
I36	Glossary of Weather Terms

AC 00-45 Aviation Weather Services

I40	The Aviation Weather Service Program
I41	Surface Aviation Weather Reports
I42	Pilot and Radar Reports and Satellite Pictures
I43	Aviation Weather Forecasts
I44	Surface Analysis Chart
I45	Weather Depiction Chart
I46	Radar Summary Chart
I47	Significant Weather Prognostics

AIM Aeronautical Information Manual

J01	Air Navigation Radio Aids
J03	Airport Lighting Aids
J05	Airport Marking Aids and Signs
J08	Controlled Airspace
J09	Special Use Airspace
J10	Other Airspace Areas
J11	Service Available to Pilots
J12	Radio Communications Phraseology and Techniques
J13	Airport Operations
J14	ATC Clearance/Separations
J15	Preflight
J22	Emergency Procedures Available to Pilots
J25	Meteorology
J27	Wake Turbulence
J31	Fitness for Flight
J33	Pilot Controller Glossary
J34	Airport/Facility Directory
J37	Sectional Chart

AC 67-2 Medical Handbook for Pilots

J52	Hypoxia
J53	Hyperventilation
J58	Carbon Monoxide
J60	Night Flight
J61	Cockpit Lighting
J62	Disorientation (Vertigo)

Additional Advisory Circulars

M52	AC 00-2, Advisory Circular Checklist
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American Soaring Handbook — Soaring Society of America

N03	Ground Launch
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Soaring Flight Manual — Jeppesen-Sanderson, Inc.

N20	Sailplane Aerodynamics
N21	Performance Considerations
N22	Flight Instruments
N23	Weather for Soaring
N27	Computations for Soaring
N30	Aerotow Launch Procedures
N32	Basic Flight Maneuvers and Traffic
N34	Cross-Country Soaring

Taming The Gentle Giant — Taylor Publishing

O01	Design and Construction of Balloons
O02	Fuel Source and Supply
O03	Weight and Temperature
O05	Balloon Flight Tips

Flight Instructor Manual — Balloon Federation of America

O10	Flight Instruction Aids
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Propane Systems — Balloon Federation of America, 1991

O24	Repair and Maintenance
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Powerline Excerpts — Balloon Federation of America

O30	Excerpts
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Goodyear Airship Operations Manual

P01	Buoyancy
P03	Free Ballooning
P04	Aerostatics

SUPPLEMENTAL STUDY REFERENCE MATERIALS**FAA Accident Prevention Program Bulletins**

V01	FAA-P-8740-2, Density Altitude
V02	FAA-P-8740-5, Weight and Balance
V03	FAA-P-8740-12, Thunderstorms
V04	FAA-P-8740-19, Flying Light Twins Safely
V05	FAA-P-8740-23, Planning Your Takeoff
V06	FAA-P-8740-24, Tips on Winter Flying
V07	FAA-P-8740-25, Always Leave Yourself an Out
V08	FAA-P-8740-30, How to Obtain a Good Weather Briefing
V09	FAA-P-8740-40, Wind Shear
V10	FAA-P-8740-41, Medical Facts For Pilots
V11	FAA-P-8740-44, Impossible Turns
V12	FAA-P-8740-48, On Landings, Part I
V13	FAA-P-8740-49, On Landings, Part II
V14	FAA-P-8740-50, On Landings, Part III
V15	FAA-P-8740-51, How to Avoid a Midair Collision
V16	FAA-P-8740-52, The Silent Emergency

Note: AC 00-2, Advisory Circular Checklist, transmits the status of all FAA advisory circulars (ACs), as well as FAA internal publications and miscellaneous flight information such as Aeronautical Information Manual (AIM), Airport/Facility Directory, practical test standards, knowledge test guides, and other material directly related to airman certificates and ratings. To obtain a free copy of AC 00-2, send your request to:

U.S. Department of Transportation
General Services Section, M-45.3
Washington, DC 20590

Computer Testing Designees

The following is a list of the computer testing designees authorized to give FAA knowledge tests. This list should be helpful in choosing where to register for a test or for requesting additional information.

Aviation Business Services
1-800-947-4228
outside U.S. (415) 259-8550

Drake Prometric
1-800-359-3278
outside U.S. (612) 896-7702

Sylvan Learning Systems, Inc.
1-800-967-1100
outside U.S. (410) 880-0880, Extension 8890

The latest listing of computer testing center locations may be obtained through FedWorld, (703) 321-8020, in the FAA library file named TST_SITE. For technical assistance, contact the FedWorld help desk at (703) 487-4608.

Private Pilot

Knowledge Test Questions

Note: Figures referred to in these FAA questions can be found in the separate booklet, *Computerized Testing Supplement for Recreational Pilot and Private Pilot* (#ASA-CT-8080-2A).

3001. With respect to the certification of airmen, which is a category of aircraft?

- A—Gyroplane, helicopter, airship, free balloon.
- B—Airplane, rotorcraft, glider, lighter-than-air.
- C—Single-engine land and sea, multiengine land and sea.

3002. With respect to the certification of airmen, which is a class of aircraft?

- A—Airplane, rotorcraft, glider, lighter-than-air.
- B—Single-engine land and sea, multiengine land and sea.
- C—Lighter-than-air, airship, hot air balloon, gas balloon.

3003. With respect to the certification of aircraft, which is a category of aircraft?

- A—Normal, utility, acrobatic.
- B—Airplane, rotorcraft, glider.
- C—Landplane, seaplane.

3004. With respect to the certification of aircraft, which is a class of aircraft?

- A—Airplane, helicopter, glider, hot air balloon.
- B—Normal, utility, acrobatic, limited.
- C—Transport, restricted, provisional.

3005. The definition of nighttime is

- A—sunset to sunrise.
- B—1 hour after sunset to 1 hour before sunrise.
- C—the time between the end of evening civil twilight and the beginning of morning civil twilight.

3006. Which V-speed represents maneuvering speed?

- A— V_A .
- B— V_{LO} .
- C— V_{NE} .

3007. Which V-speed represents maximum flap extended speed?

- A— V_{FE} .
- B— V_{LOF} .
- C— V_{FC} .

3008. Which V-speed represents maximum landing gear extended speed?

- A— V_{LE} .
- B— V_{LO} .
- C— V_{FE} .

3009. V_{NO} is defined as the

- A—normal operating range.
- B—never-exceed speed.
- C—maximum structural cruising speed.

3010. V_{SO} is defined as the

- A—stalling speed or minimum steady flight speed in the landing configuration.
- B—stalling speed or minimum steady flight speed in a specified configuration.
- C—stalling speed or minimum takeoff safety speed.

3011. Which would provide the greatest gain in altitude in the shortest distance during climb after takeoff?

- A— V_Y .
- B— V_X .
- C— V_X .

3012. After takeoff, which airspeed would the pilot use to gain the most altitude in a given period of time?

- A— V_Y .
- B— V_X .
- C— V_A .

3013. Preventive maintenance has been performed on an aircraft. What paperwork is required?

- A—A full, detailed description of the work done must be entered in the airframe logbook.
- B—The date the work was completed, and the name of the person who did the work must be entered in the airframe and engine logbook.
- C—The signature, certificate number, and kind of certificate held by the person approving the work and a description of the work must be entered in the aircraft maintenance records.

3014. Which operation would be described as preventive maintenance?

- A—Servicing landing gear wheel bearings.
- B—Alteration of main seat support brackets.
- C—Engine adjustments to allow automotive gas to be used.

3015. Which operation would be described as preventive maintenance?

- A—Repair of landing gear brace struts.
- B—Replenishing hydraulic fluid.
- C—Repair of portions of skin sheets by making additional seams.

3016. What document(s) must be in your personal possession while operating as pilot in command of an aircraft?

- A—Certificates showing accomplishment of a checkout in the aircraft and a current biennial flight review.
- B—A pilot certificate with an endorsement showing accomplishment of an annual flight review and a pilot logbook showing recency of experience.
- C—An appropriate pilot certificate and an appropriate current medical certificate.

3017. When must a current pilot certificate be in the pilot's personal possession?

- A—When acting as a crew chief during launch and recovery.
- B—Only when passengers are carried.
- C—Anytime when acting as pilot in command or as a required crewmember.

3018. Private pilots acting as pilot in command, or in any other capacity as a required pilot flight crewmember, must have in their personal possession while aboard the aircraft a current

- A—logbook endorsement to show that a flight review has been satisfactorily accomplished.
- B—medical certificate and an appropriate pilot certificate.
- C—endorsement on the pilot certificate to show that a flight review has been satisfactorily accomplished.

3019. Each person who holds a pilot certificate or a medical certificate shall present it for inspection upon the request of the Administrator, the National Transportation Safety Board, or any

- A—authorized representative of the Department of Transportation.
- B—person in a position of authority.
- C—federal, state, or local law enforcement officer.

3020. A Third-Class Medical Certificate is issued on August 10, this year. To exercise the privileges of a Private Pilot Certificate, the medical certificate will be valid until midnight on

- A—August 10, 2 years later.
- B—August 31, 2 years later.
- C—August 31, 3 years later.

3021. A Third-Class Medical Certificate is issued on May 3, this year. To exercise the privileges of a Private Pilot Certificate, the medical certificate will be valid until midnight on

- A—May 3, 1 year later.
- B—May 31, 1 year later.
- C—May 31, 2 years later.

3022. For private pilot operations, a Second-Class Medical Certificate issued on July 15, this year, will expire at midnight on

- A—July 15, 2 years later.
- B—July 31, 1 year later.
- C—July 31, 2 years later.

3023. For private pilot operations, a First-Class Medical Certificate issued on October 21, this year, will expire at midnight on

- A—October 21, 2 years later.
- B—October 31, next year.
- C—October 31, 2 years later.

3024. The pilot in command is required to hold a type rating in which aircraft?

- A—Aircraft operated under an authorization issued by the Administrator.
- B—Aircraft having a gross weight of more than 12,500 pounds.
- C—Aircraft involved in ferry flights, training flights, or test flights.

3025. What is the definition of a high-performance airplane?

- A—An airplane with 180 horsepower, or retractable landing gear, flaps, and a fixed-pitch propeller.
- B—An airplane with more than 200 horsepower, or retractable landing gear, flaps, and a controllable propeller.
- C—An airplane with a normal cruise speed in excess of 200 knots, flaps, and a controllable propeller.

3026. Before a person holding a Private Pilot Certificate may act as pilot in command of a high-performance airplane, that person must have

- A—passed a flight test in that airplane from an FAA inspector.
- B—an endorsement in that person's logbook that he/she is competent to act as pilot in command.
- C—received flight instruction from an authorized flight instructor who then endorses that person's logbook.

3027. In order to act as pilot in command of a high-performance airplane, a pilot must have

- A—made three solo takeoffs and landings in a high-performance airplane.
- B—received flight instruction in an airplane that has more than 200 horsepower, or retractable landing gear, flaps, and a controllable propeller.
- C—passed a flight test in a high-performance airplane.

3028. To act as pilot in command of an aircraft carrying passengers, a pilot must show by logbook endorsement the satisfactory completion of a flight review or completion of a pilot proficiency check within the preceding

- A—6 calendar months.
- B—12 calendar months.
- C—24 calendar months.

3029. If recency of experience requirements for night flight are not met and official sunset is 1830, the latest time passengers may be carried is

- A—1829.
- B—1859.
- C—1929.

3030. To act as pilot in command of an aircraft carrying passengers, the pilot must have made at least three takeoffs and three landings in an aircraft of the same category, class, and if a type rating is required, of the same type, within the preceding

- A—90 days.
- B—12 calendar months.
- C—24 calendar months.

3031. To act as pilot in command of an aircraft carrying passengers, the pilot must have made three takeoffs and three landings within the preceding 90 days in an aircraft of the same

- A—make and model.
- B—category and class, but not type.
- C—category, class, and type, if a type rating is required.

3032. The takeoffs and landings required to meet the recency of experience requirements for carrying passengers in a tailwheel airplane

- A—may be touch and go or full stop.
- B—must be touch and go.
- C—must be to a full stop.

3033. The three takeoffs and landings that are required to act as pilot in command at night must be done during the time period from

- A—sunset to sunrise.
- B—1 hour after sunset to 1 hour before sunrise.
- C—the end of evening civil twilight to the beginning of morning civil twilight.

3034. To meet the recency of experience requirements to act as pilot in command carrying passengers at night, a pilot must have made at least three takeoffs and three landings to a full stop within the preceding 90 days in

- A—the same category and class of aircraft to be used.
- B—the same type of aircraft to be used.
- C—any aircraft.

3035. If a certificated pilot changes permanent mailing address and fails to notify the FAA Airmen Certification Branch of the new address, the pilot is entitled to exercise the privileges of the pilot certificate for a period of only

- A—30 days after the date of the move.
- B—60 days after the date of the move.
- C—90 days after the date of the move.

3036. A certificated private pilot may not act as pilot in command of an aircraft towing a glider unless there is entered in the pilot's logbook a minimum of

- A—100 hours of pilot flight time in any aircraft.
- B—100 hours of pilot flight time in powered aircraft.
- C—200 hours of pilot flight time in powered aircraft.

3037. To act as pilot in command of an aircraft towing a glider, a person is required to have made within the preceding 12 months

- A—at least three flights as observer in a glider being towed by an aircraft.
- B—at least three flights in a powered glider.
- C—at least three actual or simulated glider tows while accompanied by a qualified pilot.

3038. A recreational pilot acting as pilot in command must have in his/her personal possession while aboard the aircraft

- A—a current logbook endorsement to show that a flight review has been satisfactorily accomplished.
- B—the current and appropriate pilot and medical certificates.
- C—the pilot logbook to show recent experience requirements to serve as pilot in command have been met.

3039. A Third-Class Medical Certificate was issued on August 10, this year. To exercise the privileges of a Recreational Pilot Certificate, the medical certificate will expire at midnight on

- A—August 10, 2 years later.
- B—August 31, 2 years later.
- C—August 31, 3 years later.

3040. If a recreational pilot had a flight review on August 8, this year, when is the next flight review required?

- A—August 8, 2 years later.
- B—August 31, next year.
- C—August 31, 2 years later.

3041. Each recreational pilot is required to have

- A—a biennial flight review.
- B—an annual flight review.
- C—a semiannual flight review.

3042. If a recreational pilot had a flight review on August 8, this year, when is the next flight review required?

- A—August 8, next year.
- B—August 31, next year.
- C—August 31, 2 years later.

3043. How many passengers is a recreational pilot allowed to carry on board?

- A—One.
- B—Two.
- C—Three.

3044. According to regulations pertaining to privileges and limitations, a recreational pilot may

- A—be paid for the operating expenses of a flight.
- B—share the operating expenses of a flight with a passenger.
- C—not be paid in any manner for the operating expenses of a flight.

3045. In regard to privileges and limitations, a recreational pilot may

- A—fly for compensation or hire.
- B—share the operating expenses of the flight with the passenger.
- C—not be paid in any manner for the operating expenses of a flight.

3046. What is the maximum distance recreational pilots may fly from the airport/heliport at which they received instruction?

- A—25 nautical miles.
- B—50 nautical miles.
- C—100 nautical miles.

3047. A recreational pilot may act as pilot in command of an aircraft that is certificated for a maximum of how many occupants?

- A—Two.
- B—Three.
- C—Four.

3048. A recreational pilot may act as pilot in command of an aircraft with a maximum engine horsepower of

- A—160.
- B—180.
- C—200.

3049. What exception, if any, permits a recreational pilot to act as pilot in command of an aircraft carrying a passenger for hire?

- A—If the passenger pays no more than the operating expenses.
- B—If a donation is made to a charitable organization for the flight.
- C—There is no exception.

3050. May a recreational pilot act as pilot in command of an aircraft in furtherance of a business?

- A—Yes, if the flight is only incidental to that business.
- B—Yes, providing the aircraft does not carry a person or property for compensation or hire.
- C—No, it is not allowed.

3051. With respect to daylight hours, what is the earliest time a recreational pilot may take off?

- A—One hour before sunrise.
- B—At sunrise.
- C—At the beginning of morning civil twilight.

3052. If sunset is 2021 and the end of evening civil twilight is 2043, when must a recreational pilot terminate the flight?

- A—2021.
- B—2043.
- C—2121.

3053. When may a recreational pilot operate to or from an airport that lies within Class C airspace?

- A—Anytime the control tower is in operation.
- B—When the ceiling is at least 1,000 feet and the surface visibility is at least 3 miles.
- C—For the purpose of obtaining an additional certificate or rating while under the supervision of an authorized flight instructor.

3054. Under what conditions may a recreational pilot operate at an airport that lies within Class D airspace and that has a part-time control tower in operation?

- A—When the tower is in operation, the ceiling is at least 2,500 feet, and the visibility is at least 3 miles.
- B—When the tower is in operation, the ceiling is at least 3,000 feet, and the visibility is more than 1 mile.
- C—When the tower is closed, the ceiling is at least 1,000 feet, and the visibility is at least 3 miles.

3055. When may a recreational pilot fly above 10,000 feet MSL?

- A—When 2,000 feet AGL or below.
- B—When 2,500 feet AGL or below.
- C—When outside of controlled airspace.

3056. During daytime, what is the minimum flight or surface visibility required for recreational pilots in Class G airspace below 10,000 feet MSL?

- A—1 mile.
- B—3 miles.
- C—5 miles.

3057. During daytime, what is the minimum flight visibility required for recreational pilots in controlled airspace below 10,000 feet MSL?

- A—1 mile.
- B—3 miles.
- C—5 miles.

3058. Under what conditions, if any, may a recreational pilot demonstrate an aircraft in flight to a prospective buyer?

- A—The buyer pays all the operating expenses.
- B—The flight is not outside the United States.
- C—None.

3059. When, if ever, may a recreational pilot act as pilot in command in an aircraft towing a banner?

- A—If the pilot has logged 100 hours of flight time in powered aircraft.
- B—If the pilot has an endorsement in his/her pilot logbook from an authorized flight instructor.
- C—It is not allowed.

3060. When must a recreational pilot have a pilot-in-command flight check?

- A—Every 400 hours.
- B—Every 180 days.
- C—If the pilot has less than 400 total flight hours and has not flown as pilot in command in an aircraft within the preceding 180 days.

3061. A recreational pilot may fly as sole occupant of an aircraft at night while under the supervision of a flight instructor provided the flight or surface visibility is at least

- A—3 miles.
- B—4 miles.
- C—5 miles.

3062. Prior to becoming certified as a private pilot with a glider rating, the pilot must have at least

- A—passed a third-class medical exam.
- B—obtained a statement from a designated medical examiner.
- C—made a statement certifying that he/she has no known medical deficiency that would make him/her unable to act as pilot.

3063. Prior to becoming certified as a private pilot with a balloon rating, the pilot must have at least

- A—passed a third-class medical exam.
- B—obtained a statement from a designated medical examiner.
- C—made a statement certifying that he/she has no known medical deficiency that would make him/her unable to act as pilot.

3064. In regard to general privileges and limitations, a private pilot may

- A—act as pilot in command of an aircraft carrying a passenger for compensation if the flight is in connection with a business or employment.
- B—share the operating expenses of a flight with a passenger.
- C—not be paid in any manner for the operating expenses of a flight.

3065. According to regulations pertaining to general privileges and limitations, a private pilot may

- A—be paid for the operating expenses of a flight if at least three takeoffs and three landings were made by the pilot within the preceding 90 days.
- B—share the operating expenses of a flight with the passengers.
- C—not be paid in any manner for the operating expenses of a flight.

3066. What exception, if any, permits a private pilot to act as pilot in command of an aircraft carrying passengers who pay for the flight?

- A—If the passengers pay all the operating expenses.
- B—If a donation is made to a charitable organization for the flight.
- C—There is no exception.

3067. The width of a Federal Airway from either side of the centerline is

- A—4 nautical miles.
- B—6 nautical miles.
- C—8 nautical miles.

3068. Unless otherwise specified, Federal Airways include that Class E airspace extending upward from

- A—700 feet above the surface up to and including 17,999 feet MSL.
- B—1,200 feet above the surface up to and including 17,999 feet MSL.
- C—the surface up to and including 18,000 feet MSL.

3069. Normal VFR operations in Class D airspace with an operating control tower require the ceiling and visibility to be at least

- A—1,000 feet and 1 mile.
- B—1,000 feet and 3 miles.
- C—2,500 feet and 3 miles.

3070. The final authority as to the operation of an aircraft is the

- A—Federal Aviation Administration.
- B—pilot in command.
- C—aircraft manufacturer.

3071. The person directly responsible for the pre-launch briefing of passengers for a flight is the

- A—safety officer.
- B—pilot in command.
- C—ground crewmember.

3072. If an in-flight emergency requires immediate action, the pilot in command may

- A—deviate from the FAR's to the extent required to meet the emergency, but must submit a written report to the Administrator within 24 hours.
- B—deviate from the FAR's to the extent required to meet that emergency.
- C—not deviate from the FAR's unless prior to the deviation approval is granted by the Administrator.

3073. When must a pilot who deviates from a regulation during an emergency send a written report of that deviation to the Administrator?

- A—Within 7 days.
- B—Within 10 days.
- C—Upon request.

3074. Who is responsible for determining if an aircraft is in condition for safe flight?

- A—A certificated aircraft mechanic.
- B—The pilot in command.
- C—The owner or operator.

3075. Where may an aircraft's operating limitations be found?

- A—On the Airworthiness Certificate.
- B—In the current, FAA-approved flight manual, approved manual material, markings, and placards, or any combination thereof.
- C—In the aircraft airframe and engine logbooks.

3076. Under what conditions may objects be dropped from an aircraft?

- A—Only in an emergency.
- B—If precautions are taken to avoid injury or damage to persons or property on the surface.
- C—If prior permission is received from the Federal Aviation Administration.

3077. A person may not act as a crewmember of a civil aircraft if alcoholic beverages have been consumed by that person within the preceding

- A—8 hours.
- B—12 hours.
- C—24 hours.

3078. Under what condition, if any, may a pilot allow a person who is obviously under the influence of drugs to be carried aboard an aircraft?

- A—In an emergency or if the person is a medical patient under proper care.
- B—Only if the person does not have access to the cockpit or pilot's compartment.
- C—Under no condition.

3079. No person may attempt to act as a crewmember of a civil aircraft with

- A—.008 percent by weight or more alcohol in the blood.
- B—.004 percent by weight or more alcohol in the blood.
- C—.04 percent by weight or more alcohol in the blood.

3080. Which preflight action is specifically required of the pilot prior to each flight?

- A—Check the aircraft logbooks for appropriate entries.
- B—Become familiar with all available information concerning the flight.
- C—Review wake turbulence avoidance procedures.

3081. Preflight action, as required for all flights away from the vicinity of an airport, shall include

- A—the designation of an alternate airport.
- B—a study of arrival procedures at airports/ heliports of intended use.
- C—an alternate course of action if the flight cannot be completed as planned.

3082. In addition to other preflight actions for a VFR flight away from the vicinity of the departure airport, regulations specifically require the pilot in command to

- A—review traffic control light signal procedures.
- B—check the accuracy of the navigation equipment and the emergency locator transmitter (ELT).
- C—determine runway lengths at airports of intended use and the aircraft's takeoff and landing distance data.

3083. Flight crewmembers are required to keep their safety belts and shoulder harnesses fastened during

- A—takeoffs and landings.
- B—all flight conditions.
- C—flight in turbulent air.

3084. Which best describes the flight conditions under which flight crewmembers are specifically required to keep their safety belts and shoulder harnesses fastened?

- A—Safety belts during takeoff and landing; shoulder harnesses during takeoff and landing.
- B—Safety belts during takeoff and landing; shoulder harnesses during takeoff and landing and while en route.
- C—Safety belts during takeoff and landing and while en route; shoulder harnesses during takeoff and landing.

3085. With respect to passengers, what obligation, if any, does a pilot in command have concerning the use of safety belts?

- A—The pilot in command must instruct the passengers to keep their safety belts fastened for the entire flight.
- B—The pilot in command must brief the passengers on the use of safety belts and notify them to fasten their safety belts during taxi, takeoff, and landing.
- C—The pilot in command has no obligation in regard to passengers' use of safety belts.

3086. With certain exceptions, safety belts are required to be secured about passengers during

- A—taxi, takeoffs, and landings.
- B—all flight conditions.
- C—flight in turbulent air.

3087. Safety belts are required to be properly secured about which persons in an aircraft and when?

- A—Pilots only, during takeoffs and landings.
- B—Passengers, during taxi, takeoffs, and landings only.
- C—Each person on board the aircraft during the entire flight.

3088. No person may operate an aircraft in formation flight

- A—over a densely populated area.
- B—in Class D airspace under special VFR.
- C—except by prior arrangement with the pilot in command of each aircraft.

3089. Which aircraft has the right-of-way over all other air traffic?

- A—A balloon.
- B—An aircraft in distress.
- C—An aircraft on final approach to land.

3090. What action is required when two aircraft of the same category converge, but not head-on?

- A—The faster aircraft shall give way.
- B—The aircraft on the left shall give way.
- C—Each aircraft shall give way to the right.

3091. Which aircraft has the right-of-way over the other aircraft listed?

- A—Glider.
- B—Airship.
- C—Aircraft refueling other aircraft.

3092. An airplane and an airship are converging. If the airship is left of the airplane's position, which aircraft has the right-of-way?

- A—The airship.
- B—The airplane.
- C—Each pilot should alter course to the right.

3093. Which aircraft has the right-of-way over the other aircraft listed?

- A—Airship.
- B—Aircraft towing other aircraft.
- C—Gyroplane.

3094. What action should the pilots of a glider and an airplane take if on a head-on collision course?

- A—The airplane pilot should give way to the left.
- B—The glider pilot should give way to the right.
- C—Both pilots should give way to the right.

3095. When two or more aircraft are approaching an airport for the purpose of landing, the right-of-way belongs to the aircraft

- A—that has the other to its right.
- B—that is the least maneuverable.
- C—at the lower altitude, but it shall not take advantage of this rule to cut in front of or to overtake another.

3096. A seaplane and a motorboat are on crossing courses. If the motorboat is to the left of the seaplane, which has the right-of-way?

- A—The motorboat.
- B—The seaplane.
- C—Both should alter course to the right.

3097. Unless otherwise authorized, what is the maximum indicated airspeed at which a person may operate an aircraft below 10,000 feet MSL?

- A—200 knots.
- B—250 knots.
- C—288 knots.

3098. Unless otherwise authorized, the maximum indicated airspeed at which aircraft may be flown when at or below 2,500 feet AGL and within 4 nautical miles of the primary airport of Class C airspace is

- A—200 knots.
- B—230 knots.
- C—250 knots.

3099. When flying in the airspace underlying Class B airspace, the maximum speed authorized is

- A—200 knots.
- B—230 knots.
- C—250 knots.

3100. When flying in a VFR corridor designated through Class B airspace, the maximum speed authorized is

- A—180 knots.
- B—200 knots.
- C—250 knots.

3101. Except when necessary for takeoff or landing, what is the minimum safe altitude for a pilot to operate an aircraft anywhere?

- A—An altitude allowing, if a power unit fails, an emergency landing without undue hazard to persons or property on the surface.
- B—An altitude of 500 feet above the surface and no closer than 500 feet to any person, vessel, vehicle, or structure.
- C—An altitude of 500 feet above the highest obstacle within a horizontal radius of 1,000 feet.

3102. Except when necessary for takeoff or landing, what is the minimum safe altitude required for a pilot to operate an aircraft over congested areas?

- A—An altitude of 1,000 feet above any person, vessel, vehicle, or structure.
- B—An altitude of 500 feet above the highest obstacle within a horizontal radius of 1,000 feet of the aircraft.
- C—An altitude of 1,000 feet above the highest obstacle within a horizontal radius of 2,000 feet of the aircraft.

3103. Except when necessary for takeoff or landing, what is the minimum safe altitude required for a pilot to operate an aircraft over other than a congested area?

- A—An altitude allowing, if a power unit fails, an emergency landing without undue hazard to persons or property on the surface.
- B—An altitude of 500 feet AGL, except over open water or a sparsely populated area, which requires 500 feet from any person, vessel, vehicle, or structure.
- C—An altitude of 500 feet above the highest obstacle within a horizontal radius of 1,000 feet.

3104. Except when necessary for takeoff or landing, an aircraft may not be operated closer than what distance from any person, vessel, vehicle, or structure?

- A—500 feet.
- B—700 feet.
- C—1,000 feet.

3105. If an altimeter setting is not available before flight, to which altitude should the pilot adjust the altimeter?

- A—The elevation of the nearest airport corrected to mean sea level.
- B—The elevation of the departure area.
- C—Pressure altitude corrected for nonstandard temperature.

3106. Prior to takeoff, the altimeter should be set to which altitude or altimeter setting?

- A—The current local altimeter setting, if available, or the departure airport elevation.
- B—The corrected density altitude of the departure airport.
- C—The corrected pressure altitude for the departure airport.

3107. At what altitude shall the altimeter be set to 29.92, when climbing to cruising flight level?

- A—14,500 feet MSL.
- B—18,000 feet MSL.
- C—24,000 feet MSL.

3108. When an ATC clearance has been obtained, no pilot in command may deviate from that clearance, unless that pilot obtains an amended clearance. The one exception to this regulation is

- A—when the clearance states "at pilot's discretion."
- B—an emergency.
- C—if the clearance contains a restriction.

3109. When would a pilot be required to submit a detailed report of an emergency which caused the pilot to deviate from an ATC clearance?

- A—When requested by ATC.
- B—Immediately.
- C—Within 7 days.

3110. What action, if any, is appropriate if the pilot deviates from an ATC instruction during an emergency and is given priority?

- A—Take no special action since you are pilot in command.
- B—File a detailed report within 48 hours to the chief of the appropriate ATC facility, if requested.
- C—File a report to the FAA Administrator, as soon as possible.

3111. A steady green light signal directed from the control tower to an aircraft in flight is a signal that the pilot

- A—is cleared to land.
- B—should give way to other aircraft and continue circling.
- C—should return for landing.

3112. Which light signal from the control tower clears a pilot to taxi?

- A—Flashing green.
- B—Steady green.
- C—Flashing white.

3113. If the control tower uses a light signal to direct a pilot to give way to other aircraft and continue circling, the light will be

- A—flashing red.
- B—steady red.
- C—alternating red and green.

3114. A flashing white light signal from the control tower to a taxiing aircraft is an indication to

- A—taxi at a faster speed.
- B—taxi only on taxiways and not cross runways.
- C—return to the starting point on the airport.

3115. An alternating red and green light signal directed from the control tower to an aircraft in flight is a signal to

- A—hold position.
- B—exercise extreme caution.
- C—not land; the airport is unsafe.

3116. While on final approach for landing, an alternating green and red light followed by a flashing red light is received from the control tower. Under these circumstances, the pilot should

- A—discontinue the approach, fly the same traffic pattern and approach again, and land.
- B—exercise extreme caution and abandon the approach, realizing the airport is unsafe for landing.
- C—abandon the approach, circle the airport to the right, and expect a flashing white light when the airport is safe for landing.

3117. A blue segmented circle on a Sectional Chart depicts which class airspace?

- A—Class B.
- B—Class C.
- C—Class D.

3118. Airspace at an airport with a part-time control tower is classified as Class D airspace only

- A—when the weather minimums are below basic VFR.
- B—when the associated control tower is in operation.
- C—when the associated Flight Service Station is in operation.

3119. Unless otherwise authorized, two-way radio communications with Air Traffic Control are required for landings or takeoffs

- A—at all tower controlled airports regardless of weather conditions.
- B—at all tower controlled airports only when weather conditions are less than VFR.
- C—at all tower controlled airports within Class D airspace only when weather conditions are less than VFR.

3120. Each pilot of an aircraft approaching to land on a runway served by a visual approach slope indicator (VASI) shall

- A—maintain a 3° glide to the runway.
- B—maintain an altitude at or above the glide slope.
- C—stay high until the runway can be reached in a power-off landing.

3121. When approaching to land on a runway served by a visual approach slope indicator (VASI), the pilot shall

- A—maintain an altitude that captures the glide slope at least 2 miles downwind from the runway threshold.
- B—maintain an altitude at or above the glide slope.
- C—remain on the glide slope and land between the two-light bar.

3122. Which is appropriate for a helicopter approaching an airport for landing?

- A—Remain below the airplane traffic pattern altitude.
- B—Avoid the flow of fixed-wing traffic.
- C—Fly right-hand traffic.

3123. Which is the correct traffic pattern departure procedure to use at a noncontrolled airport?

- A—Depart in any direction consistent with safety, after crossing the airport boundary.
- B—Make all turns to the left.
- C—Comply with any FAA traffic pattern established for the airport.

3124. Two-way radio communication must be established with the Air Traffic Control facility having jurisdiction over the area prior to entering which class airspace?

- A—Class C.
- B—Class E.
- C—Class G.

3125. What minimum radio equipment is required for operation within Class C airspace?

- A—Two-way radio communications equipment and a 4096-code transponder.
- B—Two-way radio communications equipment, a 4096-code transponder, and DME.
- C—Two-way radio communications equipment, a 4096-code transponder, and an encoding altimeter.

3126. What minimum pilot certification is required for operation within Class B airspace?

- A—Recreational Pilot Certificate.
- B—Private Pilot Certificate or Student Pilot Certificate with appropriate logbook endorsements.
- C—Private Pilot Certificate with an instrument rating.

3127. What minimum pilot certification is required for operation within Class B airspace?

- A—Private Pilot Certificate or Student Pilot Certificate with appropriate logbook endorsements.
- B—Commercial Pilot Certificate.
- C—Private Pilot Certificate with an instrument rating.

3128. What minimum radio equipment is required for VFR operation within Class B airspace?

- A—Two-way radio communications equipment and a 4096-code transponder.
- B—Two-way radio communications equipment, a 4096-code transponder, and an encoding altimeter.
- C—Two-way radio communications equipment, a 4096-code transponder, an encoding altimeter, and a VOR or TACAN receiver.

3129. An operable 4096-code transponder and Mode C encoding altimeter are required in

- A—Class B airspace and within 30 miles of the Class B primary airport.
- B—Class D airspace.
- C—Class E airspace below 10,000 feet MSL.

3130. In which type of airspace are VFR flights prohibited?

- A—Class A.
- B—Class B.
- C—Class C.

3131. What is the specific fuel requirement for flight under VFR during daylight hours in an airplane?

- A—Enough to complete the flight at normal cruising speed with adverse wind conditions.
- B—Enough to fly to the first point of intended landing and to fly after that for 30 minutes at normal cruising speed.
- C—Enough to fly to the first point of intended landing and to fly after that for 45 minutes at normal cruising speed.

3132. What is the specific fuel requirement for flight under VFR at night in an airplane?

- A—Enough to complete the flight at normal cruising speed with adverse wind conditions.
- B—Enough to fly to the first point of intended landing and to fly after that for 30 minutes at normal cruising speed.
- C—Enough to fly to the first point of intended landing and to fly after that for 45 minutes at normal cruising speed.

3133. No person may begin a flight in a rotorcraft under VFR unless there is enough fuel to fly to the first point of intended landing and, assuming normal cruising speed, to fly thereafter for at least

- A—20 minutes.
- B—30 minutes.
- C—1 hour.

3134. What minimum visibility and clearance from clouds are required for a recreational pilot in Class G airspace at 1,200 feet AGL or below during daylight hours?

- A—1 mile visibility and clear of clouds.
- B—3 miles visibility and clear of clouds.
- C—3 miles visibility, 500 feet below the clouds.

3135. Outside controlled airspace, the minimum flight visibility requirement for a recreational pilot flying VFR above 1,200 feet AGL and below 10,000 feet MSL during daylight hours is

- A—1 mile.
- B—3 miles.
- C—5 miles.

3136. During operations within controlled airspace at altitudes of less than 1,200 feet AGL, the minimum horizontal distance from clouds requirement for VFR flight is

- A—1,000 feet.
- B—1,500 feet.
- C—2,000 feet.

3137. What minimum visibility and clearance from clouds are required for VFR operations in Class G airspace at 700 feet AGL or below during daylight hours?

- A—1 mile visibility and clear of clouds.
- B—1 mile visibility, 500 feet below, 1,000 feet above, and 2,000 feet horizontal clearance from clouds.
- C—3 miles visibility and clear of clouds.

3138. What minimum flight visibility is required for VFR flight operations on an airway below 10,000 feet MSL?

- A—1 mile.
- B—3 miles.
- C—4 miles.

3139. The minimum distance from clouds required for VFR operations on an airway below 10,000 feet MSL is

- A—remain clear of clouds.
- B—500 feet below, 1,000 feet above, and 2,000 feet horizontally.
- C—500 feet above, 1,000 feet below, and 2,000 feet horizontally.

3140. During operations within controlled airspace at altitudes of more than 1,200 feet AGL, but less than 10,000 feet MSL, the minimum distance above clouds requirement for VFR flight is

- A—500 feet.
- B—1,000 feet.
- C—1,500 feet.

3141. VFR flight in controlled airspace above 1,200 feet AGL and below 10,000 feet MSL requires a minimum visibility and vertical cloud clearance of

- A—3 miles, and 500 feet below or 1,000 feet above the clouds in controlled airspace.
- B—5 miles, and 1,000 feet below or 1,000 feet above the clouds at all altitudes.
- C—5 miles, and 1,000 feet below or 1,000 feet above the clouds only in Class A airspace.

3142. During operations outside controlled airspace at altitudes of more than 1,200 feet AGL, but less than 10,000 feet MSL, the minimum flight visibility for VFR flight at night is

- A—1 mile.
- B—3 miles.
- C—5 miles.

3143. Outside controlled airspace, the minimum flight visibility requirement for VFR flight above 1,200 feet AGL and below 10,000 feet MSL during daylight hours is

- A—1 mile.
- B—3 miles.
- C—5 miles.

3144. During operations outside controlled airspace at altitudes of more than 1,200 feet AGL, but less than 10,000 feet MSL, the minimum distance below clouds requirement for VFR flight at night is

- A—500 feet.
- B—1,000 feet.
- C—1,500 feet.

3145. The minimum flight visibility required for VFR flights above 10,000 feet MSL and more than 1,200 feet AGL in controlled airspace is

- A—1 mile.
- B—3 miles.
- C—5 miles.

3146. For VFR flight operations above 10,000 feet MSL and more than 1,200 feet AGL, the minimum horizontal distance from clouds required is

- A—1,000 feet.
- B—2,000 feet.
- C—1 mile.

3147. During operations at altitudes of more than 1,200 feet AGL and at or above 10,000 feet MSL, the minimum distance above clouds requirement for VFR flight is

- A—500 feet.
- B—1,000 feet.
- C—1,500 feet.

3148. No person may take off or land an aircraft under basic VFR at an airport that lies within Class D airspace unless the

- A—flight visibility at that airport is at least 1 mile.
- B—ground visibility at that airport is at least 1 mile.
- C—ground visibility at that airport is at least 3 miles.

3149. The basic VFR weather minimums for operating an aircraft within Class D airspace are

- A—500-foot ceiling and 1 mile visibility.
- B—1,000-foot ceiling and 3 miles visibility.
- C—clear of clouds and 2 miles visibility.

3150. A special VFR clearance authorizes the pilot of an aircraft to operate VFR while within Class D airspace when the visibility is

- A—less than 1 mile and the ceiling is less than 1,000 feet.
- B—at least 1 mile and the aircraft can remain clear of clouds.
- C—at least 3 miles and the aircraft can remain clear of clouds.

3151. What is the minimum weather condition required for airplanes operating under special VFR in Class D airspace?

- A—1 mile flight visibility.
- B—1 mile flight visibility and 1,000-foot ceiling.
- C—3 miles flight visibility and 1,000-foot ceiling.

3152. Under what conditions, if any, may a private pilot operate a helicopter under special VFR at night within Class D airspace?

- A—The helicopter must be fully instrument equipped and the pilot must be instrument rated.
- B—The flight visibility must be at least 1 mile.
- C—There are no conditions; regulations permit this.

3153. What are the minimum requirements for airplane operations under special VFR in Class D airspace at night?

- A—The airplane must be under radar surveillance at all times while in Class D airspace.
- B—The airplane must be equipped for IFR with an altitude reporting transponder.
- C—The pilot must be instrument rated, and the airplane must be IFR equipped.

3154. No person may operate an airplane within Class D airspace at night under special VFR unless the

- A—flight can be conducted 500 feet below the clouds.
- B—airplane is equipped for instrument flight.
- C—flight visibility is at least 3 miles.

3155. Which cruising altitude is appropriate for a VFR flight on a magnetic course of 135°?

- A—Even thousandths.
- B—Even thousandths plus 500 feet.
- C—Odd thousandths plus 500 feet.

3156. Which VFR cruising altitude is acceptable for a flight on a Victor Airway with a magnetic course of 175°? The terrain is less than 1,000 feet.

- A—4,500 feet.
- B—5,000 feet.
- C—5,500 feet.

3157. Which VFR cruising altitude is appropriate when flying above 3,000 feet AGL on a magnetic course of 185°?

- A—4,000 feet.
- B—4,500 feet.
- C—5,000 feet.

3158. Each person operating an aircraft at a VFR cruising altitude shall maintain an odd-thousand plus 500-foot altitude while on a

- A—magnetic heading of 0° through 179°.
- B—magnetic course of 0° through 179°.
- C—true course of 0° through 179°.

3159. In addition to a valid Airworthiness Certificate, what documents or records must be aboard an aircraft during flight?

- A—Aircraft engine and airframe logbooks, and owner's manual.
- B—Radio operator's permit, and repair and alteration forms.
- C—Operating limitations and Registration Certificate.

3160. When must batteries in an emergency locator transmitter (ELT) be replaced or recharged, if rechargeable?

- A—After any inadvertent activation of the ELT.
- B—When the ELT has been in use for more than 1 cumulative hour.
- C—When the ELT can no longer be heard over the airplane's communication radio receiver.

3161. When are non-rechargeable batteries of an emergency locator transmitter (ELT) required to be replaced?

- A—Every 24 months.
- B—When 50 percent of their useful life expires.
- C—At the time of each 100-hour or annual inspection.

3162. Except in Alaska, during what time period should lighted position lights be displayed on an aircraft?

- A—End of evening civil twilight to the beginning of morning civil twilight.
- B—1 hour after sunset to 1 hour before sunrise.
- C—Sunset to sunrise.

3163. When operating an aircraft at cabin pressure altitudes above 12,500 feet MSL up to and including 14,000 feet MSL, supplemental oxygen shall be used during

- A—the entire flight time at those altitudes.
- B—that flight time in excess of 10 minutes at those altitudes.
- C—that flight time in excess of 30 minutes at those altitudes.

3164. Unless each occupant is provided with supplemental oxygen, no person may operate a civil aircraft of U.S. registry above a maximum cabin pressure altitude of

- A—12,500 feet MSL.
- B—14,000 feet MSL.
- C—15,000 feet MSL.

3165. An operable 4096-code transponder with an encoding altimeter is required in which airspace?

- A—Class A, Class B (and within 30 miles of the Class B primary airport), and Class C.
- B—Class D and Class E (below 10,000 feet MSL).
- C—Class D and Class G (below 10,000 feet MSL).

3166. With certain exceptions, all aircraft within 30 miles of a Class B primary airport from the surface upward to 10,000 feet MSL must be equipped with

- A—an operable VOR or TACAN receiver and an ADF receiver.
- B—instruments and equipment required for IFR operations.
- C—an operable transponder having either Mode S or 4096-code capability with Mode C automatic altitude reporting capability.

3167. No person may operate an aircraft in acrobatic flight when

- A—flight visibility is less than 5 miles.
- B—over any congested area of a city, town, or settlement.
- C—less than 2,500 feet AGL.

3168. In which controlled airspace is acrobatic flight prohibited?

- A—Class D airspace, Class E airspace designated for Federal Airways.
- B—All Class E airspace below 1,500 feet AGL.
- C—All Class G airspace.

3169. What is the lowest altitude permitted for acrobatic flight?

- A—1,000 feet AGL.
- B—1,500 feet AGL.
- C—2,000 feet AGL.

3170. No person may operate an aircraft in acrobatic flight when the flight visibility is less than

- A—3 miles.
- B—5 miles.
- C—7 miles.

3171. A chair-type parachute must have been packed by a certificated and appropriately rated parachute rigger within the preceding

- A—60 days.
- B—90 days.
- C—120 days.

3172. An approved chair-type parachute may be carried in an aircraft for emergency use if it has been packed by an appropriately rated parachute rigger within the preceding

- A—120 days.
- B—180 days.
- C—365 days.

3173. With certain exceptions, when must each occupant of an aircraft wear an approved parachute?

- A—When a door is removed from the aircraft to facilitate parachute jumpers.
- B—When intentionally pitching the nose of the aircraft up or down 30° or more.
- C—When intentionally banking in excess of 30°.

3174. The minimum allowable strength of a towline used for an aerotow of a glider having a certificated gross weight of 700 pounds is

- A—560 pounds.
- B—700 pounds.
- C—1,000 pounds.

3175. The minimum allowable strength of a towline used for an aerotow of a glider having a certificated gross weight of 1,040 pounds is

- A—502 pounds.
- B—832 pounds.
- C—1,040 pounds.

3176. For the aerotow of a glider that weighs 700 pounds, which towrope tensile strength would require the use of safety links at each end of the rope?

- A—850 pounds.
- B—1,040 pounds.
- C—1,450 pounds.

3177. When using a towline having a breaking strength more than twice the maximum certificated operating weight of the glider, an approved safety link must be installed at what point(s)?

- A—Only the point where the towline is attached to the glider.
- B—The point where the towline is attached to the glider and the point of attachment of the towline to the towplane.
- C—Only the point where the towline is attached to the towplane.

3178. Which is normally prohibited when operating a restricted category civil aircraft?

- A—Flight under instrument flight rules.
- B—Flight over a densely populated area.
- C—Flight within Class D airspace.

3179. Unless otherwise specifically authorized, no person may operate an aircraft that has an experimental certificate

- A—beneath the floor of Class B airspace.
- B—over a densely populated area or in a congested airway.
- C—from the primary airport within Class D airspace.

3180. The responsibility for ensuring that an aircraft is maintained in an airworthy condition is primarily that of the

- A—pilot in command.
- B—owner or operator.
- C—mechanic who performs the work.

3181. The responsibility for ensuring that maintenance personnel make the appropriate entries in the aircraft maintenance records indicating the aircraft has been approved for return to service lies with the

- A—owner or operator.
- B—pilot in command.
- C—mechanic who performed the work.

3182. Completion of an annual inspection and the return of the aircraft to service should always be indicated by

- A—the relicensing date on the Registration Certificate.
- B—an appropriate notation in the aircraft maintenance records.
- C—an inspection sticker placed on the instrument panel that lists the annual inspection completion date.

3183. If an alteration or repair substantially affects an aircraft's operation in flight, that aircraft must be test flown by an appropriately-rated pilot and approved for return to service prior to being operated

- A—by any private pilot.
- B—with passengers aboard.
- C—for compensation or hire.

3184. Before passengers can be carried in an aircraft that has been altered in a manner that may have appreciably changed its flight characteristics, it must be flight tested by an appropriately-rated pilot who holds at least a

- A—Commercial Pilot Certificate with an instrument rating.
- B—Private Pilot Certificate.
- C—Commercial Pilot Certificate and a mechanic's certificate.

3185. An aircraft's annual inspection was performed on July 12, this year. The next annual inspection will be due no later than

- A—July 1, next year.
- B—July 13, next year.
- C—July 31, next year.

3186. To determine the expiration date of the last annual aircraft inspection, a person should refer to the

- A—Airworthiness Certificate.
- B—Registration Certificate.
- C—aircraft maintenance records.

3187. How long does the Airworthiness Certificate of an aircraft remain valid?

- A—As long as the aircraft has a current Registration Certificate.
- B—Indefinitely, unless the aircraft suffers major damage.
- C—As long as the aircraft is maintained and operated as required by Federal Aviation Regulations.

3188. What aircraft inspections are required for rental aircraft that are also used for flight instruction?

- A—Annual and 100-hour inspections.
- B—Biannual and 100-hour inspections.
- C—Annual and 50-hour inspections.

3189. An aircraft had a 100-hour inspection when the tachometer read 1259.6. When is the next 100-hour inspection due?

- A—1349.6 hours.
- B—1359.6 hours.
- C—1369.6 hours.

3190. A 100-hour inspection was due at 3302.5 hours on the tachometer. The 100-hour inspection was actually done at 3309.5 hours. When is the next 100-hour inspection due?

- A—3312.5 hours.
- B—3402.5 hours.
- C—3409.5 hours.

3191. No person may use an ATC transponder unless it has been tested and inspected within at least the preceding

- A—6 calendar months.
- B—12 calendar months.
- C—24 calendar months.

3192. Maintenance records show the last transponder inspection was performed on September 1, 1993. The next inspection will be due no later than

- A—September 30, 1994.
- B—September 1, 1995.
- C—September 30, 1995.

3193. Which records or documents shall the owner or operator of an aircraft keep to show compliance with an applicable Airworthiness Directive?

- A—Aircraft maintenance records.
- B—Airworthiness Certificate and Pilot's Operating Handbook.
- C—Airworthiness and Registration Certificates.

3194. If an aircraft is involved in an accident which results in substantial damage to the aircraft, the nearest NTSB field office should be notified

- A—immediately.
- B—within 48 hours.
- C—within 7 days.

3195. Which incident requires an immediate notification to the nearest NTSB field office?

- A—A forced landing due to engine failure.
- B—Landing gear damage, due to a hard landing.
- C—Flight control system malfunction or failure.

3196. Which incident would necessitate an immediate notification to the nearest NTSB field office?

- A—An in-flight generator/alternator failure.
- B—An in-flight fire.
- C—An in-flight loss of VOR receiver capability.

3197. Which incident requires an immediate notification be made to the nearest NTSB field office?

- A—An overdue aircraft that is believed to be involved in an accident.
- B—An in-flight radio communications failure.
- C—An in-flight generator or alternator failure.

3198. May aircraft wreckage be moved prior to the time the NTSB takes custody?

- A—Yes, but only if moved by a federal, state, or local law enforcement officer.
- B—Yes, but only to protect the wreckage from further damage.
- C—No, it may not be moved under any circumstances.

3199. The operator of an aircraft that has been involved in an accident is required to file an accident report within how many days?

- A—5.
- B—7.
- C—10.

3200. The operator of an aircraft that has been involved in an incident is required to submit a report to the nearest field office of the NTSB

- A—within 7 days.
- B—within 10 days.
- C—when requested.

3201. The four forces acting on an airplane in flight are

- A—lift, weight, thrust, and drag.
- B—lift, weight, gravity, and thrust.
- C—lift, gravity, power, and friction.

3202. When are the four forces that act on an airplane in equilibrium?

- A—During unaccelerated flight.
- B—When the aircraft is accelerating.
- C—When the aircraft is at rest on the ground.

3203. (Refer to Figure 1.) The acute angle A is the angle of

- A—incidence.
- B—attack.
- C—dihedral.

3204. The term “angle of attack” is defined as the angle

- A—between the wing chord line and the relative wind.
- B—between the airplane's climb angle and the horizon.
- C—formed by the longitudinal axis of the airplane and the chord line of the wing.

3205. What is the relationship of lift, drag, thrust, and weight when the airplane is in straight-and-level flight?

- A—Lift equals weight and thrust equals drag.
- B—Lift, drag, and weight equal thrust.
- C—Lift and weight equal thrust and drag.

3206. How will frost on the wings of an airplane affect takeoff performance?

- A—Frost will disrupt the smooth flow of air over the wing, adversely affecting its lifting capability.
- B—Frost will change the camber of the wing, increasing its lifting capability.
- C—Frost will cause the airplane to become airborne with a higher angle of attack, decreasing the stall speed.

3207. In what flight condition is torque effect the greatest in a single-engine airplane?

- A—Low airspeed, high power, high angle of attack.
- B—Low airspeed, low power, low angle of attack.
- C—High airspeed, high power, high angle of attack.

3208. The left turning tendency of an airplane caused by P-factor is the result of the

- A—clockwise rotation of the engine and the propeller turning the airplane counter-clockwise.
- B—propeller blade descending on the right, producing more thrust than the ascending blade on the left.
- C—gyroscopic forces applied to the rotating propeller blades acting 90° in advance of the point the force was applied.

3209. When does P-factor cause the airplane to yaw to the left?

- A—When at low angles of attack.
- B—When at high angles of attack.
- C—When at high airspeeds.

3210. An airplane said to be inherently stable will

- A—be difficult to stall.
- B—require less effort to control.
- C—not spin.

3211. What determines the longitudinal stability of an airplane?

- A—The location of the CG with respect to the center of lift.
- B—The effectiveness of the horizontal stabilizer, rudder, and rudder trim tab.
- C—The relationship of thrust and lift to weight and drag.

3212. What causes an airplane (except a T-tail) to pitch nosedown when power is reduced and controls are not adjusted?

- A—The CG shifts forward when thrust and drag are reduced.
- B—The downwash on the elevators from the propeller slipstream is reduced and elevator effectiveness is reduced.
- C—When thrust is reduced to less than weight, lift is also reduced and the wings can no longer support the weight.

3213. What is the purpose of the rudder on an airplane?

- A—To control yaw.
- B—To control overbanking tendency.
- C—To control roll.

3214. (Refer to Figure 2.) If an airplane weighs 2,300 pounds, what approximate weight would the airplane structure be required to support during a 60° banked turn while maintaining altitude?

- A—2,300 pounds.
- B—3,400 pounds.
- C—4,600 pounds.

3215. (Refer to Figure 2.) If an airplane weighs 3,300 pounds, what approximate weight would the airplane structure be required to support during a 30° banked turn while maintaining altitude?

- A—1,200 pounds.
- B—3,100 pounds.
- C—3,960 pounds.

3216. (Refer to Figure 2.) If an airplane weighs 4,500 pounds, what approximate weight would the airplane structure be required to support during a 45° banked turn while maintaining altitude?

- A—4,500 pounds.
- B—6,750 pounds.
- C—7,200 pounds.

3217. The amount of excess load that can be imposed on the wing of an airplane depends upon the

- A—position of the CG.
- B—speed of the airplane.
- C—abruptness at which the load is applied.

3218. Which basic flight maneuver increases the load factor on an airplane as compared to straight-and-level flight?

- A—Climbs.
- B—Turns.
- C—Stalls.

3219. One of the main functions of flaps during approach and landing is to

- A—decrease the angle of descent without increasing the airspeed.
- B—permit a touchdown at a higher indicated airspeed.
- C—increase the angle of descent without increasing the airspeed.

3220. What is one purpose of wing flaps?

- A—To enable the pilot to make steeper approaches to a landing without increasing the airspeed.
- B—To relieve the pilot of maintaining continuous pressure on the controls.
- C—To decrease wing area to vary the lift.

3221. Excessively high engine temperatures will

- A—cause damage to heat-conducting hoses and warping of the cylinder cooling fins.
- B—cause loss of power, excessive oil consumption, and possible permanent internal engine damage.
- C—not appreciably affect an aircraft engine.

3222. If the engine oil temperature and cylinder head temperature gauges have exceeded their normal operating range, the pilot may have been operating with

- A—the mixture set too rich.
- B—higher-than-normal oil pressure.
- C—too much power and with the mixture set too lean.

3223. One purpose of the dual ignition system on an aircraft engine is to provide for

- A—improved engine performance.
- B—uniform heat distribution.
- C—balanced cylinder head pressure.

3224. On aircraft equipped with fuel pumps, the practice of running a fuel tank dry before switching tanks is considered unwise because

- A—the engine-driven fuel pump or electric fuel boost pump may draw air into the fuel system and cause vapor lock.
- B—the engine-driven fuel pump is lubricated by fuel and operating on a dry tank may cause pump failure.
- C—any foreign matter in the tank will be pumped into the fuel system.

3225. The operating principle of float-type carburetors is based on the

- A—automatic metering of air at the venturi as the aircraft gains altitude.
- B—difference in air pressure at the venturi throat and the air inlet.
- C—increase in air velocity in the throat of a venturi causing an increase in air pressure.

3226. The basic purpose of adjusting the fuel/air mixture at altitude is to

- A—decrease the amount of fuel in the mixture in order to compensate for increased air density.
- B—decrease the fuel flow in order to compensate for decreased air density.
- C—increase the amount of fuel in the mixture to compensate for the decrease in pressure and density of the air.

3227. During the run-up at a high-elevation airport, a pilot notes a slight engine roughness that is not affected by the magneto check but grows worse during the carburetor heat check. Under these circumstances, what would be the most logical initial action?

- A—Check the results obtained with a leaner setting of the mixture.
- B—Taxi back to the flight line for a maintenance check.
- C—Reduce manifold pressure to control detonation.

3228. While cruising at 9,500 feet MSL, the fuel/air mixture is properly adjusted. What will occur if a descent to 4,500 feet MSL is made without readjusting the mixture?

- A—The fuel/air mixture may become excessively lean.
- B—There will be more fuel in the cylinders than is needed for normal combustion, and the excess fuel will absorb heat and cool the engine.
- C—The excessively rich mixture will create higher cylinder head temperatures and may cause detonation.

3229. Which condition is most favorable to the development of carburetor icing?

- A—Any temperature below freezing and a relative humidity of less than 50 percent.
- B—Temperature between 32 and 50°F and low humidity.
- C—Temperature between 20 and 70°F and high humidity.

3230. The possibility of carburetor icing exists even when the ambient air temperature is as

- A—high as 70°F and the relative humidity is high.
- B—high as 95°F and there is visible moisture.
- C—low as 0°F and the relative humidity is high.

3231. If an aircraft is equipped with a fixed-pitch propeller and a float-type carburetor, the first indication of carburetor ice would most likely be

- A—a drop in oil temperature and cylinder head temperature.
- B—engine roughness.
- C—loss of RPM.

3232. Applying carburetor heat will

- A—result in more air going through the carburetor.
- B—enrich the fuel/air mixture.
- C—not affect the fuel/air mixture.

3233. What change occurs in the fuel/air mixture when carburetor heat is applied?

- A—A decrease in RPM results from the lean mixture.
- B—The fuel/air mixture becomes richer.
- C—The fuel/air mixture becomes leaner.

3234. Generally speaking, the use of carburetor heat tends to

- A—decrease engine performance.
- B—increase engine performance.
- C—have no effect on engine performance.

3235. The presence of carburetor ice in an aircraft equipped with a fixed-pitch propeller can be verified by applying carburetor heat and noting

- A—an increase in RPM and then a gradual decrease in RPM.
- B—a decrease in RPM and then a constant RPM indication.
- C—a decrease in RPM and then a gradual increase in RPM.

3236. With regard to carburetor ice, float-type carburetor systems in comparison to fuel injection systems are generally considered to be

- A—more susceptible to icing.
- B—equally susceptible to icing.
- C—susceptible to icing only when visible moisture is present.

3237. If the grade of fuel used in an aircraft engine is lower than specified for the engine, it will most likely cause

- A—a mixture of fuel and air that is not uniform in all cylinders.
- B—lower cylinder head temperatures.
- C—detonation.

3238. Detonation occurs in a reciprocating aircraft engine when

- A—the spark plugs are fouled or shorted out or the wiring is defective.
- B—hot spots in the combustion chamber ignite the fuel/air mixture in advance of normal ignition.
- C—the unburned charge in the cylinders explodes instead of burning normally.

3239. If a pilot suspects that the engine (with a fixed-pitch propeller) is detonating during climb-out after takeoff, the initial corrective action to take would be to

- A—lean the mixture.
- B—lower the nose slightly to increase airspeed.
- C—apply carburetor heat.

3240. The uncontrolled firing of the fuel/air charge in advance of normal spark ignition is known as

- A—combustion.
- B—pre-ignition.
- C—detonation.

3241. Which would most likely cause the cylinder head temperature and engine oil temperature gauges to exceed their normal operating ranges?

- A—Using fuel that has a lower-than-specified fuel rating.
- B—Using fuel that has a higher-than-specified fuel rating.
- C—Operating with higher-than-normal oil pressure.

3242. What type fuel can be substituted for an aircraft if the recommended octane is not available?

- A—The next higher octane aviation gas.
- B—The next lower octane aviation gas.
- C—Unleaded automotive gas of the same octane rating.

3243. Filling the fuel tanks after the last flight of the day is considered a good operating procedure because this will

- A—force any existing water to the top of the tank away from the fuel lines to the engine.
- B—prevent expansion of the fuel by eliminating airspace in the tanks.
- C—prevent moisture condensation by eliminating airspace in the tanks.

3244. For internal cooling, reciprocating aircraft engines are especially dependent on

- A—a properly functioning thermostat.
- B—air flowing over the exhaust manifold.
- C—the circulation of lubricating oil.

3245. An abnormally high engine oil temperature indication may be caused by

- A—the oil level being too low.
- B—operating with a too high viscosity oil.
- C—operating with an excessively rich mixture.

3246. What effect does high density altitude, as compared to low density altitude, have on propeller efficiency and why?

- A—Efficiency is increased due to less friction on the propeller blades.
- B—Efficiency is reduced because the propeller exerts less force at high density altitudes than at low density altitudes.
- C—Efficiency is reduced due to the increased force of the propeller in the thinner air.

3247. If the pitot tube and outside static vents become clogged, which instruments would be affected?

- A—The altimeter, airspeed indicator, and turn-and-slip indicator.
- B—The altimeter, airspeed indicator, and vertical speed indicator.
- C—The altimeter, attitude indicator, and turn-and-slip indicator.

3248. Which instrument will become inoperative if the pitot tube becomes clogged?

- A—Altimeter.
- B—Vertical speed.
- C—Airspeed.

3249. Which instrument(s) will become inoperative if the static vents become clogged?

- A—Airspeed only.
- B—Altimeter only.
- C—Airspeed, altimeter, and vertical speed.

3250. (Refer to Figure 3.) Altimeter 1 indicates

- A—500 feet.
- B—1,500 feet.
- C—10,500 feet.

3251. (Refer to Figure 3.) Altimeter 2 indicates

- A—1,500 feet.
- B—4,500 feet.
- C—14,500 feet.

3252. (Refer to Figure 3.) Altimeter 3 indicates

- A—9,500 feet.
- B—10,950 feet.
- C—15,940 feet.

3253. (Refer to Figure 3.) Which altimeter(s) indicate(s) more than 10,000 feet?

- A—1, 2, and 3.
- B—1 and 2 only.
- C—1 only.

3254. Altimeter setting is the value to which the barometric pressure scale of the altimeter is set so the altimeter indicates

- A—calibrated altitude at field elevation.
- B—absolute altitude at field elevation.
- C—true altitude at field elevation.

3255. How do variations in temperature affect the altimeter?

- A—Pressure levels are raised on warm days and the indicated altitude is lower than true altitude.
- B—Higher temperatures expand the pressure levels and the indicated altitude is higher than true altitude.
- C—Lower temperatures lower the pressure levels and the indicated altitude is lower than true altitude.

3256. What is true altitude?

- A—The vertical distance of the aircraft above sea level.
- B—The vertical distance of the aircraft above the surface.
- C—The height above the standard datum plane.

3257. What is absolute altitude?

- A—The altitude read directly from the altimeter.
- B—The vertical distance of the aircraft above the surface.
- C—The height above the standard datum plane.

3258. What is density altitude?

- A—The height above the standard datum plane.
- B—The pressure altitude corrected for nonstandard temperature.
- C—The altitude read directly from the altimeter.

3259. What is pressure altitude?

- A—The indicated altitude corrected for position and installation error.
- B—The altitude indicated when the barometric pressure scale is set to 29.92.
- C—The indicated altitude corrected for nonstandard temperature and pressure.

3260. Under what condition is indicated altitude the same as true altitude?

- A—If the altimeter has no mechanical error.
- B—When at sea level under standard conditions.
- C—When at 18,000 feet MSL with the altimeter set at 29.92.

3261. If it is necessary to set the altimeter from 29.15 to 29.85, what change occurs?

- A—70-foot increase in indicated altitude.
- B—70-foot increase in density altitude.
- C—700-foot increase in indicated altitude.

3262. The pitot system provides impact pressure for which instrument?

- A—Altimeter.
- B—Vertical-speed indicator.
- C—Airspeed indicator.

3263. As altitude increases, the indicated airspeed at which a given airplane stalls in a particular configuration will

- A—decrease as the true airspeed decreases.
- B—decrease as the true airspeed increases.
- C—remain the same regardless of altitude.

3264. What does the red line on an airspeed indicator represent?

- A—Maneuvering speed.
- B—Turbulent or rough-air speed.
- C—Never-exceed speed.

3265. (Refer to Figure 4.) What is the full flap operating range for the airplane?

- A—60 to 100 MPH.
- B—60 to 208 MPH.
- C—65 to 165 MPH.

3266. (Refer to Figure 4.) What is the caution range of the airplane?

- A—0 to 60 MPH.
- B—100 to 165 MPH.
- C—165 to 208 MPH.

3267. (Refer to Figure 4.) The maximum speed at which the airplane can be operated in smooth air is

- A—100 MPH.
- B—165 MPH.
- C—208 MPH.

3268. (Refer to Figure 4.) Which color identifies the never-exceed speed?

- A—Lower limit of the yellow arc.
- B—Upper limit of the white arc.
- C—The red radial line.

3269. (Refer to Figure 4.) Which color identifies the power-off stalling speed in a specified configuration?

- A—Upper limit of the green arc.
- B—Upper limit of the white arc.
- C—Lower limit of the green arc.

3270. (Refer to Figure 4.) What is the maximum flaps-extended speed?

- A—65 MPH.
- B—100 MPH.
- C—165 MPH.

3271. (Refer to Figure 4.) Which color identifies the normal flap operating range?

- A—The lower limit of the white arc to the upper limit of the green arc.
- B—The green arc.
- C—The white arc.

3272. (Refer to Figure 4.) Which color identifies the power-off stalling speed with wing flaps and landing gear in the landing configuration?

- A—Upper limit of the green arc.
- B—Upper limit of the white arc.
- C—Lower limit of the white arc.

3273. (Refer to Figure 4.) What is the maximum structural cruising speed?

- A—100 MPH.
- B—165 MPH.
- C—208 MPH.

3274. What is an important airspeed limitation that is not color coded on airspeed indicators?

- A—Never-exceed speed.
- B—Maximum structural cruising speed.
- C—Maneuvering speed.

3275. (Refer to Figure 5.) A turn coordinator provides an indication of the

- A—movement of the aircraft about the yaw and roll axes.
- B—angle of bank up to but not exceeding 30°.
- C—attitude of the aircraft with reference to the longitudinal axis.

3276. (Refer to Figure 6.) To receive accurate indications during flight from a heading indicator, the instrument must be

- A—set prior to flight on a known heading.
- B—calibrated on a compass rose at regular intervals.
- C—periodically realigned with the magnetic compass as the gyro precesses.

3277. (Refer to Figure 7.) The proper adjustment to make on the attitude indicator during level flight is to align the

- A—horizon bar to the level-flight indication.
- B—horizon bar to the miniature airplane.
- C—miniature airplane to the horizon bar.

3278. (Refer to Figure 7.) How should a pilot determine the direction of bank from an attitude indicator such as the one illustrated?

- A—By the direction of deflection of the banking scale (A).
- B—By the direction of deflection of the horizon bar (B).
- C—By the relationship of the miniature airplane (C) to the deflected horizon bar (B).

3279. Deviation in a magnetic compass is caused by the

- A—presence of flaws in the permanent magnets of the compass.
- B—difference in the location between true north and magnetic north.
- C—magnetic fields within the aircraft distorting the lines of magnetic force.

3280. In the Northern Hemisphere, a magnetic compass will normally indicate initially a turn toward the west if

- A—a left turn is entered from a north heading.
- B—a right turn is entered from a north heading.
- C—an aircraft is accelerated while on a north heading.

3281. In the Northern Hemisphere, a magnetic compass will normally indicate initially a turn toward the east if

- A—an aircraft is decelerated while on a south heading.
- B—an aircraft is accelerated while on a north heading.
- C—a left turn is entered from a north heading.

3282. In the Northern Hemisphere, a magnetic compass will normally indicate a turn toward the north if

- A—a right turn is entered from an east heading.
- B—a left turn is entered from a west heading.
- C—an aircraft is accelerated while on an east or west heading.

3283. In the Northern Hemisphere, the magnetic compass will normally indicate a turn toward the south when

- A—a left turn is entered from an east heading.
- B—a right turn is entered from a west heading.
- C—the aircraft is decelerated while on a west heading.

3284. In the Northern Hemisphere, if an aircraft is accelerated or decelerated, the magnetic compass will normally indicate

- A—a turn momentarily.
- B—correctly when on a north or south heading.
- C—a turn toward the south.

3285. In the Northern Hemisphere, if a glider is accelerated or decelerated, the magnetic compass will normally indicate

- A—a turn toward north while decelerating on an east heading.
- B—correctly only when on a north or south heading.
- C—a turn toward south while accelerating on a west heading.

3286. During flight, when are the indications of a magnetic compass accurate?

- A—Only in straight-and-level unaccelerated flight.
- B—As long as the airspeed is constant.
- C—During turns if the bank does not exceed 18°

3287. An airplane has been loaded in such a manner that the CG is located aft of the aft CG limit. One undesirable flight characteristic a pilot might experience with this airplane would be

- A—a longer takeoff run.
- B—difficulty in recovering from a stalled condition.
- C—stalling at higher-than-normal airspeed.

3288. Loading an airplane to the most aft CG will cause the airplane to be

- A—less stable at all speeds.
- B—less stable at slow speeds, but more stable at high speeds.
- C—less stable at high speeds, but more stable at low speeds.

3289. If the outside air temperature (OAT) at a given altitude is warmer than standard, the density altitude is

- A—equal to pressure altitude.
- B—lower than pressure altitude.
- C—higher than pressure altitude.

3290. Which combination of atmospheric conditions will reduce aircraft takeoff and climb performance?

- A—Low temperature, low relative humidity, and low density altitude.
- B—High temperature, low relative humidity, and low density altitude.
- C—High temperature, high relative humidity, and high density altitude.

3291. What effect does high density altitude have on aircraft performance?

- A—It increases engine performance.
- B—It reduces climb performance.
- C—It increases takeoff performance.

3292. (Refer to Figure 8.) What is the effect of a temperature increase from 25 to 50°F on the density altitude if the pressure altitude remains at 5,000 feet?

- A—1,200-foot increase.
- B—1,400-foot increase.
- C—1,650-foot increase.

3293. (Refer to Figure 8.) Determine the pressure altitude with an indicated altitude of 1,380 feet MSL with an altimeter setting of 28.22 at standard temperature.

- A—1,250 feet MSL.
- B—1,373 feet MSL.
- C—3,010 feet MSL.

3294. (Refer to Figure 8.) Determine the density altitude for these conditions:

Altimeter setting 29.25
Runway temperature +81°F
Airport elevation 5,250 ft MSL

- A—4,600 feet MSL.
- B—5,877 feet MSL.
- C—8,500 feet MSL.

3295. (Refer to Figure 8.) Determine the pressure altitude at an airport that is 3,563 feet MSL with an altimeter setting of 29.96.

- A—3,527 feet MSL.
- B—3,556 feet MSL.
- C—3,639 feet MSL.

3296. (Refer to Figure 8.) What is the effect of a temperature increase from 30 to 50°F on the density altitude if the pressure altitude remains at 3,000 feet MSL?

- A—900-foot increase.
- B—1,100-foot decrease.
- C—1,300-foot increase.

3297. (Refer to Figure 8.) Determine the pressure altitude at an airport that is 1,386 feet MSL with an altimeter setting of 29.97.

- A—1,341 feet MSL.
- B—1,451 feet MSL.
- C—1,562 feet MSL.

3298. (Refer to Figure 8.) Determine the density altitude for these conditions:

Altimeter setting 30.35
Runway temperature +25°F
Airport elevation 3,894 ft MSL

- A—2,000 feet MSL.
- B—2,900 feet MSL.
- C—3,500 feet MSL.

3299. (Refer to Figure 8.) What is the effect of a temperature decrease and a pressure altitude increase on the density altitude from 90°F and 1,250 feet pressure altitude to 60°F and 1,750 feet pressure altitude?

- A—500-foot increase.
- B—1,300-foot decrease.
- C—1,300-foot increase.

3300. What effect, if any, does high humidity have on aircraft performance?

- A—It increases performance.
- B—It decreases performance.
- C—It has no effect on performance.

3301. What force makes an airplane turn?

- A—The horizontal component of lift.
- B—The vertical component of lift.
- C—Centrifugal force.

3302. When taxiing with strong quartering tailwinds, which aileron positions should be used?

- A—Aileron down on the downwind side.
- B—Ailerons neutral.
- C—Aileron down on the side from which the wind is blowing.

3303. Which aileron positions should a pilot generally use when taxiing in strong quartering headwinds?

- A—Aileron up on the side from which the wind is blowing.
- B—Aileron down on the side from which the wind is blowing.
- C—Ailerons neutral.

3304. Which wind condition would be most critical when taxiing a nosewheel equipped high-wing airplane?

- A—Quartering tailwind.
- B—Direct crosswind.
- C—Quartering headwind.

3305. (Refer to Figure 9, area A.) How should the flight controls be held while taxiing a tricycle-gear equipped airplane into a left quartering headwind?

- A—Left aileron up, elevator neutral.
- B—Left aileron down, elevator neutral.
- C—Left aileron up, elevator down.

3306. (Refer to Figure 9, area B.) How should the flight controls be held while taxiing a tailwheel airplane into a right quartering headwind?

- A—Right aileron up, elevator up.
- B—Right aileron down, elevator neutral.
- C—Right aileron up, elevator down.

3307. (Refer to Figure 9, area C.) How should the flight controls be held while taxiing a tailwheel airplane with a left quartering tailwind?

- A—Left aileron up, elevator neutral.
- B—Left aileron down, elevator neutral.
- C—Left aileron down, elevator down.

3308. (Refer to Figure 9, area C.) How should the flight controls be held while taxiing a tricycle-gear equipped airplane with a left quartering tailwind?

- A—Left aileron up, elevator neutral.
- B—Left aileron down, elevator down.
- C—Left aileron up, elevator down.

3309. In what flight condition must an aircraft be placed in order to spin?

- A—Partially stalled with one wing low.
- B—In a steep diving spiral.
- C—Stalled.

3310. During a spin to the left, which wing(s) is/are stalled?

- A—Both wings are stalled.
- B—Neither wing is stalled.
- C—Only the left wing is stalled.

3311. The angle of attack at which an airplane wing stalls will

- A—increase if the CG is moved forward.
- B—change with an increase in gross weight.
- C—remain the same regardless of gross weight.

3312. What is ground effect?

- A—The result of the interference of the surface of the Earth with the airflow patterns about an airplane.
- B—The result of an alteration in airflow patterns increasing induced drag about the wings of an airplane.
- C—The result of the disruption of the airflow patterns about the wings of an airplane to the point where the wings will no longer support the airplane in flight.

3313. Floating caused by the phenomenon of ground effect will be most realized during an approach to land when at

- A—less than the length of the wingspan above the surface.
- B—twice the length of the wingspan above the surface.
- C—a higher-than-normal angle of attack.

3314. What must a pilot be aware of as a result of ground effect?

- A—Wingtip vortices increase creating wake turbulence problems for arriving and departing aircraft.
- B—Induced drag decreases; therefore, any excess speed at the point of flare may cause considerable floating.
- C—A full stall landing will require less up elevator deflection than would a full stall when done free of ground effect.

3315. Ground effect is most likely to result in which problem?

- A—Settling to the surface abruptly during landing.
- B—Becoming airborne before reaching recommended takeoff speed.
- C—Inability to get airborne even though airspeed is sufficient for normal takeoff needs.

3316. During an approach to a stall, an increased load factor will cause the airplane to

- A—stall at a higher airspeed.
- B—have a tendency to spin.
- C—be more difficult to control.

3317. Angle of attack is defined as the angle between the chord line of an airfoil and the

- A—direction of the relative wind.
- B—pitch angle of an airfoil.
- C—rotor plane of rotation.

3318. (Refer to Figure 10.) During flight, if cyclic control pressure is applied which results in a maximum increase in pitch angle of the rotor blade at position A, the rotor disc will tilt

- A—forward.
- B—aft.
- C—left.

3319. The lift differential that exists between the advancing main rotor blade and the retreating main rotor blade is known as

- A—transverse flow effect.
- B—dissymmetry of lift.
- C—hunting tendency.

3320. During forward cruising flight at constant airspeed and altitude, the individual rotor blades, when compared to each other, are operating

- A—with increased lift on the retreating blade.
- B—with a decreasing angle of attack on the advancing blade.
- C—at unequal airspeed, unequal angles of attack, and equal lift moment.

3321. The upward bending of the rotor blades resulting from the combined forces of lift and centrifugal force is known as

- A—coning.
- B—blade slapping.
- C—inertia.

3322. When a blade flaps up, the CG moves closer to its axis of rotation giving that blade a tendency to

- A—decelerate.
- B—accelerate.
- C—stabilize its rotational velocity.

3323. During a hover, a helicopter tends to drift to the right. To compensate for this, some helicopters have the

- A—tail rotor tilted to the left.
- B—tail rotor tilted to the right.
- C—rotor mast rigged to the left side.

3324. Which is a result of the phenomenon of ground effect?

- A—The induced angle of attack of each rotor blade is increased.
- B—The lift vector becomes more horizontal.
- C—The angle of attack generating lift is increased.

3325. Translational lift is the result of

- A—decreased rotor efficiency.
- B—airspeed.
- C—both airspeed and groundspeed.

3326. The primary purpose of the tail rotor system is to

- A—assist in making a coordinated turn.
- B—maintain heading during forward flight.
- C—counteract the torque effect of the main rotor.

3327. If RPM is low and manifold pressure is high, what initial corrective action should be taken?

- A—Increase the throttle.
- B—Lower the collective pitch.
- C—Raise the collective pitch.

3328. The purpose of the lead-lag (drag) hinge in a three-bladed, fully articulated helicopter rotor system is to compensate for

- A—Coriolis effect.
- B—coning.
- C—geometric unbalance.

3329. High airspeeds, particularly in turbulent air, should be avoided primarily because of the possibility of

- A—an abrupt pitchup.
- B—retreating blade stall.
- C—a low-frequency vibration developing.

3330. The maximum forward speed of a gyroplane is limited by

- A—retreating blade stall.
- B—the rotor RPM red line.
- C—solidity ratio.

3331. When operating at high forward airspeeds, retreating blade stalls are more likely to occur under which condition?

- A—Low gross weight and low density altitude.
- B—High RPM and low density altitude.
- C—Steep turns in turbulent air.

3332. Ground resonance is most likely to develop when

- A—on the ground and harmonic vibrations develop between the main and tail rotors.
- B—a series of shocks causes the rotor system to become unbalanced.
- C—there is a combination of a decrease in the angle of attack on the advancing blade and an increase in the angle of attack on the retreating blade.

3333. While in level cruising flight in a helicopter, a pilot experiences low-frequency vibrations (100 to 400 cycles per minute). These vibrations are normally associated with the

- A—engine.
- B—cooling fan.
- C—main rotor.

3334. Select the helicopter component that, if defective, would cause medium-frequency vibrations.

- A—Tail rotor.
- B—Main rotor.
- C—Engine.

3335. The principal reason the shaded area of a Height vs. Velocity Chart should be avoided is

- A—turbulence near the surface can dephase the blade dampers.
- B—rotor RPM may decay before ground contact is made if an engine failure should occur.
- C—insufficient airspeed would be available to ensure a safe landing in case of an engine failure.

3336. During surface taxiing, the collective pitch is used to control

- A—drift during a crosswind.
- B—rate of speed.
- C—ground track.

3337. During surface taxiing, the cyclic pitch stick is used to control

- A—forward movement.
- B—heading.
- C—ground track.

3338. If the pilot experiences ground resonance during rotor spin-up, what action should the pilot take?

- A—Taxi to a smooth area.
- B—Close the throttle and slowly raise the spin-up lever.
- C—Make a normal takeoff immediately.

3339. What precaution should be taken while taxiing a gyroplane?

- A—The cyclic stick should be held in the neutral position at all times.
- B—Avoid abrupt control movements when blades are turning.
- C—The cyclic stick should be held slightly aft of neutral at all times.

3340. What force provides the forward motion necessary to move a glider through the air?

- A—Lift.
- B—Centripetal force.
- C—Gravity.

3341. To obtain maximum distance over the ground, the airspeed to use is the

- A—minimum control speed.
- B—best lift/drag speed.
- C—minimum sink speed.

3342. What effect would gusts and turbulence have on the load factor of a glider with changes in airspeed?

- A—Load factor decreases as airspeed increases.
- B—Load factor increases as airspeed increases.
- C—Load factor increases as airspeed decreases.

3343. (Refer to Figure 11.) Which yaw string and inclinometer illustrations indicate a slipping right turn?

- A—3 and 6.
- B—2 and 6.
- C—2 and 4.

3344. (Refer to Figure 11.) Which of the illustrations depicts the excessive use of right rudder during the entry of a right turn?

- A—2 only.
- B—2 and 4.
- C—3 and 4.

3345. A sailplane has a best glide ratio of 23:1. How many feet will the glider lose in 8 nautical miles?

- A—1,840 feet.
- B—2,100 feet.
- C—2,750 feet.

3346. A sailplane has a best glide ratio of 30:1. How many nautical miles will the glider travel while losing 2,000 feet?

- A—10 nautical miles.
- B—15 nautical miles.
- C—21 nautical miles.

3347. A sailplane has lost 2,000 feet in 9 nautical miles. The best glide ratio for this sailplane is approximately

- A—24:1.
- B—27:1.
- C—30:1.

3348. How many feet will a sailplane sink in 15 nautical miles if its lift/drag ratio is 22:1?

- A—2,700 feet.
- B—3,600 feet.
- C—4,100 feet.

3349. How many feet will a glider sink in 10 nautical miles if its lift/drag ratio is 23:1?

- A—2,400 feet.
- B—2,600 feet.
- C—4,300 feet.

3350. What is the proper airspeed to use when flying between thermals on a cross-country flight against a headwind?

- A—The best lift/drag speed increased by one-half the estimated wind velocity.
- B—The minimum sink speed increased by one-half the estimated wind velocity.
- C—The best lift/drag speed decreased by one-half the estimated wind velocity.

3351. The part of a balloon that bears the entire load is the

- A—envelope material.
- B—envelope seams.
- C—load tapes (or cords).

3352. In hot air balloons, propane is preferred to butane or other hydrocarbons because it

- A—is less volatile.
- B—is slower to vaporize.
- C—has a lower boiling point.

3353. The initial temperature at which propane boils is

- A—+32°F.
- B—-44°F.
- C—-60°F.

3354. On cold days, it may be necessary to preheat the propane tanks because

- A—the temperature of the liquid propane controls the burner pressure during combustion.
- B—there may be ice in the lines to the burner.
- C—the propane needs to be thawed from a solid to a liquid state.

3355. When ample liquid propane is available, propane will vaporize sufficiently to provide proper operation between the temperatures of

- A—+30 to +90°F.
- B—-44 to +25°F.
- C—-51 to +20°F.

3356. If ample propane is available, within which temperature range will propane vaporize sufficiently to provide enough pressure for burner operation during flight?

- A—0 to 30°F.
- B—10 to 30°F.
- C—30 to 90°F.

3357. The valve located on the top of the propane tank which opens automatically when the pressure in the tank exceeds maximum allowable pressure is the

- A—pressure release valve.
- B—metering valve.
- C—blast valve.

3358. The valve located on each tank that indicates when the tank is filled to 80 percent capacity is the

- A—main tank valve.
- B—vapor-bleed valve.
- C—pilot valve.

3359. The lifting forces which act on a hot air balloon are primarily the result of the interior air temperature being

- A—greater than ambient temperature.
- B—less than ambient temperature.
- C—equal to ambient temperature.

3360. Burner efficiency of a hot air balloon decreases approximately what percent for each 1,000 feet above MSL?

- A—4 percent.
- B—8 percent.
- C—15 percent.

3361. While in flight, ice begins forming on the outside of the fuel tank in use. This would most likely be caused by

- A—water in the fuel.
- B—a leak in the fuel line.
- C—vaporized fuel instead of liquid fuel being drawn from the tank into the main burner.

3362. For what reason is methanol added to the propane fuel of hot air balloons?

- A—To check for fuel leaks.
- B—As a fire retardant.
- C—As an anti-icing additive.

3363. On a balloon equipped with a blast valve, the blast valve is used for

- A—climbs and descents only.
- B—altitude control.
- C—emergencies only.

3364. The term “weigh-off” means to determine the

- A—static equilibrium of the balloon as loaded for flight.
- B—amount of gas required for an ascent to a preselected altitude.
- C—standard weight and balance of the balloon.

3365. What causes false lift which sometimes occurs during launch procedures?

- A—Closing the maneuvering vent too rapidly.
- B—Excessive temperature within the envelope.
- C—Venturi effect of the wind on the envelope.

3366. What is the relationship of false lift with the wind?

- A—False lift increases as the wind accelerates the balloon.
- B—False lift does not exist if the surface winds are calm.
- C—False lift decreases as the wind accelerates the balloon.

3367. What would cause a gas balloon to start a descent if a cold air mass is encountered and the envelope becomes cooled?

- A—A density differential.
- B—A barometric pressure differential.
- C—The contraction of the gas.

3368. Under which condition will an airship float in the air?

- A—When buoyant force equals horizontal equilibrium existing between propeller thrust and airship drag.
- B—When buoyant force is less than the difference between airship weight and the weight of the air volume being displaced.
- C—When buoyant force equals the difference between airship weight and the weight of the air volume being displaced.

3369. During flight in an airship, when is vertical equilibrium established?

- A—When buoyancy is greater than airship weight.
- B—When buoyancy equals airship weight.
- C—When buoyancy is less than airship weight.

3370. An airship descending through a steep temperature inversion will

- A—show no change in superheat as altitude is lost.
- B—show a decrease in superheat as altitude is lost.
- C—become progressively lighter, thus becoming increasingly more difficult to drive down.

3371. What is airship superheat?

- A—A condition of excessive exterior temperature of the envelope.
- B—The temperature of the lifting gas exceeding the red line.
- C—The difference between outside air temperature and the temperature inside the envelope.

3372. In relation to the operation of an airship, what is the definition of aerostatics?

- A—The gravitational factors involving equilibrium of a body freely suspended in the atmosphere.
- B—The science of the dynamics involved in the expansion and contraction of hydrogen gas.
- C—The expansion and contraction of the lifting gas helium.

3373. Below pressure height, each 5°F of positive superheat amounts to approximately

- A—1 percent of gross lift.
- B—2 percent of net lift.
- C—2 percent of total lift.

3374. When the airship is at pressure height and superheat increases, constant pressure must be maintained by valving

- A—gas from the envelope.
- B—air from the envelope.
- C—gas from the ballonets.

3375. How does the pilot know when pressure height has been reached?

- A—Liquid in the gas manometer will rise and the liquid in the air manometer will fall below normal levels.
- B—Liquid in the gas and air manometers will fall below the normal level.
- C—Liquid in the gas manometer will fall and the liquid in the air manometer will rise above normal levels.

3376. The pressure height of an airship is the altitude at which

- A—the airship would be unable to gain more altitude.
- B—gas pressure would reach 3 inches of water.
- C—the ballonet(s) would be empty.

3377. The maximum altitude that a rigid airship can reach (under a given atmospheric condition) and then return safely to the surface is determined by

- A—the disposable load.
- B—ballonet capacity.
- C—pressure altitude.

3378. An unbalanced condition of an airship in flight must be overcome by

- A—valving air from the ballonets.
- B—valving gas from the envelope.
- C—a negative or a positive dynamic force.

3379. Why should damper valves normally be kept closed during climbs? Because any air forced into the system would

- A—increase the amount of gas that must be exhausted to prevent the airship from ascending at an excessively high rate.
- B—increase the amount of air to be exhausted, resulting in a lower rate of ascent.
- C—decrease the purity of the gas within the envelope.

3380. To check the gas pressures (pressure height) of an airship during a climb, the air damper valves should be

- A—opened forward and closed aft.
- B—opened aft and closed forward.
- C—closed.

3381. Every physical process of weather is accompanied by, or is the result of, a

- A—movement of air.
- B—pressure differential.
- C—heat exchange.

3382. What causes variations in altimeter settings between weather reporting points?

- A—Unequal heating of the Earth's surface.
- B—Variation of terrain elevation.
- C—Coriolis force.

3383. A temperature inversion would most likely result in which weather condition?

- A—Clouds with extensive vertical development above an inversion aloft.
- B—Good visibility in the lower levels of the atmosphere and poor visibility above an inversion aloft.
- C—An increase in temperature as altitude is increased.

3384. The most frequent type of ground or surface-based temperature inversion is that which is produced by

- A—terrestrial radiation on a clear, relatively still night.
- B—warm air being lifted rapidly aloft in the vicinity of mountainous terrain.
- C—the movement of colder air under warm air, or the movement of warm air over cold air.

3385. Which weather conditions should be expected beneath a low-level temperature inversion layer when the relative humidity is high?

- A—Smooth air, poor visibility, fog, haze, or low clouds.
- B—Light wind shear, poor visibility, haze, and light rain.
- C—Turbulent air, poor visibility, fog, low stratus type clouds, and showery precipitation.

3386. What are the standard temperature and pressure values for sea level?

- A—15°C and 29.92" Hg.
- B—59°C and 1013.2 millibars.
- C—59°F and 29.92 millibars.

3387. If a pilot changes the altimeter setting from 30.11 to 29.96, what is the approximate change in indication?

- A—Altimeter will indicate .15" Hg higher.
- B—Altimeter will indicate 150 feet higher.
- C—Altimeter will indicate 150 feet lower.

3388. Under which condition will pressure altitude be equal to true altitude?

- A—When the atmospheric pressure is 29.92" Hg.
- B—When standard atmospheric conditions exist.
- C—When indicated altitude is equal to the pressure altitude.

3389. Under what condition is pressure altitude and density altitude the same value?

- A—At sea level, when the temperature is 0°F.
- B—When the altimeter has no installation error.
- C—At standard temperature.

3390. If a flight is made from an area of low pressure into an area of high pressure without the altimeter setting being adjusted, the altimeter will indicate

- A—the actual altitude above sea level.
- B—higher than the actual altitude above sea level.
- C—lower than the actual altitude above sea level.

3391. If a flight is made from an area of high pressure into an area of lower pressure without the altimeter setting being adjusted, the altimeter will indicate

- A—lower than the actual altitude above sea level.
- B—higher than the actual altitude above sea level.
- C—the actual altitude above sea level.

3392. Under what condition will true altitude be lower than indicated altitude?

- A—In colder than standard air temperature.
- B—In warmer than standard air temperature.
- C—When density altitude is higher than indicated altitude.

3393. Which condition would cause the altimeter to indicate a lower altitude than true altitude?

- A—Air temperature lower than standard.
- B—Atmospheric pressure lower than standard.
- C—Air temperature warmer than standard.

3394. Which factor would tend to increase the density altitude at a given airport?

- A—An increase in barometric pressure.
- B—An increase in ambient temperature.
- C—A decrease in relative humidity.

3395. The wind at 5,000 feet AGL is southwesterly while the surface wind is southerly. This difference in direction is primarily due to

- A—stronger pressure gradient at higher altitudes.
- B—friction between the wind and the surface.
- C—stronger Coriolis force at the surface.

3396. What condition does a rising barometer indicate for balloon operations?

- A—Decreasing clouds and wind.
- B—Chances of thunderstorms.
- C—Approaching frontal activity.

3397. What is meant by the term "dewpoint"?

- A—The temperature at which condensation and evaporation are equal.
- B—The temperature at which dew will always form.
- C—The temperature to which air must be cooled to become saturated.

3398. The amount of water vapor which air can hold depends on the

- A—dewpoint.
- B—air temperature.
- C—stability of the air.

3399. Clouds, fog, or dew will always form when

- A—water vapor condenses.
- B—water vapor is present.
- C—relative humidity reaches 100 percent.

3400. What are the processes by which moisture is added to unsaturated air?

- A—Evaporation and sublimation.
- B—Heating and condensation.
- C—Supersaturation and evaporation.

3401. Which conditions result in the formation of frost?

- A—The temperature of the collecting surface is at or below freezing when small droplets of moisture fall on the surface.
- B—The temperature of the collecting surface is at or below the dewpoint of the adjacent air and the dewpoint is below freezing.
- C—The temperature of the surrounding air is at or below freezing when small drops of moisture fall on the collecting surface.

3402. The presence of ice pellets at the surface is evidence that there

- A—are thunderstorms in the area.
- B—has been cold frontal passage.
- C—is a temperature inversion with freezing rain at a higher altitude.

3403. What measurement can be used to determine the stability of the atmosphere?

- A—Atmospheric pressure.
- B—Actual lapse rate.
- C—Surface temperature.

3404. What would decrease the stability of an air mass?

- A—Warming from below.
- B—Cooling from below.
- C—Decrease in water vapor.

3405. What is a characteristic of stable air?

- A—Stratiform clouds.
- B—Unlimited visibility.
- C—Cumulus clouds.

3406. Moist, stable air flowing upslope can be expected to

- A—produce stratus type clouds.
- B—cause showers and thunderstorms.
- C—develop convective turbulence.

3407. If an unstable air mass is forced upward, what type clouds can be expected?

- A—Stratus clouds with little vertical development.
- B—Stratus clouds with considerable associated turbulence.
- C—Clouds with considerable vertical development and associated turbulence.

3408. What feature is associated with a temperature inversion?

- A—A stable layer of air.
- B—An unstable layer of air.
- C—Chinook winds on mountain slopes.

3409. What is the approximate base of the cumulus clouds if the surface air temperature at 1,000 feet MSL is 70°F and the dewpoint is 48°F?

- A—4,000 feet MSL.
- B—5,000 feet MSL.
- C—6,000 feet MSL.

3410. At approximately what altitude above the surface would the pilot expect the base of cumuliform clouds if the surface air temperature is 82°F and the dewpoint is 38°F?

- A—9,000 feet AGL.
- B—10,000 feet AGL.
- C—11,000 feet AGL.

3411. What early morning weather observations indicate the possibility of good weather conditions for balloon flight most of the day?

- A—Clear skies and surface winds, 10 knots or less.
- B—Low moving, scattered cumulus clouds and surface winds, 5 knots or less.
- C—Overcast with stratus clouds and surface winds, 5 knots or less.

3412. What are characteristics of a moist, unstable air mass?

- A—Cumuliform clouds and showery precipitation.
- B—Poor visibility and smooth air.
- C—Stratiform clouds and showery precipitation.

3413. What are characteristics of unstable air?

- A—Turbulence and good surface visibility.
- B—Turbulence and poor surface visibility.
- C—Nimbostratus clouds and good surface visibility.

3414. A stable air mass is most likely to have which characteristic?

- A—Showery precipitation.
- B—Turbulent air.
- C—Smooth air.

3415. The suffix “nimbus,” used in naming clouds, means

- A—a cloud with extensive vertical development.
- B—a rain cloud.
- C—a middle cloud containing ice pellets.

3416. Clouds are divided into four families according to their

- A—outward shape.
- B—height range.
- C—composition.

3417. An almond or lens-shaped cloud which appears stationary, but which may contain winds of 50 knots or more, is referred to as

- A—an inactive frontal cloud.
- B—a funnel cloud.
- C—a lenticular cloud.

3418. Crests of standing mountain waves may be marked by stationary, lens-shaped clouds known as

- A—mammato cumulus clouds.
- B—standing lenticular clouds.
- C—roll clouds.

3419. What clouds have the greatest turbulence?

- A—Towering cumulus.
- B—Cumulonimbus.
- C—Nimbostratus.

3420. What cloud types would indicate convective turbulence?

- A—Cirrus clouds.
- B—Nimbostratus clouds.
- C—Towering cumulus clouds.

3421. The boundary between two different air masses is referred to as a

- A—frontolysis.
- B—frontogenesis.
- C—front.

3422. One of the most easily recognized discontinuities across a front is

- A—a change in temperature.
- B—an increase in cloud coverage.
- C—an increase in relative humidity.

3423. One weather phenomenon which will always occur when flying across a front is a change in the

- A—wind direction.
- B—type of precipitation.
- C—stability of the air mass.

3424. Steady precipitation preceding a front is an indication of

- A—stratiform clouds with moderate turbulence.
- B—cumuliform clouds with little or no turbulence.
- C—stratiform clouds with little or no turbulence.

3425. Possible mountain wave turbulence could be anticipated when winds of 40 knots or greater blow

- A—across a mountain ridge, and the air is stable.
- B—down a mountain valley, and the air is unstable.
- C—parallel to a mountain peak, and the air is stable.

3426. Where does wind shear occur?

- A—Only at higher altitudes.
- B—Only at lower altitudes.
- C—At all altitudes, in all directions.

3427. When may hazardous wind shear be expected?

- A—When stable air crosses a mountain barrier where it tends to flow in layers forming lenticular clouds.
- B—In areas of low-level temperature inversion, frontal zones, and clear air turbulence.
- C—Following frontal passage when stratocumulus clouds form indicating mechanical mixing.

3428. A pilot can expect a wind-shear zone in a temperature inversion whenever the windspeed at 2,000 to 4,000 feet above the surface is at least

- A—10 knots.
- B—15 knots.
- C—25 knots.

3429. One in-flight condition necessary for structural icing to form is

- A—small temperature/dewpoint spread.
- B—stratiform clouds.
- C—visible moisture.

3430. In which environment is aircraft structural ice most likely to have the highest accumulation rate?

- A—Cumulus clouds with below freezing temperatures.
- B—Freezing drizzle.
- C—Freezing rain.

3431. Why is frost considered hazardous to flight?

- A—Frost changes the basic aerodynamic shape of the airfoils, thereby decreasing lift.
- B—Frost slows the airflow over the airfoils, thereby increasing control effectiveness.
- C—Frost spoils the smooth flow of air over the wings, thereby decreasing lifting capability.

3432. How does frost affect the lifting surfaces of an airplane on takeoff?

- A—Frost may prevent the airplane from becoming airborne at normal takeoff speed.
- B—Frost will change the camber of the wing, increasing lift during takeoff.
- C—Frost may cause the airplane to become airborne with a lower angle of attack at a lower indicated airspeed.

3433. The conditions necessary for the formation of cumulonimbus clouds are a lifting action and

- A—unstable air containing an excess of condensation nuclei.
- B—unstable, moist air.
- C—either stable or unstable air.

3434. What feature is normally associated with the cumulus stage of a thunderstorm?

- A—Roll cloud.
- B—Continuous updraft.
- C—Frequent lightning.

3435. Which weather phenomenon signals the beginning of the mature stage of a thunderstorm?

- A—The appearance of an anvil top.
- B—Precipitation beginning to fall.
- C—Maximum growth rate of the clouds.

3436. What conditions are necessary for the formation of thunderstorms?

- A—High humidity, lifting force, and unstable conditions.
- B—High humidity, high temperature, and cumulus clouds.
- C—Lifting force, moist air, and extensive cloud cover.

3437. During the life cycle of a thunderstorm, which stage is characterized predominately by downdrafts?

- A—Cumulus.
- B—Dissipating.
- C—Mature.

3438. Thunderstorms reach their greatest intensity during the

- A—mature stage.
- B—downdraft stage.
- C—cumulus stage.

3439. Thunderstorms which generally produce the most intense hazard to aircraft are

- A—squall line thunderstorms.
- B—steady-state thunderstorms.
- C—warm front thunderstorms.

3440. A nonfrontal, narrow band of active thunderstorms that often develop ahead of a cold front is a known as a

- A—prefrontal system.
- B—squall line.
- C—dry line.

3441. If there is thunderstorm activity in the vicinity of an airport at which you plan to land, which hazardous atmospheric phenomenon might be expected on the landing approach?

- A—Precipitation static.
- B—Wind-shear turbulence.
- C—Steady rain.

3442. Upon encountering severe turbulence, which flight condition should the pilot attempt to maintain?

- A—Constant altitude and airspeed.
- B—Constant angle of attack.
- C—Level flight attitude.

3443. What situation is most conducive to the formation of radiation fog?

- A—Warm, moist air over low, flatland areas on clear, calm nights.
- B—Moist, tropical air moving over cold, offshore water.
- C—The movement of cold air over much warmer water.

3444. If the temperature/dewpoint spread is small and decreasing, and the temperature is 62°F, what type weather is most likely to develop?

- A—Freezing precipitation.
- B—Thunderstorms.
- C—Fog or low clouds.

3445. In which situation is advection fog most likely to form?

- A—A warm, moist air mass on the windward side of mountains.
- B—An air mass moving inland from the coast in winter.
- C—A light breeze blowing colder air out to sea.

3446. What types of fog depend upon wind in order to exist?

- A—Radiation fog and ice fog.
- B—Steam fog and ground fog.
- C—Advection fog and upslope fog.

3447. Low-level turbulence can occur and icing can become hazardous in which type of fog?

- A—Rain-induced fog.
- B—Upslope fog.
- C—Steam fog.

3448. The development of thermals depends upon

- A—a counterclockwise circulation of air.
- B—temperature inversions.
- C—solar heating.

3449. Which is considered to be the most hazardous condition when soaring in the vicinity of thunderstorms?

- A—Static electricity.
- B—Lightning.
- C—Wind shear and turbulence.

3450. Convective circulation patterns associated with sea breezes are caused by

- A—warm, dense air moving inland from over the water.
- B—water absorbing and radiating heat faster than the land.
- C—cool, dense air moving inland from over the water.

3451. During which period is a sea breeze front most suitable for soaring flight?

- A—Shortly after sunrise.
- B—During the early forenoon.
- C—During the afternoon.

3452. Which weather phenomenon is always associated with a thunderstorm?

- A—Lightning.
- B—Heavy rain.
- C—Hail.

3453. Individual forecasts for specific routes of flight can be obtained from which weather source?

- A—Transcribed Weather Broadcasts (TWEB's).
- B—Terminal Forecasts.
- C—Area Forecasts.

3454. Transcribed Weather Broadcasts (TWEB's) may be monitored by tuning the appropriate radio receiver to certain

- A—airport advisory frequencies.
- B—VOR and NDB frequencies.
- C—ATIS frequencies.

3455. When telephoning a weather briefing facility for preflight weather information, pilots should state

- A—the aircraft identification or the pilot's name.
- B—true airspeed.
- C—fuel on board.

3456. To get a complete weather briefing for the planned flight, the pilot should request

- A—a general briefing.
- B—an abbreviated briefing.
- C—a standard briefing.

3457. Which type weather briefing should a pilot request, when departing within the hour, if no preliminary weather information has been received?

- A—Outlook briefing.
- B—Abbreviated briefing.
- C—Standard briefing.

3458. Which type of weather briefing should a pilot request to supplement mass disseminated data?

- A—An outlook briefing.
- B—A supplemental briefing.
- C—An abbreviated briefing.

3459. To update a previous weather briefing, a pilot should request

- A—an abbreviated briefing.
- B—a standard briefing.
- C—an outlook briefing.

3460. A weather briefing that is provided when the information requested is 6 or more hours in advance of the proposed departure time is

- A—an outlook briefing.
- B—a forecast briefing.
- C—a prognostic briefing.

3461. When requesting weather information for the following morning, a pilot should request

- A—an outlook briefing.
- B—a standard briefing.
- C—an abbreviated briefing.

3462. (Refer to Figure 12.) Which of the reporting stations have VFR weather?

- A—All.
- B—INK, BOI, and JFK.
- C—INK, BOI, and LAX.

3463. Ceiling is defined as the height above the Earth's surface of the

- A—lowest reported obscuration and the highest layer of clouds reported as overcast.
- B—lowest layer of clouds or obscuring phenomena reported as broken, overcast, and not classified as thin or partial.
- C—lowest layer of clouds reported as scattered, broken, or thin.

3464. (Refer to Figure 12.) The wind direction and velocity at JFK is from

- A—180° true at 4 knots.
- B—180° magnetic at 4 knots.
- C—040° true at 18 knots.

3465. (Refer to Figure 12.) What are the wind conditions at Wink, Texas (INK)?

- A—Calm.
- B—110° at 12 knots, peak gusts 18 knots.
- C—111° at 2 knots, peak gusts 18 knots.

3466. (Refer to Figure 12.) The remarks section for MDW has RF2 and RB12 listed. These two entries mean

- A—rain and fog have reduced visibility to 2 miles and rain began at 1812Z.
- B—rain and fog are obscuring two-tenths of the sky and rain began at 1812Z.
- C—freezing rain has reduced visibility to 2 miles and the barometer has risen .12" Hg.

3467. (Refer to Figure 12.) What are the current conditions depicted for Chicago Midway Airport (MDW)?

- A—Sky partially obscured, measured ceiling 700 overcast, visibility 1-1/2, heavy rain, fog.
- B—Thin overcast, measured 700 ceiling overcast, visibility 1-1/2, heavy rain, fog.
- C—Sky partially obscured, measured ceiling 700 overcast, visibility 11, occasionally 2, with rain and heavy fog.

3468. (Refer to Figure 13.) According to the weather briefing, the most ideal time to launch balloons is

- A—as soon as possible after 1300Z.
- B—at 1500Z when the ground will be partially shaded.
- C—at 2000Z when there is enough wind for cross-country.

3469. (Refer to Figure 13.) According to the weather briefing, good balloon weather will begin to deteriorate

- A—soon after 1300Z as the wind starts to increase.
- B—about 1500Z when the lower scattered clouds begin to form.
- C—at 2000Z due to sharp increase in wind conditions.

3470. (Refer to Figure 13.) What effect do the clouds mentioned in the weather briefing have on soaring conditions?

- A—All thermals stop at the base of the clouds.
- B—Thermals persist to the tops of the clouds at 25,000 feet.
- C—The scattered clouds indicate thermals at least to the tops of the lower clouds.

3471. (Refer to Figure 13.) At what time will thermals begin to form?

- A—Between 1300Z and 1500Z while the sky is clear.
- B—By 1500Z (midmorning) when scattered clouds begin to form.
- C—About 2000Z (early afternoon) when the wind begins to increase.

3472. (Refer to Figure 14.) The base and tops of the overcast layer reported by a pilot are

- A—1,800 feet MSL and 5,500 feet MSL.
- B—5,500 feet AGL and 7,200 feet MSL.
- C—7,200 feet MSL and 8,900 feet MSL.

3473. (Refer to Figure 14.) The wind and temperature at 12,000 feet MSL as reported by a pilot are

- A—009° at 121 MPH and 90°F.
- B—090° at 21 knots and -9°F.
- C—090° at 21 knots and -9°C.

3474. (Refer to Figure 14.) If the terrain elevation is 1,295 feet MSL, what is the height above ground level of the base of the ceiling?

- A—505 feet AGL.
- B—1,295 feet AGL.
- C—6,586 feet AGL.

3475. (Refer to Figure 14.) The intensity of the turbulence reported at a specific altitude is

- A—moderate at 5,500 feet and at 7,200 feet.
- B—moderate from 5,500 feet to 7,200 feet.
- C—light to moderate from 7,200 feet to 8,900 feet.

3476. (Refer to Figure 14.) The intensity and type of icing reported by a pilot is

- A—light to moderate.
- B—light to moderate clear.
- C—moderate rime.

3477. Which weather reports and forecasts are most important for local area balloon operations?

- A—Winds Aloft Forecasts and Radar Summary Charts.
- B—Winds Aloft Forecasts and Surface Analysis Charts.
- C—Winds Aloft Forecasts and Surface Aviation Weather Reports.

3478. From which primary source should information be obtained regarding expected weather at the estimated time of arrival if your destination has no Terminal Forecast?

- A—Low-Level Prognostic Chart.
- B—Weather Depiction Chart.
- C—Area Forecast.

3479. (Refer to Figure 15.) What ceiling is forecast for GAG between 1600Z and 0100Z?

- A—6,000 scattered, chance 10,000 broken.
- B—10,000 scattered, chance 2,500 scattered.
- C—10,000 broken, chance 5,000 broken.

3480. (Refer to Figure 15.) What wind conditions are expected at HBR at 1600Z?

- A—Calm.
- B—150° at 15 knots.
- C—300° at 10 knots.

3481. (Refer to Figure 15.) What is the outlook for weather conditions at MLC?

- A—Ceilings 2,000 to 3,000 feet with southerly winds.
- B—Ceiling 700 feet, sky obscured, visibility 1/2 mile in thundershowers.
- C—Ceilings 1,000 to 3,000 feet with thunderstorms and rain showers.

3482. (Refer to Figure 15.) The wind condition in the Terminal Forecast 6-hour categorical outlook for PNC is for

- A—velocities of 25 knots or stronger.
- B—a wind shift from south to northwest.
- C—the wind to change from a gusty condition to calm.

3483. (Refer to Figure 15.) When is the wind forecast to shift at TUL?

- A—1500Z.
- B—By 2300Z.
- C—Between 2300Z and 0900Z the next day.

3484. (Refer to Figure 15.) According to the Terminal Forecast for OKC, the cold front should pass through

- A—between 1800Z and 2100Z.
- B—by 2100Z.
- C—after 2100Z.

3485. (Refer to Figure 15.) What type conditions are forecast for OKC from 1800Z to 2100Z?

- A—MVFR and VFR.
- B—MVFR, VFR, and IFR.
- C—VFR and IFR.

3486. (Refer to Figure 15.) What is the outlook for the weather condition at TUL?

- A—VFR with wind 25 knots or more with the direction not forecast.
- B—Chance of ceilings 3,000 feet broken, visibility 5 miles in thunderstorms and rain showers.
- C—VFR with winds 320° at 15 knots with gusts to 25 knots.

3487. To best determine general forecast weather conditions over several states, the pilot should refer to

- A—Area Forecasts.
- B—Weather Depiction Charts.
- C—Satellite Maps.

3488. (Refer to Figure 16.) What is the forecast ceiling and visibility for Tennessee from 2300Z through 0500Z?

- A—500 feet to less than 1,000 feet, and 1 mile to less than 3 miles.
- B—1,000 to 3,000 feet, and 3 to 5 miles.
- C—3,000 feet or greater, and 5 miles or greater.

3489. To determine the freezing level and areas of probable icing aloft, the pilot should refer to the

- A—Radar Summary Chart.
- B—Weather Depiction Chart.
- C—Area Forecast.

3490. The section of the Area Forecast entitled "SIG CLDS AND WX" contains a summary of

- A—cloudiness and weather significant to flight operations broken down by states or other geographical areas.
- B—forecast sky cover, cloud tops, visibility, and obstructions to vision along specific routes.
- C—weather advisories still in effect at the time of issue.

3491. (Refer to Figure 16.) What hazards are forecast in the Area Forecast for TN, AL, and the coastal waters?

- A—Thunderstorms with severe or greater turbulence, severe icing, and low-level wind shear.
- B—Moderate rime icing above the freezing level to 10,000 feet.
- C—Moderate turbulence from 25,000 to 38,000 feet due to the jetstream.

3492. (Refer to Figure 16.) What type obstructions to vision, if any, are forecast for the entire area from 2300Z until 0500Z the next day?

- A—None of any significance, VFR is forecast.
- B—Visibility 3 to 5 miles in fog.
- C—Visibility below 3 miles in fog over south-central Texas.

3493. (Refer to Figure 16.) What sky condition and type obstructions to vision are forecast for all the area except TN from 1040Z until 2300Z?

- A—Ceilings 3,000 to 5,000 feet broken, visibility 3 to 5 miles in fog.
- B—8,000 feet scattered to clear except visibility below 3 miles in fog until 1500Z over south-central Texas.
- C—Generally ceilings 3,000 to 8,000 feet to clear with visibility sometimes below 3 miles in fog.

3494. To obtain a continuous transcribed weather briefing, including winds aloft and route forecasts for a cross-country flight, a pilot should monitor a

- A—Transcribed Weather Broadcast (TWEB) on an ADF radio receiver.
- B—VHF radio receiver tuned to an Automatic Terminal Information Service (ATIS) frequency.
- C—regularly scheduled weather broadcast on a VOR frequency.

3495. What is indicated when a current CONVECTIVE SIGMET forecasts thunderstorms?

- A—Moderate thunderstorms covering 30 percent of the area.
- B—Moderate or severe turbulence.
- C—Thunderstorms obscured by massive cloud layers.

3496. What information is contained in a CONVECTIVE SIGMET?

- A—Tornadoes, embedded thunderstorms, and hail 3/4 inch or greater in diameter.
- B—Severe icing, severe turbulence, or widespread dust storms lowering visibility to less than 3 miles.
- C—Surface winds greater than 40 knots or thunderstorms equal to or greater than video integrator processor (VIP) level 4.

3497. SIGMET's are issued as a warning of weather conditions hazardous to which aircraft?

- A—Small aircraft only.
- B—Large aircraft only.
- C—All aircraft.

3498. Which in-flight advisory would contain information on severe icing?

- A—Convective SIGMET.
- B—SIGMET.
- C—AIRMET.

3499. AIRMET's are issued as a warning of weather conditions particularly hazardous to which aircraft?

- A—Small single-engine aircraft.
- B—Large multiengine aircraft.
- C—All aircraft.

3500. (Refer to Figure 17.) What wind is forecast for STL at 6,000 feet?

- A—210° magnetic at 13 knots.
- B—230° true at 25 knots.
- C—232° true at 5 knots.

3501. (Refer to Figure 17.) What wind is forecast for STL at 18,000 feet?

- A—230° true at 56 knots.
- B—235° true at 06 knots.
- C—235° magnetic at 06, peak gusts to 16 knots.

3502. (Refer to Figure 17.) Determine the wind and temperature aloft forecast for DEN at 30,000 feet.

- A—023° magnetic at 53 knots, temperature 47°C.
- B—230° true at 53 knots, temperature -47°C.
- C—235° true at 34 knots, temperature -7°C.

3503. (Refer to Figure 17.) Determine the wind and temperature aloft forecast for 3,000 feet at MKC.

- A—050° true at 7 knots, temperature missing.
- B—360° magnetic at 5 knots, temperature -7°C.
- C—360° true at 50 knots, temperature +7°C.

3504. (Refer to Figure 17.) What wind is forecast for STL at 34,000 feet?

- A—007° magnetic at 30 knots.
- B—073° true at 6 knots.
- C—230° true at 106 knots.

3505. What values are used for Winds Aloft Forecasts?

- A—Magnetic direction and knots.
- B—Magnetic direction and miles per hour.
- C—True direction and knots.

3506. When the term "light and variable" is used in reference to a Winds Aloft Forecast, the coded group and windspeed is

- A—0000 and less than 7 knots.
- B—9900 and less than 5 knots.
- C—9999 and less than 10 knots.

3507. (Refer to Figure 18.) What is the status of the front that extends from New Mexico to Indiana?

- A—Stationary.
- B—Occluded.
- C—Retreating.

3508. (Refer to Figure 18.) The IFR weather in eastern Texas is due to

- A—intermittent rain.
- B—fog.
- C—dust devils.

3509. (Refer to Figure 18.) Of what value is the Weather Depiction Chart to the pilot?

- A—For determining general weather conditions on which to base flight planning.
- B—For a forecast of cloud coverage, visibilities, and frontal activity.
- C—For determining frontal trends and air mass characteristics.

3510. (Refer to Figure 18.) The marginal weather in southeast New Mexico is due to

- A—reported thunderstorms.
- B—600-foot overcast ceilings.
- C—low visibility.

3511. (Refer to Figure 18.) What weather phenomenon is causing IFR conditions along the coast of Oregon and California?

- A—Squall line activity.
- B—Low ceilings.
- C—Heavy rain showers.

3512. (Refer to Figure 18.) According to the Weather Depiction Chart, the weather for a flight from central Arkansas to southeast Alabama is

- A—broken clouds at 2,500 feet.
- B—visibility from 3 to 5 miles.
- C—broken to scattered clouds at 25,000 feet.

3513. Radar weather reports are of special interest to pilots because they indicate

- A—large areas of low ceilings and fog.
- B—location of precipitation along with type, intensity, and trend.
- C—location of broken to overcast clouds.

3514. What information is provided by the Radar Summary Chart that is not shown on other weather charts?

- A—Lines and cells of hazardous thunderstorms.
- B—Ceilings and precipitation between reporting stations.
- C—Types of clouds between reporting stations.

3515. (Refer to Figure 19, area A.) What is the direction and speed of movement of the radar return?

- A—020° at 20 knots.
- B—East at 15 knots.
- C—Northeast at 22 knots.

3516. (Refer to Figure 19, area C.) What type of weather is occurring in the radar return?

- A—Continuous rain.
- B—Heavy rain showers.
- C—Rain showers increasing in intensity.

3517. (Refer to Figure 19, area D.) What is the direction and speed of movement of the radar return?

- A—Southeast at 30 knots.
- B—Northeast at 20 knots.
- C—West at 30 knots.

3518. (Refer to Figure 19, area D.) The top of the precipitation is

- A—2,000 feet.
- B—20,000 feet.
- C—30,000 feet.

3519. (Refer to Figure 19, area B.) What does the dashed line enclose?

- A—Areas of heavy rain.
- B—Severe weather watch area.
- C—Areas of hail 1/4 inch in diameter.

3520. (Refer to Figure 20.) How are Significant Weather Prognostic Charts best used by a pilot?

- A—For overall planning at all altitudes.
- B—For determining areas to avoid (freezing levels and turbulence).
- C—For analyzing current frontal activity and cloud coverage.

3521. (Refer to Figure 20.) Interpret the weather symbol depicted in southern California on the 12-hour Significant Weather Prognostic Chart.

- A—Moderate turbulence, surface to 18,000 feet.
- B—Thunderstorm tops at 18,000 feet.
- C—Base of clear air turbulence, 18,000 feet.

3522. (Refer to Figure 20.) What weather is forecast for the Gulf Coast area just ahead of the cold front during the first 12 hours?

- A—Ceiling 1,000 to 3,000 feet and/or visibility 3 to 5 miles with intermittent thundershowers and rain showers.
- B—IFR with moderate or greater turbulence over the coastal areas.
- C—Rain and thunderstorms moving northeastward ahead of the front.

3523. (Refer to Figure 20.) The low pressure associated with the cold front in the western states is forecast to move

- A—east at 30 knots.
- B—northeast at 12 knots.
- C—southeast at 30 knots.

3524. (Refer to Figure 20.) At what altitude is the freezing level over northeastern Oklahoma on the 24-hour Significant Weather Prognostic Chart?

- A—4,000 feet.
- B—8,000 feet.
- C—10,000 feet.

3525. In addition to the standard briefing, what additional information should be asked of the weather briefer in order to evaluate soaring conditions?

- A—The upper soundings to determine the thermal index at all soaring levels.
- B—Dry adiabatic rate of cooling to determine the height of cloud bases.
- C—Moist adiabatic rate of cooling to determine the height of cloud tops.

3526. When telephoning a weather briefing facility for preflight weather information, pilots should

- A—identify themselves as pilots.
- B—tell the number of hours they have flown within the preceding 90 days.
- C—state the number of occupants on board and the color of the aircraft.

3527. When telephoning a weather briefing facility for preflight weather information, pilots should state

- A—the full name and address of the pilot in command.
- B—the intended route, destination, and type of aircraft.
- C—the radio frequencies to be used.

3528. When telephoning a weather briefing facility for preflight weather information, pilots should state

- A—the full name and address of the formation commander.
- B—that they possess a current pilot certificate.
- C—whether they intend to fly VFR only.

3529. (Refer to Figure 21.) En route to First Flight Airport (area 5), your flight passes over Hampton Roads Airport (area 2) at 1456 and then over Chesapeake Municipal at 1501. At what time should your flight arrive at First Flight?

- A—1516.
- B—1521.
- C—1526.

3530. (Refer to Figure 21, area 3.) Determine the approximate latitude and longitude of Currituck County Airport.

- A—36°24'N – 76°01'W.
- B—36°48'N – 76°01'W.
- C—47°24'N – 75°58'W.

3531. (Refer to Figure 21.) Determine the magnetic course from First Flight Airport (area 5) to Hampton Roads Airport (area 2).

- A—312°.
- B—321°.
- C—330°.

3532. (Refer to Figure 21.) What is your approximate position on low altitude airway Victor 1, southwest of Norfolk (area 1), if the VOR receiver indicates you are on the 340° radial of Elizabeth City VOR (area 3)?

- A—15 nautical miles from Norfolk VORTAC.
- B—18 nautical miles from Norfolk VORTAC.
- C—23 nautical miles from Norfolk VORTAC.

3533. (Refer to Figure 21, area 3; and Figure 29.) The VOR is tuned to Elizabeth City VOR, and the aircraft is positioned over Shawboro. Which VOR indication is correct?

- A—5.
- B—6.
- C—8.

3534. (Refer to Figure 22.) What is the estimated time en route from Mercer County Regional Airport (area 3) to Minot International (area 1)? The wind is from 330° at 25 knots and the true airspeed is 100 knots. Add 3-1/2 minutes for departure and climb-out.

- A—44 minutes.
- B—48 minutes.
- C—52 minutes.

3535. (Refer to Figure 22, area 2.) Which airport is located at approximately 47°39'30"N latitude and 100°53'00"W longitude?

- A—Linrud.
- B—Crooked Lake.
- C—Johnson.

3536. (Refer to Figure 22, area 3.) Which airport is located at approximately 47°21'N latitude and 101°01'W longitude?

- A—Underwood.
- B—Evenson.
- C—Washburn.

3537. (Refer to Figure 22.) An airship crosses over Minot VORTAC (area 1) at 1056 and over the creek 8 nautical miles south-southeast on Victor 15 at 1108. What should be the approximate position on Victor 15 at 1211?

- A—Over Lake Nettie National Wildlife Refuge.
- B—Crossing the road east of Underwood.
- C—Over the powerlines east of Washburn Airport.

3538. (Refer to Figure 22.) Determine the magnetic heading for a flight from Mercer County Regional Airport (area 3) to Minot International (area 1). The wind is from 330° at 25 knots, the true airspeed is 100 knots, and the magnetic variation is 11° east.

- A—002°.
- B—012°.
- C—351°.

3539. (Refer to Figure 22.) What course should be selected on the omnibearing selector (OBS) to make a direct flight from Mercer County Regional Airport (area 3) to the Minot VORTAC (area 1) with a TO indication?

- A—001°.
- B—012°.
- C—181°.

3540. (Refer to Figure 23.) What is the estimated time en route from Dave Wall Field (area 1) to St. Maries Airport (area 4)? The wind is from 215° at 25 knots and the true airspeed is 125 knots.

- A—27 minutes.
- B—30 minutes.
- C—34 minutes.

3541. (Refer to Figure 23.) Determine the estimated time en route for a flight from Priest River Airport (area 1) to Shoshone County Airport (area 3). The wind is from 030 at 12 knots and the true airspeed is 95 knots. Add 2 minutes for climb-out.

- A—23 minutes.
- B—27 minutes.
- C—31 minutes.

3542. (Refer to Figure 23.) What is the estimated time en route for a flight from St. Maries Airport (area 4) to Priest River Airport (area 1)? The wind is from 300° at 14 knots and the true airspeed is 90 knots. Add 3 minutes for climb-out.

- A—38 minutes.
- B—43 minutes.
- C—48 minutes.

3543. (Refer to Figure 23, area 3.) Determine the approximate latitude and longitude of Shoshone County Airport.

- A—47°02'N—116°11'W.
- B—47°32'N—116°11'W.
- C—47°32'N—116°41'W.

3544. (Refer to Figure 23, area 2.) If a balloon is launched at Ranch Aero (Pvt) Airport with a reported wind from 220° at 5 knots, what should be its approximate position after 2 hours of flight?

- A—Near Hackney (Pvt) Airport.
- B—Crossing the railroad southwest of Granite Airport.
- C—3-1/2 miles southwest of Rathdrum.

3545. (Refer to Figure 23.) Determine the magnetic heading for a flight from Dave Wall Field (area 1) to St. Maries Airport (area 4). The wind is from 215° at 25 knots and the true airspeed is 125 knots.

- A—161°.
- B—167°.
- C—181°.

3546. (Refer to Figure 23.) What is the magnetic heading for a flight from Priest River Airport (area 1) to Shoshone County Airport (area 3)? The wind is from 030° at 12 knots and the true airspeed is 95 knots.

- A—116°.
- B—123°.
- C—130°.

3547. (Refer to Figure 23.) Determine the magnetic heading for a flight from St. Maries Airport (area 4) to Priest River Airport (area 1). The wind is from 300° at 14 knots and the true airspeed is 90 knots.

- A—319°.
- B—325°.
- C—331°.

3548. (Refer to Figure 24.) What is the estimated time en route for a flight from Allendale County Airport (area 1) to Claxton-Evans County Airport (area 2)? The wind is from 090° at 16 knots and the true airspeed is 90 knots. Add 2 minutes for climb-out.

- A—33 minutes.
- B—37 minutes.
- C—41 minutes.

3549. (Refer to Figure 24.) What is the estimated time en route for a flight from Claxton-Evans County Airport (area 2) to Hampton Varnville Airport (area 1)? The wind is from 290° at 18 knots and the true airspeed is 85 knots. Add 2 minutes for climb-out.

- A—35 minutes.
- B—39 minutes.
- C—44 minutes.

3550. (Refer to Figure 24.) Determine the compass heading for a flight from Allendale County Airport (area 1) to Claxton-Evans County Airport (area 2). The wind is from 090° at 16 knots and the true airspeed is 90 knots.

- A—200°.
- B—205°.
- C—211°.

3551. (Refer to Figure 24.) Determine the compass heading for a flight from Claxton-Evans County Airport (area 2) to Hampton Varnville Airport (area 1). The wind is from 290° at 18 knots and the true airspeed is 85 knots.

- A—034°.
- B—038°.
- C—042°.

3552. (Refer to Figure 24.) What is the approximate position of the aircraft if the VOR receivers indicate the 310° radial of Savannah VORTAC (area 3) and the 190° radial of Allendale VOR (area 1)?

- A—Town of Guyton.
- B—Town of Springfield.
- C—3 miles east of Marlow.

3553. (Refer to Figure 24.) On what radial should the VOR receiver (OBS) be set to navigate direct from Hampton Varnville Airport (area 1) to Savannah VORTAC (area 3)?

- A—005°.
- B—185°.
- C—200°.

3554. (Refer to Figure 24.) While en route on Victor 185, a flight crosses the 248° radial of Allendale VOR at 0951 and then crosses the 216° radial of Allendale VOR at 1000. What is the estimated time of arrival at Savannah VORTAC?

- A—1023.
- B—1028.
- C—1036.

3555. (Refer to Figure 25.) Estimate the time en route from Majors Airport (area 1) to Winnsboro Airport (area 2). The wind is from 340° at 12 knots and the true airspeed is 36 knots.

- A—55 minutes.
- B—59 minutes.
- C—63 minutes.

3556. (Refer to Figure 25). Determine the magnetic course from Airpark East Airport (area 1) to Winnsboro Airport (area 2). Magnetic variation is 6°30'E.

- A—075°.
- B—082°.
- C—091°.

3557. (Refer to Figure 25.) An airship passes over the Quitman VORTAC at 0940 and then over the intersection of the powerline and Victor 114 at 0948. Approximately what time should the flight arrive over the Blue Ridge VORTAC?

- A—1104.
- B—1109.
- C—1117.

3558. (Refer to Figure 25.) Determine the magnetic heading for a flight from Majors Airport (area 1) to Winnsboro Airport (area 2). The wind is from 340° at 12 knots, the true airspeed is 36 knots, and the magnetic variation is 6°30'E.

- A—078°.
- B—091°.
- C—101°.

3559. (Refer to Figure 25.) What is the approximate position of the aircraft if the VOR receivers indicate the 250° radial of Sulphur Springs VORTAC (area 2) and the 130° radial of Blue Ridge VORTAC (area 1)?

- A—Caddo Mills Airport.
- B—Meadowview Airport.
- C—3 miles southeast of Caddo Mills Airport.

3560. (Refer to Figure 25.) On what course should the VOR receiver (OBS) be set in order to navigate direct from Majors Airport (area 1) to Quitman VORTAC (area 2)?

- A—101°.
- B—108°.
- C—281°.

3561. (Refer to Figure 25, area 1; and Figure 29.) The VOR is tuned to Blue Ridge VORTAC, and the aircraft is positioned over the town of Lone Oak, southeast of Majors Airport. Which VOR indication is correct?

- A—1.
- B—4.
- C—7.

3562. (Refer to Figure 26.) What is the estimated time en route for a flight from Denton Muni (area 1) to Addison (area 2)? The wind is from 200° at 20 knots, the true airspeed is 110 knots, and the magnetic variation is 7° east.

- A—13 minutes.
- B—16 minutes.
- C—19 minutes.

3563. (Refer to Figure 26.) Estimate the time en route from Addison (area 2) to Redbird (area 3). The wind is from 300° at 15 knots, the true airspeed is 120 knots, and the magnetic variation is 7° east.

- A—8 minutes.
- B—11 minutes.
- C—14 minutes.

3564. (Refer to Figure 26.) Determine the magnetic heading for a flight from Redbird (area 3) to Fort Worth Meacham (area 4). The wind is from 030° at 10 knots, the true airspeed is 35 knots, and the magnetic variation is 7° east.

- A—266°.
- B—298°.
- C—312°.

3565. (Refer to Figure 26.) Determine the magnetic heading for a flight from Fort Worth Meacham (area 4) to Denton Muni (area 1). The wind is from 330° at 25 knots, the true airspeed is 110 knots, and the magnetic variation is 7° east.

- A—003°.
- B—017°.
- C—023°.

3566. (Refer to Figure 26, area 5.) The VOR is tuned to the Dallas/Fort Worth VORTAC. The omnibearing selector (OBS) is set on 253°, with a TO indication, and a right course deviation indicator (CDI) deflection. What is the aircraft's position from the VORTAC?

- A—East-northeast.
- B—North-northeast.
- C—West-southwest.

3567. (Refer to Figure 27, area 2.) What is the approximate latitude and longitude of Cooperstown Airport?

- A—47°25'N – 98°06'W.
- B—47°25'N – 99°54'W.
- C—47°55'N – 98°06'W.

3568. (Refer to Figure 27.) Determine the magnetic course from Breckheimer (Pvt) Airport (area 1) to Jamestown Airport (area 4).

- A—013°.
- B—021°.
- C—181°.

3569. (Refer to Figure 27, area 5.) A balloon drifts over the town of Eckelson on a magnetic course of 282° at 10 MPH. If wind conditions remain constant, where will the balloon be after 2 hours 30 minutes?

- A—3 miles south-southwest of Buchanan.
- B—Over Buchanan.
- C—Over the tower southwest of Fried.

3570. (Refer to Figure 27, areas 4 and 3; and Figure 29.) The VOR is tuned to Jamestown VOR, and the aircraft is positioned over the town of Wimbledon. Which VOR indication is correct?

- A—1.
- B—4.
- C—6.

3571. (Refer to Figure 28.) An aircraft departs an airport in the eastern daylight time zone at 0945 EDT for a 2-hour flight to an airport located in the central daylight time zone. The landing should be at what coordinated universal time?

- A—1345Z.
- B—1445Z.
- C—1545Z.

3572. (Refer to Figure 28.) An aircraft departs an airport in the central standard time zone at 0930 CST for a 2-hour flight to an airport located in the mountain standard time zone. The landing should be at what time?

- A—0930 MST.
- B—1030 MST.
- C—1130 MST.

3573. (Refer to Figure 28.) An aircraft departs an airport in the central standard time zone at 0845 CST for a 2-hour flight to an airport located in the mountain standard time zone. The landing should be at what coordinated universal time?

- A—1345Z.
- B—1445Z.
- C—1645Z.

3574. (Refer to Figure 28.) An aircraft departs an airport in the mountain standard time zone at 1615 MST for a 2-hour 15-minute flight to an airport located in the Pacific standard time zone. The estimated time of arrival at the destination airport should be

- A—1630 PST.
- B—1730 PST.
- C—1830 PST.

3575. (Refer to Figure 28.) An aircraft departs an airport in the Pacific standard time zone at 1030 PST for a 4-hour flight to an airport located in the central standard time zone. The landing should be at what coordinated universal time?

- A—2030Z.
- B—2130Z.
- C—2230Z.

3576. (Refer to Figure 28.) An aircraft departs an airport in the mountain standard time zone at 1515 MST for a 2-hour 30-minute flight to an airport located in the Pacific standard time zone. What is the estimated time of arrival at the destination airport?

- A—1645 PST.
- B—1745 PST.
- C—1845 PST.

3577. (Refer to Figure 29, illustration 1.) The VOR receiver has the indications shown. What is the aircraft's position relative to the station?

- A—North.
- B—East.
- C—South.

3578. (Refer to Figure 29, illustration 3.) The VOR receiver has the indications shown. What is the aircraft's position relative to the station?

- A—East.
- B—Southeast.
- C—West.

3579. (Refer to Figure 29, illustration 8.) The VOR receiver has the indications shown. What radial is the aircraft crossing?

- A—030°.
- B—210°.
- C—300°.

3580. (Refer to Figure 30, illustration 1.) Determine the magnetic bearing TO the station.

- A—030°.
- B—180°.
- C—210°.

3581. (Refer to Figure 30, illustration 2.) What magnetic bearing should the pilot use to fly TO the station?

- A—010°.
- B—145°.
- C—190°.

3582. (Refer to Figure 30, illustration 2.) Determine the approximate heading to intercept the 180° bearing TO the station.

- A—040°.
- B—160°.
- C—220°.

3583. (Refer to Figure 30, illustration 3.) What is the magnetic bearing FROM the station?

- A—025°.
- B—115°.
- C—295°.

3584. (Refer to Figure 30.) Which ADF indication represents the aircraft tracking TO the station with a right crosswind?

- A—1.
- B—2.
- C—4.

3585. (Refer to Figure 30, illustration 1.) What outbound bearing is the aircraft crossing?

- A—030°.
- B—150°.
- C—180°.

3586. (Refer to Figure 30, illustration 1.) What is the relative bearing TO the station?

- A—030°.
- B—210°.
- C—240°.

3587. (Refer to Figure 30, illustration 2.) What is the relative bearing TO the station?

- A—190°.
- B—235°.
- C—315°.

3588. (Refer to Figure 30, illustration 4.) What is the relative bearing TO the station?

- A—020°.
- B—060°.
- C—340°.

3589. (Refer to Figure 31, illustration 1.) The relative bearing TO the station is

- A—045°.
- B—180°.
- C—315°.

3590. (Refer to Figure 31, illustration 2.) The relative bearing TO the station is

- A—090°.
- B—180°.
- C—270°.

3591. (Refer to Figure 31, illustration 3.) The relative bearing TO the station is

- A—090°.
- B—180°.
- C—270°.

3592. (Refer to Figure 31, illustration 4.) On a magnetic heading of 320°, the magnetic bearing TO the station is

- A—005°.
- B—185°.
- C—225°.

3593. (Refer to Figure 31, illustration 5.) On a magnetic heading of 035°, the magnetic bearing TO the station is

- A—035°.
- B—180°.
- C—215°.

3594. (Refer to Figure 31, illustration 6.) On a magnetic heading of 120°, the magnetic bearing TO the station is

- A—045°.
- B—165°.
- C—270°.

3595. (Refer to Figure 31, illustration 6.) If the magnetic bearing TO the station is 240°, the magnetic heading is

- A—045°.
- B—105°.
- C—195°.

3596. (Refer to Figure 31, illustration 7.) If the magnetic bearing TO the station is 030°, the magnetic heading is

- A—060°.
- B—120°.
- C—270°.

3597. (Refer to Figure 31, illustration 8.) If the magnetic bearing TO the station is 135°, the magnetic heading is

- A—135°.
- B—270°.
- C—360°.

3598. When the course deviation indicator (CDI) needle is centered during an omnireceiver check using a VOR test signal (VOT), the omnibearing selector (OBS) and the TO/FROM indicator should read

- A—180° FROM, only if the pilot is due north of the VOT.
- B—0° TO or 180° FROM, regardless of the pilot's position from the VOT.
- C—0° FROM or 180° TO, regardless of the pilot's position from the VOT.

3599. (Refer to Figure 26, area 4.) The floor of Class B airspace overlying Hicks Airport (T67) north-northwest of Fort Worth Meacham Field is

- A—at the surface.
- B—3,200 feet MSL.
- C—4,000 feet MSL.

3600. (Refer to Figure 26, area 2.) The floor of Class B airspace at Addison Airport is

- A—at the surface.
- B—3,000 feet MSL.
- C—3,100 feet MSL.

3601. (Refer to Figure 21.) What hazards to aircraft may exist in warning areas such as Warning W-50B?

- A—Unusual, often invisible, hazards such as aerial gunnery or guided missiles over international waters.
- B—High volume of pilot training or unusual type of aerial activity.
- C—Heavy military aircraft traffic in the approach and departure area of the North Atlantic Control Area.

3602. (Refer to Figure 27.) What hazards to aircraft may exist in areas such as Devils Lake East MOA?

- A—Unusual, often invisible, hazards to aircraft such as artillery firing.
- B—High density military training activities.
- C—Parachute jump operations.

3603. (Refer to Figure 22.) What type military flight operations should a pilot expect along IR 644?

- A—IFR training flights above 1,500 feet AGL at speeds in excess of 250 knots.
- B—VFR training flights above 1,500 feet AGL at speeds less than 250 knots.
- C—Instrument training flights below 1,500 feet AGL at speeds in excess of 150 knots.

3604. (Refer to Figure 21, area 3.) What is the recommended communications procedure for a landing at Currituck County Airport?

- A—Transmit intentions on 122.9 MHz when 10 miles out and give position reports in the traffic pattern.
- B—Contact Elizabeth City FSS for airport advisory service.
- C—Contact New Bern FSS for area traffic information.

3605. (Refer to Figure 22, area 2.) The CTAF/MULTICOM frequency for Garrison Municipal is

- A—122.8 MHz.
- B—122.9 MHz.
- C—123.0 MHz.

3606. (Refer to Figure 23, area 2; and Figure 32.) If Coeur D'Alene Tower is not in operation, which frequency should be used as a Common Traffic Advisory Frequency (CTAF) to self-announce position and intentions?

- A—119.1 MHz.
- B—122.1/108.8 MHz.
- C—122.8 MHz.

3607. (Refer to Figure 23, area 2; and Figure 32.) If Coeur D'Alene Tower is not in operation, which frequency should be used as a Common Traffic Advisory Frequency (CTAF) to monitor airport traffic?

- A—119.1 MHz.
- B—122.1/108.8 MHz.
- C—122.8 MHz.

3608. (Refer to Figure 23, area 2; and figure 32.) What is the correct UNICOM frequency to be used at Coeur D'Alene to request fuel?

- A—119.1 MHz.
- B—122.1/108.8 MHz.
- C—122.8 MHz.

3609. (Refer to Figure 26, area 3.) If Redbird Tower is not in operation, which frequency should be used as a Common Traffic Advisory Frequency (CTAF) to monitor airport traffic?

- A—120.3 MHz.
- B—122.95 MHz.
- C—126.35 MHz.

3610. (Refer to Figure 27, area 2.) What is the recommended communication procedure when inbound to land at Cooperstown Airport?

- A—Broadcast intentions when 10 miles out on the CTAF/MULTICOM frequency, 122.9 MHz.
- B—Contact UNICOM when 10 miles out on 122.8 MHz.
- C—Circle the airport in a left turn prior to entering traffic.

3611. (Refer to Figure 27, area 4.) The CTAF/UNICOM frequency at Jamestown Airport is

- A—122.0 MHz.
- B—123.0 MHz.
- C—123.6 MHz.

3612. (Refer to Figure 27, area 6.) What is the CTAF/UNICOM frequency at Barnes County Airport?

- A—122.0 MHz.
- B—122.8 MHz.
- C—123.6 MHz.

3613. When flying HAWK N666CB, the proper phraseology for initial contact with McAlester AFSS is

- A—"MC ALESTER RADIO, HAWK SIX SIX SIX CHARLIE BRAVO, RECEIVING ARDMORE VORTAC, OVER."
- B—"MC ALESTER STATION, HAWK SIX SIX SIX CEE BEE, RECEIVING ARDMORE VORTAC, OVER."
- C—"MC ALESTER FLIGHT SERVICE STATION, HAWK NOVEMBER SIX CHARLIE BRAVO, RECEIVING ARDMORE VORTAC, OVER."

3614. The correct method of stating 4,500 feet MSL to ATC is

- A—"FOUR THOUSAND FIVE HUNDRED."
- B—"FOUR POINT FIVE."
- C—"FORTY-FIVE HUNDRED FEET MSL."

3615. The correct method of stating 10,500 feet MSL to ATC is

- A—"TEN THOUSAND, FIVE HUNDRED FEET."
- B—"TEN POINT FIVE."
- C—"ONE ZERO THOUSAND, FIVE HUNDRED."

3616. How should contact be established with an En Route Flight Advisory Service (EFAS) station, and what service would be expected?

- A—Call EFAS on 122.2 for routine weather, current reports on hazardous weather, and altimeter settings.
- B—Call flight assistance on 122.5 for advisory service pertaining to severe weather.
- C—Call Flight Watch on 122.0 for information regarding actual weather and thunderstorm activity along proposed route.

3617. What service should a pilot normally expect from an En Route Flight Advisory Service (EFAS) station?

- A—Actual weather information and thunderstorm activity along the route.
- B—Preferential routing and radar vectoring to circumnavigate severe weather.
- C—Severe weather information, changes to flight plans, and receipt of routine position reports.

3618. (Refer to Figure 27, area 3.) When flying over Arrowwood National Wildlife Refuge, a pilot should fly no lower than

- A—2,000 feet AGL.
- B—2,500 feet AGL.
- C—3,000 feet AGL.

3619. (Refer to Figure 23, area 2 and legend 1.) For information about the parachute jumping and glider operations at Silverwood Airport, refer to

- A—notes on the border of the chart.
- B—the Airport/Facility Directory.
- C—the Notices to Airmen (NOTAM) publication.

3620. (Refer to Figure 23, area 1.) The visibility and cloud clearance requirements to operate VFR during daylight hours over Dave Wall Field at less than 1,200 feet AGL are

- A—1 mile and clear of clouds.
- B—1 mile and 1,000 feet above, 500 feet below, and 2,000 feet horizontally from each cloud.
- C—3 miles and 1,000 feet above, 500 feet below, and 2,000 feet horizontally from each cloud.

3621. (Refer to Figure 27, area 2.) The visibility and cloud clearance requirements to operate VFR during daylight hours over the town of Cooperstown between 1,200 feet AGL and 10,000 feet MSL are

- A—1 mile and clear of clouds.
- B—1 mile and 1,000 feet above, 500 feet below, and 2,000 feet horizontally from clouds.
- C—3 miles and 1,000 feet above, 500 feet below, and 2,000 feet horizontally from clouds.

3622. (Refer to Figure 27, area 1.) Identify the airspace over Lowe Airport that exists from the surface to 14,500 feet MSL.

- A—Class G airspace – surface to 14,500 feet MSL.
- B—Class G airspace – surface to 3,500 feet MSL; Class E airspace – 3,500 feet MSL to 14,500 feet MSL.
- C—Class G airspace – surface to 3,500 feet MSL; Class E airspace – 3,500 feet MSL to 10,000 feet MSL; Class G airspace – 10,000 feet MSL to 14,500 feet MSL.

3623. (Refer to Figure 27, area 6.) The airspace overlying and within 5 miles of Barnes County Airport is

- A—Class D airspace from the surface to the floor of the overlying Class E airspace.
- B—Class E airspace from the surface to 1,200 feet MSL.
- C—Class G airspace from the surface to 700 feet AGL.

3624. (Refer to Figure 26, area 7.) The airspace overlying McKinney Muni is uncontrolled from the surface to

- A—700 feet AGL.
- B—1,700 feet MSL.
- C—4,000 feet AGL.

3625. (Refer to Figure 26, area 4.) The airspace directly overlying Fort Worth Meacham is

- A—Class B airspace to 10,000 feet MSL.
- B—Class C airspace to 5,000 feet MSL.
- C—Class D airspace to 3,200 feet MSL.

3626. (Refer to Figure 24, area 3.) What is the floor of the Savannah Class C airspace at the outer circle?

- A—1,200 feet AGL.
- B—1,300 feet MSL.
- C—1,700 feet MSL.

3627. (Refer to Figure 21, area 1.) What minimum radio equipment is required to land and take off at Norfolk International?

- A—Mode C transponder and omnireceiver.
- B—Mode C transponder and two-way radio.
- C—Mode C transponder, omnireceiver, and DME.

3628. (Refer to Figure 26.) At which airports is fixed-wing Special VFR not authorized?

- A—Fort Worth Meacham and Fort Worth Spinks.
- B—Dallas-Fort Worth International and Dallas Love Field.
- C—Addison and Redbird.

3629. (Refer to Figure 23, area 3.) The vertical limits of that portion of Class E airspace designated as a Federal Airway over Magee Airport are

- A—1,200 feet AGL to 10,000 feet MSL.
- B—7,500 feet MSL to 12,500 feet MSL.
- C—7,500 feet MSL to 17,999 feet MSL.

3630. (Refer to Figure 22.) On what frequency can a pilot receive Hazardous Inflight Weather Advisory Service (HIWAS) in the vicinity of area 1?

- A—117.1 MHz.
- B—118.0 MHz.
- C—122.0 MHz.

3631. (Refer to Figure 21, area 5.) The CAUTION box denotes what hazard to aircraft?

- A—Guy wires extending from radio or TV towers.
- B—Tail bridge over the inlet to the body of water.
- C—Cable extending from radar-outfitted balloons.

3632. (Refer to Figure 21, area 2.) The flag symbol at Lake Drummond represents a

- A—compulsory reporting point for Norfolk Class C airspace.
- B—compulsory reporting point for Hampton Roads Airport.
- C—visual checkpoint used to identify position for initial callup to Norfolk Approach Control.

3633. (Refer to Figure 21, area 2.) The elevation of the Chesapeake Municipal Airport is

- A—20 feet.
- B—36 feet.
- C—360 feet.

3634. (Refer to Figure 22.) The terrain elevation of the light tan area between Minot (area 1) and Audubon Lake (area 2) varies from

- A—sea level to 2,000 feet MSL.
- B—2,000 feet to 2,500 feet MSL.
- C—2,000 feet to 2,700 feet MSL.

3635. (Refer to Figure 22.) Which public use airports depicted are indicated as having fuel?

- A—Minot and Mercer County Regional Airport.
- B—Minot and Garrison.
- C—Mercer County Regional Airport and Garrison.

3636. (Refer to Figure 24.) The flag symbols at Statesboro Airport, Claxton-Evans County Airport, and Ridgeland Airport are

- A—outer boundaries of Savannah Class C airspace.
- B—airports with special traffic patterns.
- C—visual checkpoints to identify position for initial callup prior to entering Savannah Class C airspace.

3637. (Refer to Figure 24, area 3.) What is the height of the lighted obstacle approximately 7 nautical miles southwest of Savannah International?

- A—1,500 feet AGL.
- B—1,532 feet AGL.
- C—1,549 feet AGL.

3638. (Refer to Figure 24, area 3.) The top of the lighted stack approximately 12 nautical miles from the Savannah VORTAC on the 350° radial is

- A—305 feet AGL.
- B—400 feet AGL.
- C—430 feet AGL.

3639. (Refer to Figure 25, area 1.) What minimum altitude is necessary to vertically clear the obstacle on the northeast side of Airpark East Airport by 500 feet?

- A—1,010 feet MSL.
- B—1,273 feet MSL.
- C—1,283 feet MSL.

3640. (Refer to Figure 25, area 2.) What minimum altitude is necessary to vertically clear the obstacle on the southeast side of Winnsboro Airport by 500 feet?

- A—823 feet MSL.
- B—1,013 feet MSL.
- C—1,403 feet MSL.

3641. (Refer to Figure 26, area 2.) The control tower frequency for Addison Airport is

- A—122.95 MHz.
- B—126.0 MHz.
- C—133.4 MHz.

3642. (Refer to Figure 26, area 8.) What minimum altitude is required to fly over the Cedar Hill TV towers in the congested area south of NAS Dallas?

- A—2,533 feet MSL.
- B—2,849 feet MSL.
- C—3,349 feet MSL.

3643. (Refer to Figure 26, area 5.) The navigation facility at Dallas-Ft. Worth International (DFW) is a

- A—VOR.
- B—VORTAC.
- C—VOR/DME.

3644. (Refer to Figure 21, area 4.) A balloon launched at the town of Edenton drifts northeasterly along the railroad. What minimum altitude must it maintain to clear all of the obstacles in the vicinity of Hertford by at least 500 feet?

- A—805 feet MSL.
- B—1,000 feet MSL.
- C—1,015 feet MSL.

3645. (Refer to Figure 22, area 1.) A balloon launched at Flying S Ranch Airport drifts southward towards the lighted obstacle. If the altimeter was set at 0 feet upon launch, what should it indicate if the balloon is to clear the obstacle at 500 feet above the top?

- A—1,531 feet AGL.
- B—1,809 feet AGL.
- C—2,340 feet AGL.

3646. (Refer to Figure 23, area 1.) A balloon, launched at CX Airport located near the east end of Lake Pend Oreille, drifts south-southwest. What is the approximate elevation of the highest terrain for 20 miles along its path?

- A—2,000 – 4,000 feet MSL.
- B—4,000 – 6,000 feet MSL.
- C—6,000 – 7,000 feet MSL.

3647. (Refer to Figure 21.) Over which area should a glider pilot expect to find the best lift under normal conditions?

- A—6.
- B—7.
- C—8.

3648. (Refer to Figure 27.) If a glider is launched over Barnes County Airport (area 6) with sufficient altitude to glide to Jamestown Airport (area 4), how long will it take for the flight at an average of 40 MPH groundspeed?

- A—20 minutes.
- B—27 minutes.
- C—46 minutes.

3649. (Refer to Figure 25, area 1.) A glider is launched over Caddo Mills Airport with sufficient altitude to glide to Airpark East Airport, south of Caddo Mills. How long will it take for the flight at an average of 35 MPH groundspeed?

- A—27 minutes.
- B—29 minutes.
- C—31 minutes.

3650. (Refer to Figure 27, areas 5 and 6.) What minimum altitude should be used for a go-ahead point at Eckelson in order to arrive at Barnes County Airport at 1,000 feet AGL if the glide ratio is 22:1 in no wind conditions? Use the recommended safety factor.

- A—5,959 feet MSL.
- B—7,960 feet MSL.
- C—9,359 feet MSL.

3651. What action can a pilot take to aid in cooling an engine that is overheating during a climb?

- A—Reduce rate of climb and increase airspeed.
- B—Reduce climb speed and increase RPM.
- C—Increase climb speed and increase RPM.

3652. What is one procedure to aid in cooling an engine that is overheating?

- A—Enrich the fuel mixture.
- B—Increase the RPM.
- C—Reduce the airspeed.

3653. How is engine operation controlled on an engine equipped with a constant-speed propeller?

- A—The throttle controls power output as registered on the manifold pressure gauge and the propeller control regulates engine RPM.
- B—The throttle controls power output as registered on the manifold pressure gauge and the propeller control regulates a constant blade angle.
- C—The throttle controls engine RPM as registered on the tachometer and the mixture control regulates the power output.

3654. What is an advantage of a constant-speed propeller?

- A—Permits the pilot to select and maintain a desired cruising speed.
- B—Permits the pilot to select the blade angle for the most efficient performance.
- C—Provides a smoother operation with stable RPM and eliminates vibrations.

3655. A precaution for the operation of an engine equipped with a constant-speed propeller is to

- A—avoid high RPM settings with high manifold pressure.
- B—avoid high manifold pressure settings with low RPM.
- C—always use a rich mixture with high RPM settings.

3656. What should be the first action after starting an aircraft engine?

- A—Adjust for proper RPM and check for desired indications on the engine gauges.
- B—Place the magneto or ignition switch momentarily in the OFF position to check for proper grounding.
- C—Test each brake and the parking brake.

3657. Should it become necessary to handprop an airplane engine, it is extremely important that a competent pilot

- A—call "contact" before touching the propeller.
- B—be at the controls in the cockpit.
- C—be in the cockpit and call out all commands.

3658. In regard to preflighting an aircraft, what is the minimum expected of a pilot prior to every flight?

- A—Drain fuel from each quick drain.
- B—Perform a walk-around inspection of the aircraft.
- C—Check the required documents aboard the aircraft.

3659. Why is the use of a written checklist recommended for preflight inspection and engine start?

- A—To ensure that all necessary items are checked in a logical sequence.
- B—For memorizing the procedures in an orderly sequence.
- C—To instill confidence in the passengers.

3660. What special check should be made on an aircraft during preflight after it has been stored an extended period of time?

- A—ELT batteries and operation.
- B—Condensation in the fuel tanks.
- C—Damage or obstructions caused by animals, birds, or insects.

3661. Which items are included in the empty weight of an aircraft?

- A—Unusable fuel and undrainable oil.
- B—Only the airframe, powerplant, and optional equipment.
- C—Full fuel tanks and engine oil to capacity.

3662. An aircraft is loaded 110 pounds over maximum certificated gross weight. If fuel (gasoline) is drained to bring the aircraft weight within limits, how much fuel should be drained?

- A—15.7 gallons.
- B—16.2 gallons.
- C—18.4 gallons.

3663. If an aircraft is loaded 90 pounds over maximum certificated gross weight and fuel (gasoline) is drained to bring the aircraft weight within limits, how much fuel should be drained?

- A—10 gallons.
- B—12 gallons.
- C—15 gallons.

3664. GIVEN:

	WEIGHT (LB)	ARM (IN)	MOMENT (LB-IN)
Empty weight	1,495.0	101.4	151,593.0
Pilot and passengers	380.0	64.0	24,320
Fuel (30 gal usable – no reserve)	2055	94.01	193,193

The CG is located how far aft of datum?

- A—CG 92.44.
- B—CG 94.01.
- C—CG 119.8.

3665. (Refer to Figures 33 and 34.) Determine if the airplane weight and balance is within limits.

Front seat occupants	340 lb
Rear seat occupants	295 lb
Fuel (main wing tanks)	44 gal
Baggage	56 lb

- A—20 pounds overweight, CG aft of aft limits.
- B—20 pounds overweight, CG within limits.
- C—20 pounds overweight, CG forward of forward limits.

3666. (Refer to Figures 33 and 34.) What is the maximum amount of baggage that can be carried when the airplane is loaded as follows?

Front seat occupants	387 lb
Rear seat occupants	293 lb
Fuel	35 gal

- A—45 pounds.
- B—63 pounds.
- C—220 pounds.

3667. (Refer to Figures 33 and 34.) Calculate the weight and balance and determine if the CG and the weight of the airplane are within limits.

Front seat occupants	350 lb
Rear seat occupants	325 lb
Baggage	27 lb
Fuel	35 gal

- A—CG 81.7, out of limits forward.
- B—CG 83.4, within limits.
- C—CG 84.1, within limits.

3668. (Refer to Figures 33 and 34.) Determine if the airplane weight and balance is within limits.

Front seat occupants	415 lb
Rear seat occupants	110 lb
Fuel, main tanks	44 gal
Fuel, aux. tanks	19 gal
Baggage	32 lb

- A—19 pounds overweight, CG within limits.
- B—19 pounds overweight, CG out of limits forward.
- C—Weight within limits, CG out of limits.

3669. (Refer to Figure 35.) What is the maximum amount of baggage that may be loaded aboard the airplane for the CG to remain within the moment envelope?

	WEIGHT (LB)	MOM/1000
Empty weight	1,350	51.5
Pilot and front passenger	250	—
Rear passengers	400	—
Baggage	—	—
Fuel, 30 gal	—	—
Oil, 8 qt	—	-0.2

- A—105 pounds.
- B—110 pounds.
- C—120 pounds.

3670. (Refer to Figure 35.) Calculate the moment of the airplane and determine which category is applicable.

	WEIGHT (LB)	MOM/1000
Empty weight	1,350	51.5
Pilot and front passenger	310	—
Rear passengers	96	—
Fuel, 38 gal	—	—
Oil, 8 qt	—	-0.2

- A—79.2, utility category.
- B—80.8, utility category.
- C—81.2, normal category.

3671. (Refer to Figure 35.) What is the maximum amount of fuel that may be aboard the airplane on takeoff if loaded as follows?

	WEIGHT (LB)	MOM/1000
Empty weight	1,350	51.5
Pilot and front passenger	340	—
Rear passengers	310	—
Baggage	45	—
Oil, 8 qt	—	—

- A—24 gallons.
- B—32 gallons.
- C—40 gallons.

3672. (Refer to Figure 35.) Determine the moment with the following data:

	WEIGHT (LB)	MOM/1000
Empty weight	1,350	51.5
Pilot and front passenger	340	—
Fuel (std tanks)	Capacity	—
Oil, 8 qt	—	—

- A—69.9 pound-inches.
- B—74.9 pound-inches.
- C—77.6 pound-inches.

3673. (Refer to Figure 35.) Determine the aircraft loaded moment and the aircraft category.

	WEIGHT (LB)	MOM/1000
Empty weight	1,350	51.5
Pilot and front passenger	380	—
Fuel, 48 gal	288	—
Oil, 8 qt	—	—

- A—78.2, normal category.
- B—79.2, normal category.
- C—80.4, utility category.

3674. (Refer to Figures 33 and 34.) Upon landing, the front passenger (180 pounds) departs the airplane. A rear passenger (204 pounds) moves to the front passenger position. What effect does this have on the CG if the airplane weighed 2,690 pounds and the MOM/100 was 2,260 just prior to the passenger transfer?

- A—The CG moves forward approximately 3 inches.
- B—The weight changes, but the CG is not affected.
- C—The CG moves forward approximately 0.1 inch.

3675. (Refer to Figures 33 and 34.) Which action can adjust the airplane's weight to maximum gross weight and the CG within limits for takeoff?

Front seat occupants	425 lb
Rear seat occupants	300 lb
Fuel, main tanks	44 gal

- A—Drain 12 gallons of fuel.
- B—Drain 9 gallons of fuel.
- C—Transfer 12 gallons of fuel from the main tanks to the auxiliary tanks.

3676. (Refer to Figures 33 and 34.) What effect does a 35-gallon fuel burn (main tanks) have on the weight and balance if the airplane weighed 2,890 pounds and the MOM/100 was 2,452 at takeoff?

- A—Weight is reduced by 210 pounds and the CG is aft of limits.
- B—Weight is reduced by 210 pounds and the CG is unaffected.
- C—Weight is reduced to 2,680 pounds and the CG moves forward.

3677. (Refer to Figures 33 and 34.) With the airplane loaded as follows, what action can be taken to balance the airplane?

Front seat occupants	411 lb
Rear seat occupants	100 lb
Main wing tanks	44 gal

- A—Fill the auxiliary wing tanks.
- B—Add a 100-pound weight to the baggage compartment.
- C—Transfer 10 gallons of fuel from the main tanks to the auxiliary tanks.

3678. (Refer to Figure 36.) Approximately what true airspeed should a pilot expect with 65 percent maximum continuous power at 9,500 feet with a temperature of 36°F below standard?

- A—178 MPH.
- B—181 MPH.
- C—183 MPH.

3679. (Refer to Figure 36.) What is the expected fuel consumption for a 1,000-nautical mile flight under the following conditions?

Pressure altitude	8,000 ft
Temperature	22°C
Manifold pressure	20.8" Hg
Wind	Calm

- A—60.2 gallons.
- B—70.1 gallons.
- C—73.2 gallons.

3680. (Refer to Figure 36.) What is the expected fuel consumption for a 500-nautical mile flight under the following conditions?

Pressure altitude	4,000 ft
Temperature	+29°C
Manifold pressure	21.3" Hg
Wind	Calm

- A—31.4 gallons.
- B—36.1 gallons.
- C—40.1 gallons.

3681. (Refer to Figure 36.) What fuel flow should a pilot expect at 11,000 feet on a standard day with 65 percent maximum continuous power?

- A—10.6 gallons per hour.
- B—11.2 gallons per hour.
- C—11.8 gallons per hour.

3682. (Refer to Figure 36.) Determine the approximate manifold pressure setting with 2,450 RPM to achieve 65 percent maximum continuous power at 6,500 feet with a temperature of 36°F higher than standard.

- A—19.8" Hg.
- B—20.8" Hg.
- C—21.0" Hg.

3683. (Refer to Figure 37.) What is the headwind component for a landing on Runway 18 if the tower reports the wind as 220° at 30 knots?

- A—19 knots.
- B—23 knots.
- C—26 knots.

3684. (Refer to Figure 37.) Determine the maximum wind velocity for a 45° crosswind if the maximum crosswind component for the airplane is 25 knots.

- A—25 knots.
- B—29 knots.
- C—35 knots.

3685. (Refer to Figure 37.) What is the maximum wind velocity for a 30° crosswind if the maximum crosswind component for the airplane is 12 knots?

- A—16 knots.
- B—20 knots.
- C—24 knots.

3686. (Refer to Figure 37.) With a reported wind of north at 20 knots, which runway (6, 29, or 32) is acceptable for use for an airplane with a 13-knot maximum crosswind component?

- A—Runway 6.
- B—Runway 29.
- C—Runway 32.

3687. (Refer to Figure 37.) With a reported wind of south at 20 knots, which runway (10, 14, or 24) is appropriate for an airplane with a 13-knot maximum crosswind component?

- A—Runway 10.
- B—Runway 14.
- C—Runway 24.

3688. (Refer to Figure 37.) What is the crosswind component for a landing on Runway 18 if the tower reports the wind as 220° at 30 knots?

- A—19 knots.
- B—23 knots.
- C—30 knots.

3689. (Refer to Figure 38.) Determine the total distance required to land.

OAT 32°F
Pressure altitude 8,000 ft
Weight 2,600 lb
Headwind component 20 kts
Obstacle 50 ft

- A—850 feet.
- B—1,400 feet.
- C—1,750 feet.

3690. (Refer to Figure 38.) Determine the total distance required to land.

OAT Std
Pressure altitude 2,000 ft
Weight 2,300 lb
Wind component Calm
Obstacle None

- A—850 feet.
- B—1,250 feet.
- C—1,450 feet.

3691. (Refer to Figure 38.) Determine the total distance required to land.

OAT 90°F
Pressure altitude 3,000 ft
Weight 2,900 lb
Headwind component 10 kts
Obstacle 50 ft

- A—1,450 feet.
- B—1,550 feet.
- C—1,725 feet.

3692. (Refer to Figure 38.) Determine the approximate ground roll distance after landing.

OAT 90°F
Pressure altitude 4,000 ft
Weight 2,800 lb
Tailwind component 10 kts

- A—1,575 feet.
- B—1,725 feet.
- C—1,950 feet.

3693. (Refer to Figure 39.) Determine the approximate landing ground roll distance.

Pressure altitude Sea level
Headwind 4 kts
Temperature Std

- A—356 feet.
- B—401 feet.
- C—490 feet.

3694. (Refer to Figure 39.) Determine the total distance required to land over a 50-foot obstacle.

Pressure altitude 7,500 ft
Headwind 8 kts
Temperature Std
Runway Dry grass

- A—1,004 feet.
- B—1,205 feet.
- C—1,506 feet.

3695. (Refer to Figure 39.) Determine the total distance required to land over a 50-foot obstacle.

Pressure altitude 5,000 ft
Headwind 8 kts
Temperature 41°F
Runway Hard surface

- A—837 feet.
- B—956 feet.
- C—1,076 feet.

3696. (Refer to Figure 39.) Determine the total distance required to land over a 50-foot obstacle.

Pressure altitude 5,000 ft
Headwind Calm
Temperature 101°F

- A—1,076 feet.
- B—1,291 feet.
- C—1,314 feet.

3697. (Refer to Figure 39.) Determine the approximate landing ground roll distance.

Pressure altitude 3,750 ft
Headwind 12 kts
Temperature Std

- A—338 feet.
- B—425 feet.
- C—483 feet.

3698. (Refer to Figure 39.) Determine the approximate landing ground roll distance.

Pressure altitude 1,250 ft
Headwind 8 kts
Temperature Std

- A—275 feet.
- B—366 feet.
- C—470 feet.

3699. (Refer to Figure 40.) Determine the total landing distance to clear a 50-foot obstacle in a gyroplane. The outside air temperature (OAT) is 75°F and the pressure altitude at the airport is 2,500 feet.

- A—521 feet.
- B—525 feet.
- C—529 feet.

3700. (Refer to Figure 40.) Approximately how much additional landing distance will be required for a gyroplane to clear a 50-foot obstacle with an increase in temperature from 40 to 60°F at 3,200 feet pressure altitude?

- A—4 feet.
- B—8 feet.
- C—12 feet.

3701. (Refer to Figure 40.) Determine the total landing distance to clear a 50-foot obstacle in a gyroplane. The outside air temperature (OAT) is 80°F and the pressure altitude is 3,500 feet.

- A—521 feet.
- B—526 feet.
- C—531 feet.

3702. (Refer to Figure 40.) Determine the total takeoff distance required for a gyroplane to clear a 50-foot obstacle if the temperature is 95°F and the pressure altitude is 1,700 feet.

- A—1,825 feet.
- B—1,910 feet.
- C—2,030 feet.

3703. (Refer to Figure 40.) Determine the total takeoff distance required for a gyroplane to clear a 50-foot obstacle if the temperature is standard at sea level pressure altitude.

- A—950 feet.
- B—1,090 feet.
- C—1,200 feet.

3704. (Refer to Figure 40.) Approximately how much additional takeoff distance will be required for a gyroplane to clear a 50-foot obstacle if the temperature increases from 75 to 90°F at a pressure altitude of 2,300 feet?

- A—160 feet.
- B—200 feet.
- C—2,020 feet.

3705. (Refer to Figure 41.) Determine the total distance required for takeoff to clear a 50-foot obstacle.

OAT Std
Pressure altitude 4,000 ft
Takeoff weight 2,800 lb
Headwind component Calm

- A—1,500 feet.
- B—1,750 feet.
- C—2,000 feet.

3706. (Refer to Figure 41.) Determine the total distance required for takeoff to clear a 50-foot obstacle.

OAT Std
Pressure altitude Sea level
Takeoff weight 2,700 lb
Headwind component Calm

- A—1,000 feet.
- B—1,400 feet.
- C—1,700 feet.

3707. (Refer to Figure 41.) Determine the approximate ground roll distance required for takeoff.

OAT 100°F
Pressure altitude 2,000 ft
Takeoff weight 2,750 lb
Headwind component Calm

- A—1,150 feet.
- B—1,300 feet.
- C—1,800 feet.

3708. (Refer to Figure 41.) Determine the approximate ground roll distance required for takeoff.

OAT 90°F
Pressure altitude 2,000 ft
Takeoff weight 2,500 lb
Headwind component 20 kts

- A—650 feet.
- B—850 feet.
- C—1,000 feet.

3709. FAA advisory circulars (some free, others at cost) are available to all pilots and are obtained by

- A—distribution from the nearest FAA district office.
- B—ordering those desired from the Government Printing Office.
- C—subscribing to the Federal Register.

3710. Prior to starting each maneuver, pilots should

- A—check altitude, airspeed, and heading indications.
- B—visually scan the entire area for collision avoidance.
- C—announce their intentions on the nearest CTAF.

3711. The most important rule to remember in the event of a power failure after becoming airborne is to

- A—immediately establish the proper gliding attitude and airspeed.
- B—quickly check the fuel supply for possible fuel exhaustion.
- C—determine the wind direction to plan for the forced landing.

3712. What is the most effective way to use the eyes during night flight?

- A—Look only at far away, dim lights.
- B—Scan slowly to permit offcenter viewing.
- C—Concentrate directly on each object for a few seconds.

3713. The best method to use when looking for other traffic at night is to

- A—look to the side of the object and scan slowly.
- B—scan the visual field very rapidly.
- C—look to the side of the object and scan rapidly.

3714. The most effective method of scanning for other aircraft for collision avoidance during nighttime hours is to use

- A—regularly spaced concentration on the 3-, 9-, and 12-o'clock positions.
- B—a series of short, regularly spaced eye movements to search each 30-degree sector.
- C—peripheral vision by scanning small sectors and utilizing offcenter viewing.

3715. During a night flight, you observe a steady red light and a flashing red light ahead and at the same altitude. What is the general direction of movement of the other aircraft?

- A—The other aircraft is crossing to the left.
- B—The other aircraft is crossing to the right.
- C—The other aircraft is approaching head-on.

3716. During a night flight, you observe a steady white light and a flashing red light ahead and at the same altitude. What is the general direction of movement of the other aircraft?

- A—The other aircraft is flying away from you.
- B—The other aircraft is crossing to the left.
- C—The other aircraft is crossing to the right.

3717. During a night flight, you observe steady red and green lights ahead and at the same altitude. What is the general direction of movement of the other aircraft?

- A—The other aircraft is crossing to the left.
- B—The other aircraft is flying away from you.
- C—The other aircraft is approaching head-on.

3718. Airport taxiway edge lights are identified at night by

- A—white directional lights.
- B—blue omnidirectional lights.
- C—alternate red and green lights.

3719. VFR approaches to land at night should be accomplished

- A—at a higher airspeed.
- B—with a steeper descent.
- C—the same as during daytime.

3720. (Refer to Figure 44.) What action, if any, should be taken for lateral balance if the helicopter is loaded as follows?

Gross weight 1,800 lb
 Pilot 140 lb, 13.5 in. left of "0" MOM arm
 Copilot 180 lb, 13.5 in. right of "0" MOM arm

- A—Add 10 pounds of weight to the pilot's side.
- B—Decrease the gross weight 50 pounds.
- C—No action is required.

3721. (Refer to Figure 44.) What action should be taken for lateral balance if the helicopter is loaded as follows?

Gross weight 1,800 lb
 Pilot 100 lb, 13.5 in. left of "0" MOM arm
 Copilot 200 lb, 13.5 in. right of "0" MOM arm

- A—Add 50 pounds of weight to the pilot's side.
- B—Decrease the gross weight 50 pounds.
- C—No action is required.

3722. (Refer to Figure 42.) Determine the weight and balance of the helicopter.

	WEIGHT (LB)	ARM (IN)	MOMENT (LB-IN)
Empty weight	1,495.0	101.4	151,593.0
Pilot and one passenger	350.0	64.0	—
Fuel (40 gal usable)	—	96.0	—

- A—Over gross weight limit, but within CG limit.
- B—Within gross weight limit and at the aft CG limit.
- C—Over gross weight limit and exceeds the aft CG limit.

3723. (Refer to Figure 43.) Determine if the helicopter's CG is within limits.

	WEIGHT (LB)	MOMENT (1000)
Empty weight (including oil)	1,025	102,705
Pilot and passenger	345	—
Fuel, 35 gal	—	—

- A—Out of limits forward.
- B—Within limits.
- C—Out of limits aft.

3724. (Refer to Figure 43.) What effect does adding a 185-pound passenger have on the CG, if prior to boarding the passenger, the helicopter weighed 1,380 pounds and the moment is 136,647.5 pound-inches?

- A—The CG is moved forward 1.78 inches.
- B—The CG is moved aft 1.78 inches.
- C—The CG is moved forward 2.36 inches.

3725. How is the CG of the helicopter affected after a fuel burn of 20 gallons?

Gross weight prior to fuel burn 2,050 lb
 Moment 195,365 lb-in
 Fuel arm 96.9 in

- A—CG shifts forward 1.0 inch.
- B—CG shifts forward 0.1 inch.
- C—CG shifts aft 1.0 inch.

3726. (Refer to Figure 43.) How is the CG of the helicopter affected when all of the auxiliary fuel is burned off?

Gross weight prior to fuel burn 1,660 lb
 Moment 159,898.5 lb-in

- A—CG moves aft 0.12 inch.
- B—CG moves forward 0.78 inch.
- C—CG moves forward 1.07 inches.

3727. (Refer to Figure 44.) Calculate the weight and balance of the helicopter, and determine if the CG is within limits.

	WEIGHT (LB)	ARM (IN)	MOMENT (100)
Empty weight	1,495.0	101.4	1,515.93
Oil, 8 qt	—	100.5	—
Fuel, 40 gal	—	96.0	—
Pilot	160.0	64.0	—

- A—CG 90.48 inches, out of limits forward.
- B—CG 95.32 inches, within limits.
- C—CG 97.58 inches, within limits.

3728. (Refer to Figure 44.) Determine if the helicopter weight and balance is within limits.

	WEIGHT (LB)	ARM (IN)	MOMENT (100)
Empty weight	1,495.0	101.4	1,515.93
Oil, 8 qt	—	100.5	—
Fuel, 40 gal	—	96.0	—
Pilot and copilot	300.0	64.0	—

- A—CG 95.2 inches, within limits.
- B—CG 95.3 inches, weight and CG out of limits.
- C—CG 95.4 inches, within limits.

3729. (Refer to Figures 45 and 46.) What is the new CG of the gyroplane after a 10-gallon fuel burn if the original weight was 1,450 pounds and the MOM/1000 was 108 pound-inches?

- A—Out of limits forward.
- B—Out of limits aft.
- C—Within limits near the forward limit.

3730. (Refer to Figures 45 and 46.) What is the condition of the weight and balance of the gyroplane as loaded?

	WEIGHT (LB)	MOMENT (1000)
Empty weight	1,074	85.6
Oil, 6 qt	—	1.0
Pilot and passenger	247	—
Fuel, 12 gal	—	—
Baggage	95	—

- A—Within limits.
- B—Overweight.
- C—Out of limits aft.

3731. (Refer to Figures 45 and 46.) Approximately how much baggage, if any, may be carried in the gyroplane, without exceeding weight and balance limits?

	WEIGHT (LB)	MOMENT (1000)
Empty weight	1,074	85.6
Oil, 6 qt	—	1.0
Fuel, Full	—	—
Pilot (FWD)	224	—

- A—None, overweight.
- B—70 pounds.
- C—100 pounds.

3732. (Refer to Figure 47.) What is the best rate-of-climb speed for the helicopter?

- A—24 MPH.
- B—40 MPH.
- C—57 MPH.

3733. With calm wind conditions, which flight operation would require the most power?

- A—A right-hovering turn.
- B—A left-hovering turn.
- C—Hovering out of ground effect.

3734. If the pilot were to make a near-vertical power approach into a confined area with the airspeed near zero, what hazardous condition may develop?

- A—Ground resonance when ground contact is made.
- B—A settling-with-power condition.
- C—Blade stall vibration could develop.

3735. (Refer to Figure 47.) The airspeed range to avoid while flying in ground effect is

- A—25 – 40 MPH.
- B—25 – 57 MPH.
- C—40 MPH and above.

3736. (Refer to Figure 47.) Which airspeed/altitude combination should be avoided during helicopter operations?

- A—30 MPH/200 feet AGL.
- B—50 MPH/300 feet AGL.
- C—60 MPH/20 feet AGL.

3737. (Refer to Figure 47.) Which airspeed/altitude combination should be avoided during helicopter operations?

- A—20 MPH/200 feet AGL.
- B—35 MPH/175 feet AGL.
- C—40 MPH/75 feet AGL.

3738. If anti-torque failure occurred during the landing touchdown, what could be done to help straighten out a left yaw prior to touchdown?

- A—A flare to zero airspeed and a vertical descent to touchdown should be made.
- B—Apply available throttle to help swing the nose to the right just prior to touchdown.
- C—A normal running landing should be made.

3739. Which flight technique is recommended for use during hot weather?

- A—Use minimum allowable RPM and maximum allowable manifold pressure during all phases of flight.
- B—During hovering flight, maintain minimum engine RPM during left pedal turns, and maximum engine RPM during right pedal turns.
- C—During takeoff, accelerate slowly into forward flight.

3740. Under what condition should a helicopter pilot consider using a running takeoff?

- A—When gross weight or density altitude prevents a sustained hover at normal hovering altitude.
- B—When a normal climb speed is assured between 10 and 20 feet.
- C—When the additional airspeed can be quickly converted to altitude.

3741. What action should the pilot take if engine failure occurs at altitude?

- A—Open the throttle as the collective pitch is raised.
- B—Reduce cyclic back stick pressure during turns.
- C—Lower the collective pitch control, as necessary, to maintain rotor RPM.

3742. Which is a precaution to be observed during an autorotative descent?

- A—Normally, the airspeed is controlled with the collective pitch.
- B—Normally, only the cyclic control is used to make turns.
- C—Do not allow the rate of descent to get too low at zero airspeed.

3743. The proper action to initiate a quick stop is to apply

- A—forward cyclic and lower the collective pitch.
- B—aft cyclic and raise the collective pitch.
- C—aft cyclic and lower the collective pitch.

3744. What is the procedure for a slope landing?

- A—When the downslope skid is on the ground, hold the collective pitch at the same position.
- B—Minimum RPM shall be held until the full weight of the helicopter is on the skid.
- C—When parallel to the slope, slowly lower the up-slope skid to the ground prior to lowering the downslope skid.

3745. Takeoff from a slope is normally accomplished by

- A—moving the cyclic in a direction away from the slope.
- B—bringing the helicopter to a level attitude before completely leaving the ground.
- C—moving the cyclic stick to a full up position as the helicopter nears a level attitude.

3746. Which action would be appropriate for confined area operations?

- A—Takeoffs and landings must be made into the wind.
- B—Plan the flightpath over areas suitable for a forced landing.
- C—A very steep angle of descent should be used to land on the selected spot.

3747. If possible, when departing a confined area, what type of takeoff is preferred?

- A—A normal takeoff from a hover.
- B—A vertical takeoff.
- C—A normal takeoff from the surface.

3748. Which is a correct general rule for pinnacle and ridgeline operations?

- A—Gaining altitude on takeoff is more important than gaining airspeed.
- B—The approach path to a ridgeline is usually perpendicular to the ridge.
- C—A climb to a pinnacle or ridgeline should be performed on the upwind side.

3749. Before beginning a confined area or pinnacle landing, the pilot should first

- A—execute a high reconnaissance.
- B—execute a low reconnaissance.
- C—fly around the area to discover areas of turbulence.

3750. What minimum upward current must a glider encounter to maintain altitude?

- A—At least 2 feet per second.
- B—The same as the glider's sink rate.
- C—The same as the adjacent down currents.

3751. On which side of a rocky knoll, that is surrounded by vegetation, should a pilot find the best thermals?

- A—On the side facing the Sun.
- B—On the downwind side.
- C—Exactly over the center.

3752. What is one recommended method for locating thermals?

- A—Fly an ever increasing circular path.
- B—Maintain a straight track downwind.
- C—Look for converging streamers of dust or smoke.

3753. What is a recommended procedure for entering a dust devil for soaring?

- A—Enter above 500 feet and circle the edge in the same direction as the rotation.
- B—Enter below 500 feet and circle the edge opposite the direction of rotation.
- C—Enter at or above 500 feet and circle the edge opposite the direction of rotation.

3754. What is an important precaution when soaring in a dust devil?

- A—Avoid the eye of the vortex.
- B—Avoid the clear area at the outside edge of the dust.
- C—Maintain the same direction as the rotation of the vortex.

3755. What is the best visual indication of a thermal?

- A—Fragmented cumulus clouds with concave bases.
- B—Smooth cumulus clouds with concave bases.
- C—Scattered to broken sky with cumulus clouds.

3756. How can a pilot locate bubble thermals?

- A—Look for wet areas where recent showers have occurred.
- B—Look for birds that are soaring in areas of intermittent heating.
- C—Fly the area just above the boundary of a temperature inversion.

3757. Where may the most favorable type thermals for cross-country soaring be found?

- A—Just ahead of a warm front.
- B—Along thermal streets.
- C—Under mountain waves.

3758. Where and under what condition can enough lift be found for soaring when the weather is generally stable?

- A—On the upwind side of hills or ridges with moderate winds present.
- B—In mountain waves that form on the upwind side of the mountains.
- C—Over isolated peaks when strong winds are present.

3759. To use VHF/DF facilities for assistance in locating an aircraft's position, the aircraft must have a

- A—VHF transmitter and receiver.
- B—4096-code transponder.
- C—VOR receiver and DME.

3760. A slightly high glide slope indication from a precision approach path indicator is

- A—four white lights.
- B—three white lights and one red light.
- C—two white lights and two red lights.

3761. A below glide slope indication from a tri-color VASI is a

- A—red light signal.
- B—pink light signal.
- C—green light signal.

3762. An above glide slope indication from a tri-color VASI is

- A—a white light signal.
- B—a green light signal.
- C—an amber light signal.

3763. An on glide slope indication from a tri-color VASI is

- A—a white light signal.
- B—a green light signal.
- C—an amber light signal.

3764. A below glide slope indication from a pulsating approach slope indicator is a

- A—pulsating white light.
- B—steady white light.
- C—pulsating red light.

3765. (Refer to Figure 48.) Illustration A indicates that the aircraft is

- A—below the glide slope.
- B—on the glide slope.
- C—above the glide slope.

3766. (Refer to Figure 48.) VASI lights as shown by illustration C indicate that the airplane is

- A—off course to the left.
- B—above the glide slope.
- C—below the glide slope.

3767. (Refer to Figure 48.) While on final approach to a runway equipped with a standard 2-bar VASI, the lights appear as shown by illustration D. This means that the aircraft is

- A—above the glide slope.
- B—below the glide slope.
- C—on the glide slope.

3768. To set the high intensity runway lights on medium intensity, the pilot should click the microphone seven times, then click it

- A—one time.
- B—three times.
- C—five times.

3769. An airport's rotating beacon operated during daylight hours indicates

- A—there are obstructions on the airport.
- B—that weather at the airport located in Class D airspace is below basic VFR weather minimums.
- C—the Air Traffic Control tower is not in operation.

3770. A lighted heliport may be identified by a

- A—green, yellow, and white rotating beacon.
- B—flashing yellow light.
- C—blue lighted square landing area.

3771. A military air station can be identified by a rotating beacon that emits

- A—white and green alternating flashes.
- B—two quick, white flashes between green flashes.
- C—green, yellow, and white flashes.

3772. How can a military airport be identified at night?

- A—Alternate white and green light flashes.
- B—Dual peaked (two quick) white flashes between green flashes.
- C—White flashing lights with steady green at the same location.

3773. (Refer to Figure 49.) That portion of the runway identified by the letter A may be used for

- A—landing.
- B—taxiing and takeoff.
- C—taxiing and landing.

3774. (Refer to Figure 49.) According to the airport diagram, which statement is true?

- A—Runway 30 is equipped at position E with emergency arresting gear to provide a means of stopping military aircraft.
- B—Takeoffs may be started at position A on Runway 12, and the landing portion of this runway begins at position B.
- C—The takeoff and landing portion of Runway 12 begins at position B.

3775. (Refer to Figure 49.) What is the difference between area A and area E on the airport depicted?

- A—"A" may be used for taxi and takeoff; "E" may be used only as an overrun.
- B—"A" may be used for all operations except heavy aircraft landings; "E" may be used only as an overrun.
- C—"A" may be used only for taxiing; "E" may be used for all operations except landings.

3776. (Refer to Figure 49.) Area C on the airport depicted is classified as a

- A—stabilized area.
- B—multiple heliport.
- C—closed runway.

3777. (Refer to Figure 50.) The arrows that appear on the end of the north/south runway indicate that the area

- A—may be used only for taxiing.
- B—is usable for taxiing, takeoff, and landing.
- C—cannot be used for landing, but may be used for taxiing and takeoff.

3778. The numbers 9 and 27 on a runway indicate that the runway is oriented approximately

- A—009° and 027° true.
- B—090° and 270° true.
- C—090° and 270° magnetic.

3779. The vertical limit of Class C airspace above the primary airport is normally

- A—1,200 feet AGL.
- B—3,000 feet AGL.
- C—4,000 feet AGL.

3780. The normal radius of the outer area of Class C airspace is

- A—5 nautical miles.
- B—15 nautical miles.
- C—20 nautical miles.

3781. All operations within Class C airspace must be in

- A—accordance with instrument flight rules.
- B—compliance with ATC clearances and instructions.
- C—an aircraft equipped with a 4096-code transponder with Mode C encoding capability.

3782. Under what condition may an aircraft operate from a satellite airport within Class C airspace?

- A—The pilot must file a flight plan prior to departure.
- B—The pilot must monitor ATC until clear of the Class C airspace.
- C—The pilot must contact ATC as soon as practicable after takeoff.

3783. Under what condition, if any, may pilots fly through a restricted area?

- A—When flying on airways with an ATC clearance.
- B—With the controlling agency's authorization.
- C—Regulations do not allow this.

3784. A balloon flight through a restricted area is

- A—permitted at certain times, but only with prior permission by the appropriate authority.
- B—permitted anytime, but caution should be exercised because of high-speed military aircraft.
- C—never permitted.

3785. What action should a pilot take when operating under VFR in a Military Operations Area (MOA)?

- A—Obtain a clearance from the controlling agency prior to entering the MOA.
- B—Operate only on the airways that transverse the MOA.
- C—Exercise extreme caution when military activity is being conducted.

3786. Responsibility for collision avoidance in an alert area rests with

- A—the controlling agency.
- B—all pilots.
- C—Air Traffic Control.

3787. The lateral dimensions of Class D airspace are based on

- A—the number of airports that lie within the Class D airspace.
- B—5 statute miles from the geographical center of the primary airport.
- C—the instrument procedures for which the controlled airspace is established.

3788. A non-tower satellite airport, within the same Class D airspace as that designated for the primary airport, requires radio communications be established and maintained with the

- A—satellite airport's UNICOM.
- B—associated Flight Service Station.
- C—primary airport's control tower.

3789. Prior to entering an Airport Advisory Area, a pilot should

- A—monitor ATIS for weather and traffic advisories.
- B—contact approach control for vectors to the traffic pattern.
- C—contact the local FSS for airport and traffic advisories.

3790. Select the UNICOM frequencies normally assigned to stations at landing areas used exclusively as heliports.

- A—122.75 and 123.65 MHz.
- B—123.0 and 122.95 MHz.
- C—123.05 and 123.075 MHz.

3791. Automatic Terminal Information Service (ATIS) is the continuous broadcast of recorded information concerning

- A—pilots of radar-identified aircraft whose aircraft is in dangerous proximity to terrain or to an obstruction.
- B—nonessential information to reduce frequency congestion.
- C—noncontrol information in selected high-activity terminal areas.

3792. An ATC radar facility issues the following advisory to a pilot flying on a heading of 090°:

"TRAFFIC 3 O'CLOCK, 2 MILES, WESTBOUND..."

Where should the pilot look for this traffic?

- A—East.
- B—South.
- C—West.

3793. An ATC radar facility issues the following advisory to a pilot flying on a heading of 360°:

"TRAFFIC 10 O'CLOCK, 2 MILES, SOUTHBOUND..."

Where should the pilot look for this traffic?

- A—Northwest.
- B—Northeast.
- C—Southwest.

3794. An ATC radar facility issues the following advisory to a pilot during a local flight:

"TRAFFIC 2 O'CLOCK, 5 MILES, NORTHBOUND..."

Where should the pilot look for this traffic?

- A—Between directly ahead and 90° to the left.
- B—Between directly behind and 90° to the right.
- C—Between directly ahead and 90° to the right.

3795. An ATC radar facility issues the following advisory to a pilot flying north in a calm wind:

"TRAFFIC 9 O'CLOCK, 2 MILES, SOUTHBOUND..."

Where should the pilot look for this traffic?

- A—South.
- B—North.
- C—West.

3796. Basic radar service in the terminal radar program is best described as

- A—traffic advisories and limited vectoring to VFR aircraft.
- B—mandatory radar service provided by the Automated Radar Terminal System (ARTS) program.
- C—wind-shear warning at participating airports.

3797. From whom should a departing VFR aircraft request Stage II Terminal Radar Advisory Service during ground operations?

- A—Clearance delivery.
- B—Tower, just before takeoff.
- C—Ground control, on initial contact.

3798. Stage III Service in the terminal radar program provides

- A—IFR separation (1,000 feet vertical and 3 miles lateral) between all aircraft.
- B—warning to pilots when their aircraft are in unsafe proximity to terrain, obstructions, or other aircraft.
- C—sequencing and separation for participating VFR aircraft.

3799. Which initial action should a pilot take prior to entering Class C airspace?

- A—Contact approach control on the appropriate frequency.
- B—Contact the tower and request permission to enter.
- C—Contact the FSS for traffic advisories.

3800. When making routine transponder code changes, pilots should avoid inadvertent selection of which codes?

- A—0700, 1700, 7000.
- B—1200, 1500, 7000.
- C—7500, 7600, 7700.

3801. When operating under VFR below 18,000 feet MSL, unless otherwise authorized, what transponder code should be selected?

- A—1200.
- B—7600.
- C—7700.

3802. Unless otherwise authorized, if flying a transponder-equipped aircraft, a recreational pilot should squawk which VFR code?

- A—1200.
- B—7600.
- C—7700.

3803. If Air Traffic Control advises that radar service is terminated when the pilot is departing Class C airspace, the transponder should be set to code

- A—0000.
- B—1200.
- C—4096.

3804. If the aircraft's radio fails, what is the recommended procedure when landing at a controlled airport?

- A—Observe the traffic flow, enter the pattern, and look for a light signal from the tower.
- B—Enter a crosswind leg and rock the wings.
- C—Flash the landing lights and cycle the landing gear while circling the airport.

3805. (Refer to Figure 50.) Select the proper traffic pattern and runway for landing.

- A—Left-hand traffic and Runway 18.
- B—Right-hand traffic and Runway 18.
- C—Left-hand traffic and Runway 22.

3806. (Refer to Figure 50.) If the wind is as shown by the landing direction indicator, the pilot should land on

- A—Runway 18 and expect a crosswind from the right.
- B—Runway 22 directly into the wind.
- C—Runway 36 and expect a crosswind from the right.

3807. (Refer to Figure 51.) The segmented circle indicates that the airport traffic is

- A—left-hand for Runway 35 and right-hand for Runway 17.
- B—left-hand for Runway 17 and right-hand for Runway 35.
- C—right-hand for Runway 9 and left-hand for Runway 27.

3808. (Refer to Figure 51.) The traffic patterns indicated in the segmented circle have been arranged to avoid flights over an area to the

- A—south of the airport.
- B—north of the airport.
- C—southeast of the airport.

3809. (Refer to Figure 51.) The segmented circle indicates that a landing on Runway 26 will be with a

- A—right-quartering headwind.
- B—left-quartering headwind.
- C—right-quartering tailwind.

3810. (Refer to Figure 51.) Which runway and traffic pattern should be used as indicated by the wind cone in the segmented circle?

- A—Right-hand traffic on Runway 8.
- B—Right-hand traffic on Runway 17.
- C—Left-hand traffic on Runway 35.

3811. After landing at a tower-controlled airport, when should the pilot contact ground control?

- A—When advised by the tower to do so.
- B—Prior to turning off the runway.
- C—After reaching a taxiway that leads directly to the parking area.

3812. If instructed by ground control to taxi to Runway 9, the pilot may proceed

- A—via taxiways and across runways to, but not onto, Runway 9.
- B—to the next intersecting runway where further clearance is required.
- C—via taxiways and across runways to Runway 9, where an immediate takeoff may be made.

3813. What ATC facility should the pilot contact to receive a special VFR departure clearance in Class D airspace?

- A—Automated Flight Service Station.
- B—Air Traffic Control Tower.
- C—Air Route Traffic Control Center.

3814. What procedure is recommended when climbing or descending VFR on an airway?

- A—Execute gentle banks, left and right for continuous visual scanning of the airspace.
- B—Advise the nearest FSS of the altitude changes.
- C—Fly away from the centerline of the airway before changing altitude.

3815. (Refer to Figure 52.) If more than one cruising altitude is intended, which should be entered in block 7 of the flight plan?

- A—Initial cruising altitude.
- B—Highest cruising altitude.
- C—Lowest cruising altitude.

3816. (Refer to Figure 52.) What information should be entered in block 9 for a VFR day flight?

- A—The name of the airport of first intended landing.
- B—The name of destination airport if no stopover for more than 1 hour is anticipated.
- C—The name of the airport where the aircraft is based.

3817. (Refer to Figure 52.) What information should be entered in block 12 for a VFR day flight?

- A—The estimated time en route plus 30 minutes.
- B—The estimated time en route plus 45 minutes.
- C—The amount of usable fuel on board expressed in time.

3818. How should a VFR flight plan be closed at the completion of the flight at a controlled airport?

- A—The tower will automatically close the flight plan when the aircraft turns off the runway.
- B—The pilot must close the flight plan with the nearest FSS or other FAA facility upon landing.
- C—The tower will relay the instructions to the nearest FSS when the aircraft contacts the tower for landing.

3819. When activated, an emergency locator transmitter (ELT) transmits on

- A—118.0 and 118.8 MHz.
- B—121.5 and 243.0 MHz.
- C—123.0 and 119.0 MHz.

3820. When must the battery in an emergency locator transmitter (ELT) be replaced (or recharged if the battery is rechargeable)?

- A—After one-half the battery's useful life.
- B—During each annual and 100-hour inspection.
- C—Every 24 calendar months.

3821. When may an emergency locator transmitter (ELT) be tested?

- A—Anytime.
- B—At 15 and 45 minutes past the hour.
- C—During the first 5 minutes after the hour.

3822. Which procedure is recommended to ensure that the emergency locator transmitter (ELT) has not been activated?

- A—Turn off the aircraft ELT after landing.
- B—Ask the airport tower if they are receiving an ELT signal.
- C—Monitor 121.5 before engine shutdown.

3823. Below FL180, en route weather advisories should be obtained from an FSS on

- A—122.0 MHz.
- B—122.1 MHz.
- C—123.6 MHz.

3824. Wingtip vortices are created only when an aircraft is

- A—operating at high airspeeds.
- B—heavily loaded.
- C—developing lift.

3825. The greatest vortex strength occurs when the generating aircraft is

- A—light, dirty, and fast.
- B—heavy, dirty, and fast.
- C—heavy, clean, and slow.

3826. Wingtip vortices created by large aircraft tend to

- A—sink below the aircraft generating turbulence.
- B—rise into the traffic pattern.
- C—rise into the takeoff or landing path of a crossing runway.

3827. When taking off or landing at an airport where heavy aircraft are operating, one should be particularly alert to the hazards of wingtip vortices because this turbulence tends to

- A—rise from a crossing runway into the takeoff or landing path.
- B—rise into the traffic pattern area surrounding the airport.
- C—sink into the flightpath of aircraft operating below the aircraft generating the turbulence.

3828. The wind condition that requires maximum caution when avoiding wake turbulence on landing is a

- A—light, quartering headwind.
- B—light, quartering tailwind.
- C—strong headwind.

3829. When landing behind a large aircraft, the pilot should avoid wake turbulence by staying

- A—above the large aircraft's final approach path and landing beyond the large aircraft's touchdown point.
- B—below the large aircraft's final approach path and landing before the large aircraft's touchdown point.
- C—above the large aircraft's final approach path and landing before the large aircraft's touchdown point.

3830. When departing behind a heavy aircraft, the pilot should avoid wake turbulence by maneuvering the aircraft

- A—below and downwind from the heavy aircraft.
- B—above and upwind from the heavy aircraft.
- C—below and upwind from the heavy aircraft.

3831. Pilots flying over a national wildlife refuge are requested to fly no lower than

- A—1,000 feet AGL.
- B—2,000 feet AGL.
- C—3,000 feet AGL.

3832. Large accumulations of carbon monoxide in the human body result in

- A—tightness across the forehead.
- B—loss of muscular power.
- C—an increased sense of well-being.

3833. What effect does haze have on the ability to see traffic or terrain features during flight?

- A—Haze causes the eyes to focus at infinity.
- B—The eyes tend to overwork in haze and do not detect relative movement easily.
- C—All traffic or terrain features appear to be farther away than their actual distance.

3834. The most effective method of scanning for other aircraft for collision avoidance during daylight hours is to use

- A—regularly spaced concentration on the 3-, 9-, and 12-o'clock positions.
- B—a series of short, regularly spaced eye movements to search each 10-degree sector.
- C—peripheral vision by scanning small sectors and utilizing offcenter viewing.

3835. Which technique should a pilot use to scan for traffic to the right and left during straight-and-level flight?

- A—Systematically focus on different segments of the sky for short intervals.
- B—Concentrate on relative movement detected in the peripheral vision area.
- C—Continuous sweeping of the windshield from right to left.

3836. How can you determine if another aircraft is on a collision course with your aircraft?

- A—The other aircraft will always appear to get larger and closer at a rapid rate.
- B—The nose of each aircraft is pointed at the same point in space.
- C—There will be no apparent relative motion between your aircraft and the other aircraft.

3837. An ATC clearance provides

- A—priority over all other traffic.
- B—adequate separation from all traffic.
- C—authorization to proceed under specified traffic conditions in controlled airspace.

3838. (Refer to Figure 53.) When approaching Lincoln Municipal from the west at noon for the purpose of landing, initial communications should be with

- A—Lincoln Approach Control on 124.0 MHz.
- B—Minneapolis Center on 128.75 MHz.
- C—Lincoln Tower on 118.5 MHz.

3839. (Refer to Figure 53.) Which type radar service is provided to VFR aircraft at Lincoln Municipal?

- A—Sequencing to the primary Class C airport and standard separation.
- B—Sequencing to the primary Class C airport and conflict resolution so that radar targets do not touch, or 1,000 feet vertical separation.
- C—Sequencing to the primary Class C airport, traffic advisories, conflict resolution, and safety alerts.

3840. (Refer to Figure 53.) What is the recommended communications procedure for landing at Lincoln Municipal during the hours when the tower is not in operation?

- A—Monitor airport traffic and announce your position and intentions on 118.5 MHz.
- B—Contact UNICOM on 122.95 MHz for traffic advisories.
- C—Monitor ATIS for airport conditions, then announce your position on 122.95 MHz.

3841. (Refer to Figure 53.) Where is Loup City Municipal located with relation to the city?

- A—Northeast approximately 3 miles.
- B—Northwest approximately 1 mile.
- C—East approximately 10 miles.

3842. (Refer to Figure 53.) Traffic patterns in effect at Lincoln Municipal are

- A—to the right on Runway 17L and Runway 35L; to the left on Runway 17R and Runway 35R.
- B—to the left on Runway 17L and Runway 35L; to the right on Runway 17R and Runway 35R.
- C—to the right on Runways 14 – 32.

3843. The letters VHF/DF appearing in the Airport/Facility Directory for a certain airport indicate that

- A—this airport is designated as an airport of entry.
- B—the Flight Service Station has equipment with which to determine your direction from the station.
- C—this airport has a direct-line phone to the Flight Service Station.

3844. Which statement best defines hypoxia?

- A—A state of oxygen deficiency in the body.
- B—An abnormal increase in the volume of air breathed.
- C—A condition of gas bubble formation around the joints or muscles.

3845. Rapid or extra deep breathing while using oxygen can cause a condition known as

- A—hyperventilation.
- B—aerosinusitis.
- C—aerotitis.

3846. Which would most likely result in hyperventilation?

- A—Emotional tension, anxiety, or fear.
- B—The excessive consumption of alcohol.
- C—An extremely slow rate of breathing and insufficient oxygen.

3847. A pilot should be able to overcome the symptoms or avoid future occurrences of hyperventilation by

- A—closely monitoring the flight instruments to control the airplane.
- B—slowing the breathing rate, breathing into a bag, or talking aloud.
- C—increasing the breathing rate in order to increase lung ventilation.

3848. Susceptibility to carbon monoxide poisoning increases as

- A—altitude increases.
- B—altitude decreases.
- C—air pressure increases.

3849. What preparation should a pilot make to adapt the eyes for night flying?

- A—Wear sunglasses after sunset until ready for flight.
- B—Avoid red lights at least 30 minutes before the flight.
- C—Avoid bright white lights at least 30 minutes before the flight.

3850. The danger of spatial disorientation during flight in poor visual conditions may be reduced by

- A—shifting the eyes quickly between the exterior visual field and the instrument panel.
- B—having faith in the instruments rather than taking a chance on the sensory organs.
- C—leaning the body in the opposite direction of the motion of the aircraft.

3851. A state of temporary confusion resulting from misleading information being sent to the brain by various sensory organs is defined as

- A—spatial disorientation.
- B—hyperventilation.
- C—hypoxia.

3852. Pilots are more subject to spatial disorientation if

- A—they ignore the sensations of muscles and inner ear.
- B—body signals are used to interpret flight attitude.
- C—eyes are moved often in the process of cross-checking the flight instruments.

3853. If a pilot experiences spatial disorientation during flight in a restricted visibility condition, the best way to overcome the effect is to

- A—rely upon the aircraft instrument indications.
- B—concentrate on yaw, pitch, and roll sensations.
- C—consciously slow the breathing rate until symptoms clear and then resume normal breathing rate.

3854. FAA advisory circulars containing subject matter specifically related to Airmen are issued under which subject number?

- A—60.
- B—70.
- C—90.

3855. FAA advisory circulars containing subject matter specifically related to Airspace are issued under which subject number?

- A—60.
- B—70.
- C—90.

3856. FAA advisory circulars containing subject matter specifically related to Air Traffic Control and General Operations are issued under which subject number?

- A—60.
- B—70.
- C—90.

3857. Which is an advantage of using a CG hook for a winch tow rather than the nose hook?

- A—A greater percent of the line length can be used to reach altitude.
- B—Maximum release altitude is limited.
- C—It is the safest method of launching.

3858. To stop pitch oscillation (porpoising) during a winch launch, the pilot should

- A—release back pressure and then pull back against the cycle of pitching oscillation to get in phase with the undulations.
- B—signal the ground crew to increase the speed of the tow.
- C—relax the back pressure on the control stick and shallow the angle of climb.

3859. A pilot plans to fly solo in the front seat of a two-place glider which displays the following placards on the instrument panel:

MINIMUM PILOT WEIGHT: 135 LB
MAXIMUM PILOT WEIGHT: 220 LB

NOTE: Seat ballast should be used as necessary.

The recommended towing speed for all tows is 55 – 65 knots. What action should be taken if the pilot's weight is 115 pounds?

- A—Add 20 pounds of seat ballast to the rear seat.
- B—Add 55 pounds of seat ballast to obtain the average pilot weight of 170 pounds.
- C—Add 20 pounds of seat ballast.

3860. A pilot plans to fly solo in the front seat of a two-place glider which displays the following placards on the instrument panel:

MINIMUM PILOT WEIGHT: 135 LB

MAXIMUM PILOT WEIGHT: 220 LB

NOTE: Seat ballast should be used as necessary.

The recommended towing speed for all tows is 55 - 65 knots. What action should be taken if the pilot's weight is 125 pounds?

- A—Add 10 pounds of seat ballast to the rear seat.
- B—Add 10 pounds of seat ballast.
- C—Add 45 pounds of seat ballast to obtain the average pilot weight of 170 pounds.

3861. (Refer to Figure 54.) Calculate the weight and balance of the glider, and determine if the CG is within limits.

Pilot (fwd seat) 160 lb
Passenger (aft seat) 185 lb

- A—CG 71.65 inches aft of datum — out of limits forward.
- B—CG 79.67 inches aft of datum — within limits.
- C—CG 83.43 inches aft of datum — within limits.

3862. (Refer to Figure 54.) How is the CG affected if radio and oxygen equipment weighing 35 pounds is added at station 43.8? The glider weighs 945 pounds with a moment of 78,000.2 pound-inches prior to adding the equipment.

- A—CG shifts forward 0.79 inch — out of limits forward.
- B—CG shifts forward 1.38 inches — within limits.
- C—CG shifts aft 1.38 inches — out of limits aft.

3863. (Refer to Figure 54.) What is the CG of the glider if the pilot and passenger each weigh 215 pounds?

- A—74.69 inches aft of datum — out of limits forward.
- B—81.08 inches aft of datum — within limits.
- C—81.08 inches aft of datum — over maximum gross weight.

3864. (Refer to Figure 55.) How many feet will the glider sink in 1 statute mile at 53 MPH in still air?

- A—144 feet.
- B—171 feet.
- C—211 feet.

3865. (Refer to Figure 55.) At what speed will the glider attain a sink rate of 5 feet per second in still air?

- A—75 MPH.
- B—79 MPH.
- C—84 MPH.

3866. (Refer to Figure 55.) How many feet will the glider descend at minimum sink speed for 1 statute mile in still air?

- A—132 feet.
- B—170 feet.
- C—180 feet.

3867. (Refer to Figure 55.) At what speed will the glider gain the most distance while descending 1,000 feet in still air?

- A—44 MPH.
- B—53 MPH.
- C—83 MPH.

3868. (Refer to Figure 55.) What approximate lift/drag ratio will the glider attain at 68 MPH in still air?

- A—10.5:1.
- B—21.7:1.
- C—28.5:1.

3869. (Refer to Figure 56.) Illustration 2 means

- A—release towline.
- B—ready to tow.
- C—hold position.

3870. (Refer to Figure 56.) Illustration 3 means

- A—stop operations.
- B—release towline.
- C—take up slack.

3871. (Refer to Figure 56.) Which illustration is a signal to stop operation?

- A—2.
- B—3.
- C—7.

3872. (Refer to Figure 56.) Which illustration is a signal from the sailplane for the towplane to turn right?

- A—5.
- B—6.
- C—11.

3873. (Refer to Figure 56.) Which illustration is a signal that the glider is unable to release?

- A—8.
- B—10.
- C—11.

3874. (Refer to Figure 56.) Which illustration is a signal to the towplane to reduce airspeed?

- A—7.
- B—10.
- C—12.

3875. (Refer to Figure 56.) Which illustration means the towplane cannot release?

- A—6.
- B—8.
- C—9.

3876. What corrective action should the sailplane pilot take during takeoff if the towplane is still on the ground and the sailplane is airborne and drifting to the left?

- A—Crab into the wind by holding upwind (right) rudder pressure.
- B—Crab into the wind so as to maintain a position directly behind the towplane.
- C—Establish a right wing low drift correction to remain in the flightpath of the towplane.

3877. An indication that the glider has begun a turn too soon on aerotow is that the

- A—glider's nose is pulled to the outside of the turn.
- B—towplane's nose is pulled to the outside of the turn.
- C—towplane will pitch up.

3878. The sailplane has become airborne and the towplane loses power before leaving the ground. The sailplane should release immediately,

- A—and maneuver to the right of the towplane.
- B—extend the spoilers, and land straight ahead.
- C—and maneuver to the left of the towplane.

3879. What should a glider pilot do if a towline breaks below 200 feet AGL?

- A—Turn into the wind, then back to the runway for a downwind landing.
- B—Turn away from the wind, then back to the runway for a downwind landing.
- C—Land straight ahead or make slight turns to reach a suitable landing area.

3880. A pilot unintentionally enters a steep diving spiral to the left. What is the proper way to recover from this attitude without overstressing the glider?

- A—Apply up-elevator pressure to raise the nose.
- B—Apply more up-elevator pressure and then use right aileron pressure to control the overbanking tendency.
- C—Relax the back pressure and shallow the bank; then apply up-elevator pressure until the nose has been raised to the desired position.

3881. What corrective action should be taken if, while thermalling at minimum sink speed in turbulent air, the left wing drops while turning to the left?

- A—Apply more opposite (right) aileron pressure than opposite (right) rudder pressure to counteract the overbanking tendency.
- B—Apply opposite (right) rudder pressure to slow the rate of turn.
- C—Lower the nose before applying opposite (right) aileron pressure.

3882. A sailplane pilot can differentiate between a spin and a spiral dive because in a spiral dive,

- A—the speed remains constant.
- B—the G loads increase.
- C—there is a small loss of altitude in each rotation.

3883. How are forward slips normally performed?

- A—With the direction of the slip away from any crosswind that exists.
- B—With dive brakes or spoilers fully open.
- C—With rudder and aileron deflection on the same side.

3884. What would be a proper action or procedure to use if the pilot is getting too low on a cross-country flight in a sailplane?

- A—Continue on course until descending to 1,000 feet above the ground and then plan the landing approach.
- B—Fly directly into the wind and make a straight-in approach at the end of the glide.
- C—Have a suitable landing area selected upon reaching 2,000 feet AGL, and a specific field chosen upon reaching 1,500 feet AGL.

3885. Why should propane tanks not be refueled in a closed trailer or truck?

- A—Propane vapor is one and one-half times heavier than air and will linger in the floor of the truck or trailer.
- B—The propane vapor is odorless and the refuelers may be overcome by the fumes.
- C—Propane is very cold and could cause damage to the truck or trailer.

3886. Why should special precautions be taken when filling the propane bottles?

- A—Propane is transferred from the storage tanks to the propane bottles under high pressure.
- B—During transfer, propane reaches a high temperature and can cause severe burns.
- C—Propane is super-cold and may cause severe freeze burns.

3887. What constitutes the payload of a balloon?

- A—Total gross weight.
- B—Total weight of passengers, cargo, and fuel.
- C—Weight of the aircraft and equipment.

3888. (Refer to Figure 57.) The gross weight of the balloon is 1,350 pounds and the outside air temperature (OAT) is +51°F. The maximum height would be

- A—5,000 feet.
- B—8,000 feet.
- C—10,000 feet.

3889. (Refer to Figure 57.) The gross weight of the balloon is 1,200 pounds and the maximum height the pilot needs to attain is 5,000 feet. The maximum temperature to achieve this performance is

- A—+37°F.
- B—+70°F.
- C—+97°F.

3890. (Refer to Figure 58.) Determine the maximum payload for a balloon flying at 2,500 feet at an ambient temperature of 91°F.

- A—420 pounds.
- B—465 pounds.
- C—505 pounds.

3891. (Refer to Figure 58.) What is the maximum altitude for the balloon if the gross weight is 1,100 pounds and standard temperature exists at all altitudes?

- A—1,000 feet.
- B—4,000 feet.
- C—5,500 feet.

3892. (Refer to Figure 58.) What is the maximum altitude for the balloon if the gross weight is 1,000 pounds and standard temperature exists at all altitudes?

- A—4,000 feet.
- B—5,500 feet.
- C—11,000 feet.

3893. (Refer to Figure 58.) Determine the maximum weight allowable for pilot and passenger for a flight at approximately 1,000 feet with a temperature of 68°F. Launch with 20 gallons of propane.

- A—580 pounds.
- B—620 pounds.
- C—720 pounds.

3894. (Refer to Figure 58.) What is the maximum weight allowed for pilot and passengers for a flight at 5,000 feet with a standard temperature? Launch with 20 gallons of propane.

- A—670 pounds.
- B—760 pounds.
- C—1,095 pounds.

3895. All fuel tanks should be fired during preflight to determine

- A—the burner pressure and condition of the valves.
- B—that the pilot light functions properly on each tank.
- C—if there are any leaks in the tank.

3896. What is a recommended ascent upon initial launch?

- A—Maximum ascent to altitude to avoid low-level thermals.
- B—Shallow ascent to avoid flashbacks of flames as the envelope is cooled.
- C—A moderate-rate ascent to determine wind directions at different levels.

3897. What is a potential hazard when climbing at maximum rate?

- A—The envelope may collapse.
- B—Deflation ports may be forced open.
- C—The rapid flow of air may extinguish the burner and pilot light.

3898. How should a roundout from a moderate-rate ascent to level flight be made?

- A—Reduce the amount of heat gradually as the balloon is approaching altitude.
- B—Cool the envelope by venting and add heat just before arriving at altitude.
- C—Vent at altitude and add heat upon settling back down to altitude.

3899. What is one procedure for relighting the burner while in flight?

- A—Open the regulator or blast valve full open and light the pilot light.
- B—Close the tank valves, vent the fuel lines, reopen the tank valves, and light the pilot light.
- C—Open another tank valve, open the regulator or blast valve, and light the main jets with reduced flow.

3900. The windspeed is such that it is necessary to deflate the envelope as rapidly as possible during a landing. When should the deflation port (rip panel) be opened?

- A—The instant the gondola contacts the surface.
- B—As the balloon skips off the surface the first time and the last of the ballast has been discharged.
- C—Just prior to ground contact.

3901. When landing a free balloon, what should the occupants do to minimize landing shock?

- A—Be seated on the floor of the basket.
- B—Stand with knees slightly bent, in the center of the gondola, facing the direction of movement.
- C—Stand back-to-back and hold onto the load ring.

3902. Prior to a high-wind landing, the pilot in command should brief the passengers to prepare for the landing by

- A—kneeling on the floor and facing aft.
- B—crouching on the floor and jumping out of the basket upon contact with the ground.
- C—crouching while hanging on in two places, and remaining in the basket until advised otherwise.

3903. Which precaution should be exercised if confronted with the necessity of having to land a balloon when the air is turbulent?

- A—Land in any available lake close to the upwind shore.
- B—Land in the center of the largest available field.
- C—Land in the trees to absorb shock forces, thus cushioning the landing.

3904. What action is most appropriate when an envelope over-temperature condition occurs?

- A—Throw all unnecessary equipment overboard.
- B—Descend; hover in ground effect until the envelope cools.
- C—Land as soon as practical.

3905. In addition to the required documents, what carry-on equipment should be accounted for during preflight?

- A—Flotation gear.
- B—Emergency locator transmitter.
- C—Two means of burner ignition.

3906. How should a balloon fuel system be checked for leaks prior to flight?

- A—Listen and smell.
- B—Check all connections with a lighted match.
- C—Cover all connections and tubing with soapy water.

3907. In a balloon, best fuel economy in level flight can be accomplished by

- A—riding the haze line in a temperature inversion.
- B—short blasts of heat at high frequency.
- C—long blasts of heat at low frequency.

3908. The minimum size a launch site should be is at least

- A—twice the height of the balloon.
- B—100 feet for every 1 knot of wind.
- C—500 feet on the downwind side.

3909. What is a hazard of rapid descents?

- A—Wind shear can cavitate one side of the envelope, forcing air out of the mouth.
- B—The pilot light cannot remain lit with the turbulent air over the basket.
- C—Aerodynamic forces may collapse the envelope.

3910. It may be possible to make changes in the direction of flight in a hot air balloon by

- A—flying a constant atmospheric pressure gradient.
- B—operating at different flight altitudes.
- C—operating above the friction level, if there is no gradient wind.

3911. What action should be taken if a balloon encounters unforecast weather and shifts direction abruptly while in the vicinity of a thunderstorm?

- A—Land immediately.
- B—Descend to and maintain the lowest altitude possible.
- C—Ascend to an altitude which will ensure adequate obstacle clearance in all directions.

3912. To land an airship that is 250 pounds heavy when the wind is calm, the best landing can usually be made if the airship is

- A—in trim.
- B—nose heavy approximately 20°.
- C—tail heavy approximately 20°.

3913. Which takeoff procedure is considered to be most hazardous for an airship?

- A—Maintaining only 50 percent of the maximum permissible positive angle of inclination.
- B—Failing to apply full engine power properly on all takeoffs, regardless of wind.
- C—Maintaining a negative angle of inclination during takeoff after elevator response is adequate for controllability.

3914. Which action is necessary in order to perform a normal descent in an airship?

- A—Valve gas.
- B—Valve air.
- C—Take air into the aft ballonets.

3915. If an airship should experience failure of both engines during flight and neither engine can be restarted, what initial immediate action must the pilot take?

- A—The airship must be driven down to a landing before control and envelope shape are lost.
- B—The emergency auxiliary power unit must be started for electrical power to the airscoop blowers so that ballonnet inflation can be maintained.
- C—Immediate preparations to operate the airship as a free balloon are necessary.